Internet Appendix for "Measuring Firm Capabilities from Job Postings: An Application to Climate Industrial Policy"

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October 2026

Internet Appendix

This Internet Appendix supplements the paper. Part IA.I reproduces the instructions that define environmental, social, governance, and innovation work for the labeller, together with the classes of each kind of work. Part IA.II documents how the classifier of each kind of work was trained, selected, and calibrated, and lists the terms and occupations of the measures with which the paper compares it. Part IA.III describes the data, from the postings and their link to issuers to the estimation sample, the other records and the market data, and Part IA.IV describes how the accounts record labor costs. Parts IA.V to IA.VI document the days of policy news and report the detailed results behind Sections 6 to 8 and Appendix F.

IA.I The Labelling Instructions

The language model applies a written instruction to each batch of postings. Each instruction is written as a codebook of the kind that Törnberg (2024) recommends for annotation with language models, with clear definitions of the categories, examples of text in each category, and rules for ambiguous cases, so that any coder who reads it, human or model, has what is needed to apply it consistently. The four instructions share one structure. They state the task and the purpose of the labels, and then define the capability and the evidence that establishes it, a duty assigned to the hire that names an action and its object, or a

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required skill, together with what does not count, such as the employer's values, mission statements, and boilerplate. A scoring rule distinguishes work that is the main purpose of the position (2) from a peripheral duty (1) and from none (0), and a closed list of classes, each with a short description, assigns every positive posting to one function. The social and governance instructions share a section on the boundary between the two. Protecting personal data, obtaining consent, and offering individuals a remedy count as social work, whereas security controls, cyber-incident response, and the oversight and validation of AI models count as governance work. The innovation instruction lists instead the cases that its definition could leave open, such as work on a vendor's software platform, data pipelines as against dashboards, and clinical trials of products not yet approved. Each instruction then gives between 22 and 28 calibration examples, each a short description of advertised work with the label it receives. Between 8 and 14 of them in each instruction receive no score although they come close to the definition, such as a construction labourer told to wear protective equipment, for social work, a bank teller told to follow policies, for governance work, or drawings produced to a client's specifications, for innovation work. The model receives the instruction followed by the numbered postings of a batch. This part reproduces the four instructions, for environmental, social, governance, and innovation work, as they were used, except for the formatting.

IA.I.1 Environmental Work

Task. Label each numbered job posting for the environmental work the advertised role performs and the environmental skills it requires. Score Environmental (E) using the project definitions below. Use only the posting text; do not infer duties or skills from the employer, sector or title alone. Score E on its own: it does not matter whether the same work is also engineering, innovation, safety, compliance or anything else.

Purpose. These labels measure a firm's demand for environmental capacity: how much of its hiring goes to people whose work in the job manages the firm's environmental impacts, delivers environmental services, or researches, builds, operates or finances clean technology, and to people whom the job requires to bring specific environmental skills. Judge each posting by the work the person will do and the skills the job requires, never by who the employer is, what it sells or what it says it values.

Definitions and evidence.

- **Duty:** an explicitly assigned activity or responsibility, stated in the duties, role summary, or other substantive description of the role.

- **Named duty:** identifies both an action/responsibility and its relevant object or purpose; e.g., "conduct spill inspections," "design battery-management firmware," or "size PV systems for customer homes." A keyword alone is insufficient.
- **Required E skill:** a requirement, qualification or experience line naming specific environmental or clean-technology knowledge, certification or technology the person must bring; e.g., "LEED AP required," "experience with lithium-ion cell chemistry," "knowledge of RCRA hazardous-waste rules," "GHG Protocol accounting experience." Generic lines ("passion for sustainability," "awareness of environmental regulations," "follow environmental policies") are not E skills. Licences, permits and certificates that are a legal condition of practising a trade are not E skills on their own: e.g., EPA 608/609 refrigerant certification, F-gas certificates, hazardous-materials or tanker driving endorsements, HAZWOPER training.
- **E:** work, or a required skill, that does at least one of the following:
    1. manages, measures, reduces or reports environmental impacts or resource use;
    2. delivers environmental protection, remediation or environmental services;
    3. researches, designs, develops, manufactures, installs, operates, maintains or finances low-carbon and clean technologies: renewable and low-carbon generation, energy efficiency, electrification and clean vehicles, batteries and energy storage, grid integration of renewables and storage, low-carbon hydrogen and low-carbon fuels, carbon capture;
    4. assesses climate-related risk, or manages sustainability performance, disclosure or investment on environmental matters.

Clean technology counts when the duty's or skill's object is the clean technology itself: designing battery cells, building EV charging hardware, assembling electrolysers, commissioning heat pumps. Working at a clean-technology employer is not enough. An accountant, recruiter, or IT administrator at an EV maker does not inherit E, and neither does an unrelated software engineer at a solar company.

Ignore employer values, mission statements, benefits offered to applicants (EV or cycle-to-work schemes, carbon pledges), equal-opportunity boilerplate, and generic instructions to follow environmental rules. "IT environment," "Windows," software "remediation," generic "data governance," and hospital "environmental services" (EVS: housekeeping and cleaning) do not establish E. Routine housekeeping, emptying trash or recycling bins, and handling, storing or transporting materials according to procedures as part of ordinary warehouse, cleaning, driving or production work are not E duties. Collecting, hauling, treating or disposing of waste as the role's own service is E.

Work on fossil-fuel, conventional power, automotive, or grid assets is E only when the assigned work or the required skill has an E object: emissions control, spill prevention, efficiency, electrification, or integrating renewables and storage. Ordinary sales, customer service, or appointment-setting for green products is not E. Technical sales that designs or sizes the environmental system for the customer is E.

Hydrogen is E only when it is low-carbon or replaces fossil fuels: electrolysers and hydrogen made with renewable or nuclear power, hydrogen produced with carbon capture, fuel cells, hydrogen vehicles and refuelling, and hydrogen replacing fossil fuels in industrial processes. Producing, liquefying or distributing hydrogen as an industrial gas, or using it in refining, is not E unless the posting states a low-carbon object.

Scores.

Score	Definition
0	No qualifying E duty and no required E skill; only boilerplate, setting, generic requirements, unrelated work, or unsupported implications.
1	E is peripheral: at least one qualifying E duty is peripheral to the role's main purpose, or the posting requires a specific E skill without assigning an E duty.
2	E duties constitute a main purpose or a substantive co-primary function of the role.

Use the **removal test**: would removing the E duties eliminate the role's main purpose or a substantive co-primary function? If yes, score 2. If only a supporting responsibility disappears, score 1. This is centrality, not seniority, task count, time inferred from bullet count, or the societal importance of the occupation. A required skill alone never scores 2.

Fields for every positive posting.

- `E_class`: exactly one class (below) for the dominant qualifying function. If tied, choose the clearest, most central duty; if still tied, the first qualifying duty in the posting.
- `E_evidence`: `duty` when at least one qualifying E duty is assigned; `requirement` when the only evidence is a required E skill.
- `E_role`: `operates` when all the qualifying E work is routine physical delivery to established procedures: collecting, hauling, sorting, cleaning, handling, installing, operating or maintaining equipment or plant. Use `builds` when any qualifying E work plans, designs, engineers, researches, develops, measures, monitors, assesses, manages, audits, reports or improves. A wind technician who maintains turbines operates. An engineer who designs a wind farm builds. A technician who samples and tests effluent for compliance builds (measurement).

Classes.

Class	Scope
climate_ghg	Greenhouse-gas inventories, carbon footprints, emissions-reduction targets, net-zero and abatement programmes.
air_emissions	Air quality, stack testing, continuous emissions monitoring, air permits, and emissions control and compliance of engines, vehicles and equipment.
energy	Renewable generation (solar, wind, hydro, geothermal, bioenergy): development, design, installation, commissioning, operation and procurement; energy efficiency, energy audits and energy management.
electrification_storage	Batteries and stationary energy storage; grid integration of renewables and storage, smart grids and demand response; heat pumps and electrification of buildings and industrial processes.
clean_mobility	Electric, hybrid and hydrogen vehicles and powertrains, vehicle batteries and battery management, charging infrastructure.
hydrogen_ccus_fuels	Low-carbon hydrogen (electrolysis, renewable- or nuclear-powered, or with carbon capture), electrolysers, fuel cells, hydrogen replacing fossil fuels; carbon capture, utilisation and storage, direct air capture; biofuels, renewable natural gas and sustainable aviation fuel.
nuclear	Nuclear generation, fuel and reactor technology, including small modular reactors.
water	Water supply and conservation, leakage reduction, wastewater, effluent, discharge, drainage, flood/coastal protection, and aquatic environmental management.
waste_circular	Waste collection and treatment, recycling, landfill, diversion, materials recovery, circularity, sustainable materials and ecodesign.
pollution_chemicals	Contamination, spills, environmental remediation, hazardous materials/waste, and chemical risk management.
biodiversity_land	Ecology, habitats, species, biodiversity assessment, conservation and land restoration.
env_management	Environmental management systems, ISO 14001, cross-topic environmental permitting, compliance and reporting.
green_building	Building environmental performance, energy commissioning, LEED/BREEAM, sustainable building specifications.
climate_risk	Physical and transition climate risks to organisations: scenarios, exposure, resilience and adaptation assessments.

Class	Scope
`sustainability_esg`	Corporate sustainability strategy and programmes; environmental ESG data, disclosure and ratings; green and sustainable finance and environmental investment analysis.

Use the exact listed class when it fits. Otherwise use `other:short_lowercase_name` only for a clearly qualifying duty or skill within the E definition. Do not invent a class to rescue weak evidence or an excluded activity.

Calibration examples. Examples describe duties and requirements, not title-based labels. Labels are E / class / evidence / role.

Advertised work	E / class / evidence / role
Software QA at a solar company; no clean-technology or environmental duties	0
Accountant at an electric-vehicle maker	0
Hospital EVS (environmental services) technician: clean patient rooms	0
Retail solar advisor: book appointments and sell panels to homeowners	0
Install solar PV systems as the main job	2 / energy / duty / operates
Solar consultant: design and size PV systems for customer homes	2 / energy / duty / builds
Design battery-management-system firmware for an EV battery pack	2 / clean_mobility / duty / builds
Production technician welding battery-cell interconnects on an EV pack line	2 / clean_mobility / duty / operates
Engineer converting industrial process heat from gas to electric heat pumps	2 / electrification_storage / duty / builds
Process engineer for an electrolyser manufacturing line	2 / hydrogen_ccus_fuels / duty / builds
Operator running a hydrogen electrolysis plant	2 / hydrogen_ccus_fuels / duty / operates
Operator at an industrial-gas plant producing and liquefying hydrogen; no low-carbon production stated	0
Nuclear plant systems engineer	2 / nuclear / duty / builds

Advertised work	E / class / evidence / role
Calibrate diesel engines to meet emissions standards as the main job	2 / air_emissions / duty / builds
Sustainable-finance associate screening green bonds and environmental ESG data	2 / sustainability_esg / duty / builds
Lead stack testing and emissions monitoring	2 / air_emissions / duty / builds
Roll-off driver collecting commercial waste	2 / waste_circular / duty / operates
Custodian: clean offices, empty trash and recycling bins	0
Warehouse material handler: handle hazardous materials according to company procedures	0
Driver delivering chemical products under hazardous-materials rules	0
Operate production wells; perform spill checks as one supporting duty	1 / pollution_chemicals / duty / operates
Run worker-safety programmes; also support environmental incident response	1 / pollution_chemicals / duty / builds
HVAC design engineer; no green duty; LEED AP required	1 / green_building / requirement / builds
HVAC service technician: maintain and repair units; EPA 608 Universal certification required	0
Electrical engineer for conventional distribution lines; no renewable, storage or efficiency object	0

Edge cases and final check.

- **Insufficient:** set `insufficient: true` and E to 0 only when the text cannot establish what the role does: e.g., title-only text, an unusable fragment, or non-posting content. Missing bullets or truncation alone is not enough for this flag if a substantive role summary remains. A sufficiently described role with no qualifying E duty or skill receives an ordinary 0.
- **Multiple roles:** when distinct vacancies are combined, score only the first and set `multiple_roles: true`. Multiple duties within one job do not trigger this flag.
- **Language:** apply the same criteria in every language.
- Before returning, check every positive score against an explicit duty or required skill. Check every zero against any overlooked qualifying duty or skill. Do not target a particular positive-label rate.

Output. Return only valid JSON: `{"labels":[...]}`. Include every numbered posting once, in input order. Each object contains its original number as `i` and the integer score `E`. For a positive score include `E_class`, `E_evidence` and `E_role`; omit them for zero. Include `insufficient` and `multiple_roles` only when true. No explanations or extra keys.

Example:

```
{"labels":[
 {"i":1,"E":2,"E_class":"clean_mobility","E_evidence":"duty","E_role":"builds"},
 {"i":2,"E":0},
 {"i":3,"E":1,"E_class":"green_building","E_evidence":"requirement","E_role":"builds"},
 {"i":4,"E":0,"insufficient":true}]}
```

IA.I.2 Social Work

Task. Label each numbered job posting for the social-protection work the advertised role performs. Score Social (S) using the project definitions below. Use only the posting text; do not infer duties from the employer, sector, title, or qualifications alone. Score S on its own: other functions are labelled separately, and it does not matter whether the same work also counts as environmental, governance, innovation or anything else.

Purpose. These labels measure a firm's demand for social capacity: how much of its hiring goes to people whose work in the job protects or improves the rights, safety, welfare or opportunities of the people its own operations and products affect, namely its workers, the workers in its supply chain, the communities around its operations, and its customers and users. Judge each posting by the work the person will do, never by who the employer is, what it sells or what it says it values.

Definitions and evidence.

- **Duty:** an explicitly assigned activity or responsibility, stated in the duties, role summary, or other substantive description of the role.
- **Named duty:** identifies both an action/responsibility and its relevant object or purpose; e.g., "conduct workplace hazard inspections," "manage employee grievances," or "audit suppliers against a labour code." A keyword alone is insufficient.
- **S:** work protecting or improving the rights, safety, welfare, or opportunities of workers, supply-chain workers, communities, customers, or users affected by the employer's operations or products, subject to the exclusions below.

The employer's own service is not S. Delivering the employer's ordinary service to its clients (teaching, childcare, counselling, social work, therapy, coaching, healthcare, personal care, security guarding) is not S, even when that service itself concerns people's welfare. That work is what the employer sells. S is a function that protects people from, or improves, the effects of the employer's

own operations and products. A school's safeguarding lead, a hospital's patient-safety officer and a care provider's infection-control nurse perform S; the teacher, the nurse and the care worker delivering the service do not.

Ignore employer values, mission statements, benefits offered to applicants, equal-opportunity boilerplate, and generic instructions to work safely or follow rules. "IT environment," software "remediation," and generic "data governance" do not establish S.

Generic phrases such as "ensure regulatory compliance" or "maintain quality" are insufficient alone. Count them only when surrounding text identifies the relevant assigned activity. Apply exclusions contextually, not as phrase blacklists.

Scores.

Score	Definition
0	No qualifying duty is evidenced; only boilerplate, ordinary conduct, the employer's ordinary service, unrelated work, or unsupported implications.
1	At least one qualifying duty is peripheral to the role's main purpose. One explicit duty is sufficient.
2	Qualifying duties constitute a main purpose or a substantive co-primary function of the role.

Use the **removal test**: would removing the S duties eliminate the role's main purpose or a substantive co-primary function? If yes, score 2; if only a supporting responsibility disappears, score 1. This is centrality, not seniority, task count, time inferred from bullet count, or the societal importance of the occupation. A junior safety specialist can score S:2.

Require qualifying evidence for S; do not add S because environmental or governance work involves a policy, programme, audit, or legal requirement. Safety programmes belong to S. Never assign an automatic secondary score of 1.

Classes. For each positive posting, choose exactly one class representing its dominant qualifying function. If tied, choose the class supported by the clearest, most central duty; if still tied, use the first qualifying duty in the posting. Score S using all its qualifying duties, not only the selected class.

Use the exact listed class when it fits. Otherwise use `other:short_lowercase_name` only for a clearly qualifying duty within the S definition. Do not invent a class to rescue weak evidence or an excluded activity.

Class	Scope
`ohs`	Occupational health and safety, industrial hygiene, hazard assessments, safety inspections, incident investigations, ergonomics, fire and emergency preparedness for the workplace, and safety programmes.
`dei`	Diversity, equity and inclusion programmes, pay equity, affirmative action, EEO reporting, inclusive recruitment programmes, and workplace accommodation and accessibility for disabled workers.
`labour_rights`	Collective bargaining, labour/employee relations, freedom of association, works councils and worker grievances.
`forced_child_labour`	Modern-slavery, forced/child-labour due diligence, recruitment-fee safeguards and age verification.
`human_rights`	Broader human-rights due diligence, salient-impact assessment and access to remedy.
`community`	Community engagement/investment, Indigenous rights and consultation, social impact assessment, resettlement, and the health, safety and security of communities around the employer's sites, pipelines and operations.
`supply_chain_social`	Supplier labour/human-rights audits, social codes of conduct, responsible sourcing and social capacity building.
`product_customer_safety`	Product/customer/patient safety, recalls, pharmacovigilance, clinical quality and infection-control programmes; safeguarding of children and vulnerable service users; accessibility of products and services for disabled users; platform safety, content moderation and trust-and-safety work.
`data_protection`	Individuals' privacy, consent, data-subject rights, privacy notices, personal-data breach notification/remedy, discriminatory algorithmic-bias testing and individual decision contestability.
`workforce_development`	Owned apprenticeship, reskilling and employee learning programmes or curricula; use `ohs` for dedicated safety training.
`employment_quality`	Living wages, working hours and conditions, worker well-being and worker-focused just-transition measures. Pay and benefit design counts only when its stated purpose is worker welfare, such as a living wage or decent working conditions; market benchmarking, salary surveys, incentive plans and cost control do not.

Exclude ordinary team supervision/coaching, recruitment and talent-acquisition logistics, HR generalist administration (onboarding, policy queries, personnel files), payroll, HR systems, benefits enrolment, attendance records, compensation benchmarking, customer service, sales, and sales/product/dealer training. Ordinary direct clinical or personal care is the employer's service and is excluded; dedicated patient-safety, clinical-quality, infection-control and care-staff development functions qualify.

"Work safely," "wear PPE," "report hazards," and "maintain a clean, safe workplace" do not establish S duties alone. Assigned hazard inspections, safety drills, investigations, or safety-programme implementation do. Routine physical security, guarding and loss prevention do not establish S.

Privacy, security and AI boundaries. These boundaries say what belongs to S and what belongs to governance (G), which is labelled separately.

- **Privacy and digital rights → S:** protecting personal data, consent, access/deletion rights, fair treatment, explanation, appeal and remedy. A privacy programme does not become G merely because it is managed or regulated.
- **Security controls → G:** security architecture/controls, access reviews, threat detection and cyber-incident response. Personal-data exposure alone does not establish an additional S duty; explicit privacy protection or remedy does.
- **AI/model governance → G:** oversight, accountability, risk acceptance, validation, monitoring and assurance. Mentioning the EU AI Act or NIST AI RMF is insufficient without an assigned implementation duty.
- **Bias testing → S** when it addresses discrimination or unequal treatment of people. Statistical bias, prediction error or model calibration alone is not S; it can support G when part of a defined model-risk/validation function.
- **Automated-decision review:** individual fairness, contestability and remedy → S; internal model approval, risk controls and independent assurance → G. Score both only when both functions are evidenced.

Calibration examples. Examples describe duties, not title-based labels.

Advertised work	**S**
Software QA at a solar company; no qualifying safety or social duties	0
Install solar PV generation systems as the main job	0

Advertised work	S
Run worker-safety programmes; also support environmental incident response	2
Own both environmental management and worker-safety programmes	2
Production supervisor who also leads monthly safety walks and investigates incidents on the line	1
Construction labourer told to wear PPE and follow site safety rules	0
Security/guest-service role with assigned hazard inspections and emergency drills	1
Security officer: patrol premises, monitor CCTV and control access	0
Provide ordinary nursing care, without a separate safety/quality function	0
Infection-prevention nurse running the hospital's infection-control programme	2
Run patient-safety studies; contribute evidence to regulatory submissions	2
Teacher: plan and deliver lessons and assess pupils	0
School safeguarding lead: run child-protection procedures, train staff and investigate concerns	2
Social worker or counsellor: assess clients and deliver casework or therapy	0
HR generalist: onboarding, policy queries, personnel files and benefits enrolment	0
Employee-relations specialist: investigate grievances and support works-council consultations	2
Recruiter who also runs the company's diversity-hiring programme as a supporting duty	1
Compensation analyst: benchmark salaries and design sales incentive plans	0
Community-relations manager at a mine: consult Indigenous communities and run the grievance mechanism	2
Supplier social auditor: audit factories against the labour code and follow up corrective actions	2
Trust-and-safety analyst: review flagged content and enforce platform safety policies	2

Advertised work	S
Privacy analyst: handle data-subject access requests and maintain consent records	2
Learning specialist: design and run the employee reskilling programme	2
Bank teller instructed to follow policies; no assigned control duty	0
Junior specialist whose main function is monitoring and testing compliance controls	0
Lead privacy-rights protection; perform a separate supporting compliance-assurance duty	2
Co-lead discrimination testing and AI model-risk oversight as substantive functions	2

Edge cases and final check.

- **Insufficient:** set `insufficient: true` and S to 0 only when the text cannot establish what the role does: e.g., title-only text, an unusable fragment, or non-posting content. Missing bullets or truncation alone is insufficient for this flag if a substantive role summary remains. A sufficiently described role with no qualifying duties receives an ordinary 0.
- **Multiple roles:** when distinct vacancies are combined, score only the first and set `multiple_roles: true`. Multiple duties within one job do not trigger this flag.
- **Language:** apply the same criteria in every language.
- Before returning, check every positive score against an explicit duty and every zero against any overlooked qualifying duty. Do not target a particular positive-label rate. Resolve conflicts using definitions and scoring rules.

Output. Return only valid JSON: `{"labels":[...]}`. Include every numbered posting once, in input order. Each object contains its original number as `i` and the integer score `S`. For a positive score, include `S_class`; omit it for zero. Include `insufficient` and `multiple_roles` only when true. No explanations or extra keys.

Example:

```
{"labels":[
 {"i":1,"S":2,"S_class":"data_protection"},
 {"i":2,"S":0},
 {"i":3,"S":0,"insufficient":true}]}
```

IA.I.3 Governance Work

Task. Label each numbered job posting for the governance, control and compliance work the advertised role performs. Score Governance (G) using the project definitions below. Use only the posting text; do not infer duties from the employer, sector, title, or qualifications alone. Score G on its own: other functions are labelled separately, and it does not matter whether the same work also counts as environmental, social, innovation or anything else.

Purpose. These labels measure a firm's demand for governance capacity: how much of its hiring goes to people whose work in the job establishes, operates or assures the firm's accountability, oversight, regulatory compliance and internal controls. Judge each posting by the work the person will do, never by who the employer is, what it sells or what it says it values.

Definitions and evidence.

- **Duty:** an explicitly assigned activity or responsibility, stated in the duties, role summary, or other substantive description of the role.
- **Named duty:** identifies both an action/responsibility and its relevant object or purpose; e.g., "test access controls," "investigate suspected fraud," or "prepare board minutes." A keyword alone is insufficient.
- **Control:** a defined measure that prevents, detects, corrects, or reports a compliance, integrity, security, quality, or risk-management failure. Designing, operating, testing, auditing, investigating, or reporting a control qualifies; merely obeying it does not. Junior execution can qualify without programme ownership.
- **G:** work establishing or operating organisational accountability, integrity, oversight, assurance, regulatory compliance, or the controls specified below. Routine administration and general management do not qualify.

Ignore employer values, mission statements, benefits offered to applicants, equal-opportunity boilerplate, and generic instructions to work safely or follow rules. "IT environment," software "remediation," and generic "data governance" do not establish G.

Generic phrases such as "ensure regulatory compliance" or "maintain quality" are insufficient alone. Count them only when surrounding text identifies the relevant assigned activity. Apply exclusions contextually, not as phrase blacklists.

Scores.

Score	Definition
0	No qualifying duty is evidenced; only boilerplate, ordinary conduct, unrelated work, or unsupported implications.
1	At least one qualifying duty is peripheral to the role's main purpose. One explicit duty is sufficient.
2	Qualifying duties constitute a main purpose or a substantive co-primary function of the role.

Use the **removal test**: would removing the G duties eliminate the role's main purpose or a substantive co-primary function? If yes, score 2; if only a supporting responsibility disappears, score 1. This is centrality, not seniority, task count, time inferred from bullet count, or the societal importance of the occupation. A junior control specialist can score G:2.

Require qualifying evidence for G; do not automatically add G because environmental or social work involves a policy, programme, audit, or legal requirement. Environmental permits belong to E and safety programmes to S unless additional duties independently establish G. Never assign an automatic secondary score of 1.

Classes. For each positive posting, choose exactly one class representing its dominant qualifying function. If tied, choose the class supported by the clearest, most central duty; if still tied, use the first qualifying duty in the posting. Score G using all its qualifying duties, not only the selected class.

Use the exact listed class when it fits. Otherwise use `other:short_lowercase_name` only for a clearly qualifying duty within the G definition. Do not invent a class to rescue weak evidence or an excluded activity.

Class	Scope
`anticorruption`	Anti-bribery/corruption, anti-money laundering (AML), know-your-customer (KYC), financial sanctions screening and integrity due diligence.
`fraud_prevention`	Fraud prevention, detection and investigation, fraud-risk controls, transaction and claims fraud monitoring, and payment or programme integrity.
`competition`	Antitrust, competition compliance and fair competitive conduct.
`consumer_market_conduct`	Consumer-protection controls, fair lending, investment suitability, securities/product compliance, market-abuse and trade surveillance, and sales-practice review.

Class	Scope
trade_compliance	Export controls, customs, import restrictions and trade sanctions, including ITAR.
tax_transparency	Tax strategy/governance, country-by-country reporting and transfer-pricing compliance.
ethics_conduct	Ethics programmes, codes of conduct, whistleblowing, misconduct investigations and conflicts of interest.
compliance_programme	General compliance programmes spanning several regimes: compliance monitoring and testing, compliance risk assessment, regulatory-change management, policy management and compliance reporting, when no specific listed subject dominates.
regulatory_affairs	Regulatory submissions, filings, licensing and substantive regulator liaison.
quality_systems	Quality management systems (QMS), corrective/preventive action (CAPA), quality-complaint systems, data integrity, computerised-system validation, and documented conformance inspection/audit.
financial_risk_controls	Credit/market/liquidity risk controls, financial model validation/risk management, capital adequacy, internal audit and financial-reporting controls, including SOX.
operational_risk_resilience	Enterprise and operational risk management, risk and control self-assessment, business continuity, disaster-recovery planning and operational resilience.
infosec_governance	Security policies, access/security controls, cyber-risk assessment, control assurance, and security incident detection, investigation and response.
privacy_governance	Organisational accountability for data protection: privacy compliance audits, records of processing, impact-assessment review and sign-off, and data-protection regulator liaison.
ai_governance	AI/ML accountability, approval and oversight, model inventories, AI risk assessment/validation/monitoring, independent challenge, and operational implementation of AI governance frameworks or regulation.
physical_security_compliance	Formal facility/personnel security compliance regimes, such as NISPOM, and clearance administration.
board_governance	Board and committee support, company secretarial work, statutory registers, corporate-governance codes and shareholder-meeting governance.

Class	Scope
`esg_governance`	ESG reporting/disclosure, assurance, materiality, board/executive sustainability oversight, and explicitly ESG-focused investment integration, stewardship or proxy voting.
`lobbying`	Political engagement/lobbying responsibilities, political-spend disclosure and trade-association alignment.
`supply_chain_governance`	Procurement integrity/assurance controls, supplier compliance evaluation/escalation and contractual ESG requirements.

Exclude ordinary legal advice, contract drafting and negotiation, accounting/bookkeeping, financial reporting without control responsibility, IT operations/helpdesk, software engineering, HR administration, purchasing/vendor pricing, routine document handling, and generic project scheduling, budgets or risk logs. Routine technical QA/debugging is insufficient without a defined conformance or assurance duty. General organisational efficiency is not G.

Prefer the specific subject class over a broad management or filing class: fair-lending compliance → `consumer_market_conduct`; AML transaction monitoring → `anticorruption`; card-fraud investigation → `fraud_prevention`; financial model validation → `financial_risk_controls`. Use `compliance_programme` only when no specific subject dominates. Environmental permits, such as air permits, are environmental work and do not establish G by themselves.

Privacy, security and AI boundaries. These boundaries say what belongs to G and what belongs to social (S), which is labelled separately.

- **Privacy and digital rights → S:** protecting personal data, consent, access/deletion rights, fair treatment, explanation, appeal and remedy. A privacy programme does not become G merely because it is managed or regulated.
- **Security controls → G:** security architecture/controls, access reviews, threat detection and cyber-incident response. Personal-data exposure alone does not establish an additional S duty; explicit privacy protection or remedy does.
- **AI/model governance → G:** oversight, accountability, risk acceptance, validation, monitoring and assurance. Mentioning the EU AI Act or NIST AI RMF is insufficient without an assigned implementation duty.
- **Bias testing → S** when it addresses discrimination or unequal treatment of people. Statistical bias, prediction error or model calibration alone is not S; it can support G when part of a defined model-risk/validation function.

- **Automated-decision review:** individual fairness, contestability and remedy → S; internal model approval, risk controls and independent assurance → G. Score both only when both functions are evidenced.
- **Privacy oversight:** separate organisational assurance/accountability duties may add G under `privacy_governance`. Programme administration alone does not add G.

Calibration examples. Examples describe duties, not title-based labels.

Advertised work	**G**
Software QA at a solar company; no qualifying controls or environmental duties	0
Run worker-safety programmes; also support environmental incident response	0
Security/guest-service role with assigned hazard inspections and emergency drills	0
Run patient-safety studies; contribute evidence to regulatory submissions	1
Bank teller instructed to follow policies; no assigned control duty	0
Production technician with supporting documented conformance-inspection duties	1
Quality inspector: inspect incoming parts against specifications and record nonconformances in the QMS	2
Junior specialist whose main function is monitoring and testing compliance controls	2
Fraud analyst: review flagged transactions and investigate suspected fraud	2
Internal auditor: plan and perform audits of business processes and report control findings	2
Security-operations analyst: monitor alerts and investigate and respond to cyber incidents	2
Security specialist focused on risk assessments, security plans and accreditation	2
Regulatory-affairs specialist: prepare and submit product registration dossiers	2
Company secretary: prepare board papers and minutes and maintain statutory registers	2

Advertised work	G
Business-continuity manager: maintain continuity plans and run recovery exercises	2
Accountant: journals, reconciliations and month-end close; no control responsibility	0
Accountant who also performs SOX control testing as a supporting duty	1
Lawyer drafting and negotiating commercial contracts	0
IT helpdesk: reset passwords and install software	0
Project manager maintaining the project's risk log and budget	0
Lead privacy-rights protection; perform a separate supporting compliance-assurance duty	1
Co-lead discrimination testing and AI model-risk oversight as substantive functions	2

Edge cases and final check.

- **Insufficient:** set `insufficient: true` and G to 0 only when the text cannot establish what the role does: e.g., title-only text, an unusable fragment, or non-posting content. Missing bullets or truncation alone is insufficient for this flag if a substantive role summary remains. A sufficiently described role with no qualifying duties receives an ordinary 0.
- **Multiple roles:** when distinct vacancies are combined, score only the first and set `multiple_roles: true`. Multiple duties within one job do not trigger this flag.
- **Language:** apply the same criteria in every language.
- Before returning, check every positive score against an explicit duty and every zero against any overlooked qualifying duty. Do not target a particular positive-label rate. Resolve conflicts using definitions and scoring rules.

Output. Return only valid JSON: `{"labels":[...]}`. Include every numbered posting once, in input order. Each object contains its original number as `i` and the integer score `G`. For a positive score, include `G_class`; omit it for zero. Include `insufficient` and `multiple_roles` only when true. No explanations or extra keys.

Example:

```
{"labels":[
 {"i":1,"G":1,"G_class":"privacy_governance"},
 {"i":2,"G":0},
 {"i":3,"G":0,"insufficient":true}]}
```

IA.I.4 Innovation Work

Task. Label each numbered job posting for the innovation work the advertised role actually performs. Score Innovation (I) using the project definitions below. Use only the posting text; do not infer duties from the employer, sector, title, department or qualifications alone. Judge only from the posting text.

Purpose. These labels measure a firm's demand for innovation capacity: how much of its hiring goes to people whose work in the job creates new or substantially improved products, services, software, processes, methods or other technical output for the firm. Judge each posting by the work the person will do, never by who the employer is, what it sells or what it says it values.

Definitions and evidence.

- **Duty:** an explicitly assigned activity or responsibility, stated in the duties, role summary, or other substantive description of the role. Requirements, qualifications, skills and experience lines ("experience developing X", "ability to design Y", "5+ years building Z") are not duties. Descriptions of the employer, team, platform, mission or industry are not duties unless the sentence assigns the work to this person.
- **Creation duty:** a duty that assigns the person to research, design, invent, develop, engineer, program, model, formulate, synthesise, prototype or experiment, or to test or validate something still under development.
- **One-sentence rule:** the creation verb and its object must appear in the same duty sentence. Do not assemble a qualifying duty from a verb in one sentence and an object in another.
- **I:** work that creates, or substantially improves, a product, service, software, database, design, technical process, model, algorithm, method or other technical or scientific output that is new to the employer. A creation duty qualifies only if it passes all four tests below.

The four tests.

1. **Real object.** The duty commits the person to producing something that will exist afterwards: an artifact that enters the employer's product, codebase, laboratory, plant or a client deliverable. Ordinary occupational language suffices ("design, develop and test new features" for the employer's platform). The object is **empty** when the sentence promises an outcome rather than an artifact and nothing in the posting says what would be made: "solutions", "deliverables", "initiatives", "capabilities", "enhancements", "improvements", "technology" or "project work" standing alone fail.

2. **Made by the person.** The person's own hands make part of it. Working under supervision or inside a technical team passes. The test fails when the making sits elsewhere: with external partners, vendors, suppliers, clients or contractors, or with a team the person directs, coordinates or buys from rather than joins ("lead the development of", "drive the design of", "oversee", "manage

the delivery of", "responsible for ensuring that X is built"). Weak participation verbs (assist, support, participate in, help, contribute to) pass only when the object is specific and the person sits in the technical group producing it.

3. **New to the employer.** The output is not another instance of a deliverable the employer routinely produces: the next drawing set, network design, client report or configuration is not innovation. Routineness must be **shown**, not assumed: treat the output as routine only when the posting supplies a repetition signal (executing established codes, standards, templates, agreed plans or client specifications; drawings, calculations or documentation implementing a design decided elsewhere; the firm's standard catalogue service). Words such as new, novel, custom, prototype, research, next-generation, greenfield, re-architecture or new product introduction settle it the other way. With neither signal, a duty to design or develop something passes.

4. **Durable.** The output persists and is reusable: a product, feature, formulation, compound, codebase, dataset or database, model, algorithm, method, technical process or technology. A one-off analysis, report, decision, schedule, single-project plan or training session is not durable.

Burden of proof. Score 0 only when you can name the rule that excludes the duty and point to words in the posting that trigger it. An objection resting on what the posting does not say is not sufficient: postings do not certify novelty, and the absence of the word "new" is not evidence that work is routine. None of the following alone justifies a 0: ordinary occupational vocabulary rather than a named product line; a junior, supervised or internship role; a services or consulting employer; an artifact not described in detail; a short posting. When a qualifying duty faces only a doubt without textual support, score it and set `borderline: true`. The flag records uncertainty; the score does not.

Ignore employer values, mission statements, benefits offered to applicants, equal-opportunity boilerplate, and words such as "innovative", "cutting-edge", "transformation" or "technology". An accountant, recruiter or salesperson at a research-intensive firm does not inherit I.

Scores.

Score	Definition
0	No qualifying creation duty is evidenced; only boilerplate, setting, requirements, operating work, or unsupported implications.
1	At least one qualifying creation duty is peripheral to the role's main purpose (operating, supporting, analysing, selling or managing). One explicit duty is sufficient.
2	Qualifying creation is a main purpose or a substantive co-primary function of the role.

Use the **removal test**: would removing the creation duties eliminate the role's main purpose or a substantive co-primary function? If yes, score 2; if only a supporting responsibility disappears, score

1. This is centrality, not seniority, task count, time inferred from bullet count, or the sophistication of the field. A junior developer whose job is building features scores I:2.

Classes. For each positive posting, choose exactly one class representing its dominant qualifying output. If tied, choose the class supported by the clearest, most central duty; if still tied, use the first qualifying duty in the posting. Score using all qualifying duties, not only the selected class.

Class	Scope
`research_discovery`	Research or discovery programmes: designing and running experiments, developing or optimising assays and workflows, screening, characterising and evaluating candidate compounds or materials, postdoctoral research, and phase I, II or III clinical trials of a candidate not yet approved.
`physical_product`	New or improved physical products, devices, components, materials, formulations or compounds.
`software_product`	New or improved software, applications, features or digital services offered to users, customers or patients.
`technical_process`	New or improved production, manufacturing, engineering or laboratory processes, equipment, tooling, fixtures and the code that controls them.
`data_model_method`	Internal software tools, data pipelines, warehouses and databases, models and algorithms, and new test, measurement or analytical methods.
`engineered_design`	Non-standard designs of systems, structures, facilities, networks or infrastructure, including engineered solutions designed for a client.

Boundaries.

- **Design deliverables.** Repeated production of design paperwork fails test 3 when the posting frames it as executing established codes, standards, templates, agreed plans or client specifications, or as obtaining approval of a standard package. Do not infer repetition from the employer having many projects, sites or clients.
- **Vendor platforms.** Work on a third-party platform (ServiceNow, SAP, Salesforce, Workday, an ERP, a cloud provider's services, a network vendor's stack, an identity platform) qualifies only when the duty names new code, models, data structures or applications the person writes on top of it. Configuring, implementing, deploying, integrating, migrating, customising, administering or upgrading the platform does not qualify, however senior the framing.

- **Data work.** Building data pipelines, warehouses, datasets, data models and the code that produces them qualifies (`data_model_method`). Building dashboards, reports, visualisations, reporting templates or business-intelligence layers does not, even when the verb is "develop".
- **Methods versus standards.** Developing a new test, measurement or analytical method, assay, or protocol for a novel measurement qualifies. Writing, updating or codifying standards, specifications, guidelines, best practices, SOPs or documentation, or sitting on standards committees, does not.
- **Research.** A duty to carry out a research or discovery programme qualifies whether or not the posting names the resulting artifact; standard techniques are how research is done, not a reason to exclude it. Generating evidence about a product that is already approved does not qualify: post-approval (phase IV) studies, medical affairs, regulatory submissions, health-economic and market-access evidence.
- **Clinical development.** Designing, conducting or analysing a phase I, II or III clinical trial of a drug, biologic, device or diagnostic that is not yet approved qualifies as `research_discovery`: the candidate is not yet a product, and the trial produces the knowledge required to create it. The directing rule of test 2 still applies: a role that only oversees, coordinates or outsources the trial to a contract research organisation fails.
- **Client work.** A deliverable designed for a client qualifies only if it passes test 3: non-standard rather than the firm's ordinary repeated output. Installing, configuring, operating or maintaining existing technology for a client never qualifies.
- **Financial and insurance products** qualify only when the duty assigns the design of the product, its underlying instrument, or its pricing or risk model. Owning, running or launching projects that deliver them is delegated creation.

Exclusions. Exclude: system administration, monitoring, incident response, patching, upgrades, maintenance, repair, production operations, finished-product inspection, regression and release testing, and running established models (where one duty says create **and** maintain the same object, judge it by the creation); executing an established synthesis, assay, test or production procedure to a fixed protocol; improvement achieved by scheduling, sequencing, staffing, administrative change or better management of existing operations (improvement achieved by writing code, designing equipment or changing a technical process's design qualifies); market or business analysis, routine pricing, applying an established statistical or actuarial model, sales, marketing, procurement and product support; business workflows, administrative procedures, management practices, employee training, branding and client-communication practice; and non-technical outputs of ordinary service delivery such as menus, single-venue recipes, individual treatment plans, care procedures, event formats, curricula and marketing collateral, even when called new.

Calibration examples. Examples describe duties, not title-based labels. Labels are ordered I / class.

Advertised work	I / class
Design and synthesise novel small-molecule compounds and advance candidates through preclinical studies	2 / physical_product
Develop and pilot a new laser-welding process to replace the current joining method	2 / technical_process
Monitor plant output and maintain schedules; one duty: develop a new predictive-maintenance algorithm for the compressor fleet	1 / data_model_method
Design, develop and test new features for the employer's platform through code review to release	2 / software_product
Junior engineer creating designs and final working drawings for a new terminal expansion, under supervision	2 / engineered_design
Design custom water-treatment systems for client sites where standard designs do not meet requirements	2 / engineered_design
Own the design of fulfilment and logistics systems, including site layout, equipment specification and material flow	2 / engineered_design
Build data pipelines that consolidate and cleanse partner data into the warehouse	2 / data_model_method
Develop and execute experiments and optimise workflows aimed at target identification	2 / research_discovery
Develop and modify code to improve memory calibration sequences, working with software teams	2 / technical_process
Field service engineer: install and repair equipment; also design new test fixtures for the service bench	1 / technical_process
Build new modules for the firm's platform and adapt them for client implementations	2 / software_product
Design the pricing model for a new insurance product	2 / data_model_method
Design secure cloud-based "solutions" for organisations migrating workloads; no artifact named	0
Requirements: experience designing and developing code in Dynamics AX	0
Lead development of diagnostics through collaboration with external partners who build them	0

Advertised work	I / class
Produce CAD drawings for transport projects in accordance with agreed standards and client specifications	0
Develop, maintain and enhance applications on the ServiceNow platform; no artifact beyond the platform	0
Design reporting frameworks in Qlik and Cognos and manage dashboard development	0
Participate in the industry committee that develops and maintains test standards	0
Modify operation sequences to reduce scrap and cost	0
Assist in new product development and support the quality team during launches	0
Maintain and patch existing services and respond to production incidents	0
Design and conduct the phase 2 trial of an investigational compound and analyse its efficacy data	2 / research_discovery
Run post-approval studies and medical-affairs evidence programmes for a marketed medicine	0
Oversee trial execution by contract research organisations and manage investigator relationships	0
Accountant at a research-intensive biotechnology firm	0
Own and run the project that launches a new credit-card product	0

Edge cases and final check.

- **Insufficient:** set `insufficient: true` and I to 0 only when the text cannot establish what the role does: e.g., title-only text, an unusable fragment, or non-posting content. Missing bullets or truncation alone is insufficient for this flag if a substantive role summary remains. A sufficiently described role with no qualifying duties receives an ordinary 0.
- **Multiple roles:** when distinct vacancies are combined, score only the first and set `multiple_roles: true`. Multiple duties within one job do not trigger this flag.
- **Language:** apply the same criteria in every language.
- Before returning, check every positive score against an explicit creation duty that passes all four tests, and every zero against any overlooked qualifying duty and against the burden of proof. Do not target a particular positive-label rate.

Output. Return only valid JSON: `{"labels":[...]}`. Include every numbered posting once, in input order. Each object contains its original number as `i` and the integer score `I`. For a positive score, include `I_class`; omit it for zero. Include `borderline`, `insufficient` and `multiple_roles` only when true. No explanations or extra keys.

Example:

```
{"labels":[
 {"i":1,"I":2,"I_class":"software_product"},
 {"i":2,"I":0},
 {"i":3,"I":1,"I_class":"technical_process","borderline":true},
 {"i":4,"I":0,"insufficient":true}]}
```

IA.I.5 The Classes of Each Kind of Work

Each instruction asks the language model to assign every positive posting to one class of a closed list. Tables IA.13 to IA.16 report the classes among the labelled positives, grouped by the heads of each classifier.

Table IA.13. Classes of Environmental Work

Class	Positives	%	Random
Clean technology (57%)			
energy	1,203	20.9	14
clean_mobility	749	13.0	0
electrification_storage	532	9.3	1
hydrogen_ccus_fuels	446	7.8	2
nuclear	304	5.3	5
green_building	42	0.7	2
Environmental management (43%)			
env_management	581	10.1	9
waste_circular	505	8.8	11
sustainability_esg	417	7.3	2
water	389	6.8	11
pollution_chemicals	263	4.6	4
climate_ghg	140	2.4	1
air_emissions	88	1.5	2
climate_risk	46	0.8	0
biodiversity_land	41	0.7	1
improvised other:	4	0.1	

Notes. Positives: postings that the language model scores 1 or 2, by the class of their main environmental function, pooled over the labelled samples. These samples are enriched in positives, so the percentages describe the labels rather than the population. Random: positives in the simple random sample of 4,000 postings. The two parts of environmental work used in the paper, clean technology and environmental management, are in italics, with their shares of positives. "Improvised other": classes outside the closed list, allowed only for a clearly qualifying duty; they count toward environmental management, as do all classes outside the six of clean technology. The classifier has one logistic regression for each part (Section 3.2).

Table IA.14. Classes of Social Work

Class	Positives	%	Random
Worker safety (52%)			
ohs	1,940	51.9	101
Product and patient safety (23%)			
product_customer_safety	849	22.7	45
People and rights (25%)			
labour_rights	301	8.1	4
workforce_development	208	5.6	10
dei	190	5.1	6
data_protection	122	3.3	7
community	85	2.3	3
employment_quality	17	0.5	2
supply_chain_social	12	0.3	2
human_rights	3	0.1	0
forced_child_labour	2	0.1	0
improvised other:	9	0.2	

Notes. Positives: postings that the language model scores 1 or 2, by the class of their main function, pooled over the labelled samples, which are enriched in positives, so that the percentages describe the labels rather than the population. Random: positives in the simple random sample of 4,000 postings. Classes are grouped by the three heads of the social classifier, in italics with their shares of positives; the classes outside worker safety and product and patient safety form the third head, people and rights.

Table IA.15. Classes of Governance Work

Class	Positives	%	Random
Quality assurance (35%)			
quality_systems	1,940	35.2	119
Risk and financial controls (19%)			
financial_risk_controls	793	14.4	56
operational_risk_resilience	235	4.3	24
Information security (10%)			
infosec_governance	493	8.9	46
privacy_governance	28	0.5	2
physical_security_compliance	23	0.4	1
ai_governance	1	0.0	0
Compliance and integrity (35%)			
compliance_programme	733	13.3	55
fraud_prevention	351	6.4	52
regulatory_affairs	229	4.2	19
anticorruption	188	3.4	9
consumer_market_conduct	112	2.0	10
trade_compliance	93	1.7	4
supply_chain_governance	74	1.3	8
ethics_conduct	62	1.1	1
esg_governance	40	0.7	1
tax_transparency	32	0.6	3
lobbying	18	0.3	1
board_governance	15	0.3	3
improvised other:	50	0.9	

Notes. Positives: postings that the language model scores 1 or 2, by the class of their main function, pooled over the labelled samples, which are enriched in positives, so that the percentages describe the labels rather than the population. Random: positives in the simple random sample of 4,000 postings. Classes are grouped by the four heads of the governance classifier, in italics with their shares of positives; the classes outside quality assurance, risk and financial controls, and information security and privacy form the fourth head, compliance and integrity.

Table IA.16. Classes of Innovation Work

Class	Positives	%	Random
data_model_method	1,803	31.0	109
software_product	1,688	29.0	143
physical_product	831	14.3	40
technical_process	600	10.3	22
engineered_design	548	9.4	30
research_discovery	351	6.0	7

Notes. Positives: postings that the language model scores 1 or 2, by the class of their main function, pooled over the labelled samples, which are enriched in positives, so that the percentages describe the labels rather than the population. Random: positives in the simple random sample of 4,000 postings. The innovation classifier has a single head, which cross-validation selected over separate heads.

IA.II The Classifiers

Section 3.2 describes the design of the classifier, and Appendix Table B1 reports how the four classifiers perform. This part documents how each of them was built. It describes the samples on which they were trained and evaluated, the selection of the model, the amount of training data it needs, its calibration, the threshold that turns probabilities into flags, and its robustness to the training data. It also lists the terms and occupations of the simpler measures with which the paper compares the shares.

IA.II.1 The Training and Evaluation Samples

Environmental, social, and governance work each appear in a small share of postings, from 1.6 to 10.8 percent of the simple random sample, so that a random training sample would hold few positives from which to learn. Each training sample therefore over-represents postings likely to involve the work. The training samples serve only to train the classifiers. Calibration and evaluation use samples drawn at random, so that the enrichment changes what each classifier learns from, not the probabilities it reports. Table IA.17 describes every sample.

Each training sample consists of a main draw and of smaller targeted samples. The environmental, social, and governance classifiers share a main draw of 25,000 postings from a base spread evenly over the years 2013 to 2023, drawn with more weight on the occupations that O*NET classifies as green and on postings whose embeddings resemble written descriptions of environmental work, and otherwise in proportion to the size of each occupation group. The innovation classifier has a main draw of its own, with more weight on research, engineering, and software occupations, from which the 700 postings of the innovation validation sample are set aside. The targeted samples add postings of the kinds that a main draw holds rarely, so that each classifier learns the boundaries of its definition. In total, the environmental classifier is trained on 31,995 postings, the social classifier on 28,953, the governance classifier on 27,063, and the innovation classifier on 15,390, of which between 10.0 and 28.1 percent are labelled positive (Table IA.17, Panel A).

Three kinds of samples serve to calibrate and evaluate the classifiers. The simple random sample of 4,000 postings gives every posting of the scored corpus the same chance of selection. Because environmental, social, and innovation work are rare, a score-band supplement of 2,000 postings for each, drawn at random among the postings whose raw score lies in a band that contains the threshold, adds postings where the probabilities matter most. Weighted by the inverse of their probability of selection, the random sample and a supplement form a probability sample of the corpus, on which the calibration of equation (1) is fitted. Governance work has no supplement and is calibrated on the random sample alone. The

validation samples of 800 postings, for environmental, social, and governance work, and of 700 postings, for innovation work, over-represent postings likely to involve the work, so that they are informative about the boundaries of each definition. The first is stratified by the similarity of postings to descriptions of each kind of work, by their use of ESG terms, and by borderline cases, such as postings that mention sustainability without assigning any such work, and the second by the strata of the innovation sample.

Table IA.17. The Training and Evaluation Samples

Sample	Postings	Firms	Labelled positive (%)
A. Training samples			
Environmental work	31,995	4,533	14.9
positives for clean technology	2,936		
positives for environmental management	1,832		
Social work	28,953	4,361	10.0
positives for worker safety	1,469		
positives for product and patient safety	662		
positives for people and rights	777		
Governance work	27,063	4,284	16.3
positives for quality assurance	1,638		
positives for risk and financial controls	825		
positives for information security and privacy	448		
positives for compliance and integrity	1,504		
Innovation work	15,390	3,496	28.1
positives for software and data	2,511		
positives for research, products, and processes	1,821		
B. Evaluation samples, not used for training			
Simple random sample, all four kinds of work	4,000	1,604	1.6 / 4.6 / 10.8 / 8.9
Score-band supplement, environmental work	2,000	806	35.8
Score-band supplement, social work	2,000	966	15.8
Score-band supplement, innovation work	2,000	953	32.9
Validation sample, environmental, social, and governance work	800	498	15.9 / 17.8 / 31.6
Validation sample, innovation work	700	467	21.7

Notes. Postings and firms: the postings of each sample and the firms that posted them. Labelled positive: the share of postings that the language model scores 1 or 2, among those it judges sufficient to label. Panel A: the postings on which each classifier is trained, which exclude postings judged insufficient and postings that also belong to an evaluation sample; the positives of each part of the work train the logistic regression (head) of that part. Where a row gives several rates, they are for environmental, social, governance, and innovation work, in that order. The score-band supplements are drawn within the bands 0.15 to 0.95 (environmental), 0.07 to 0.46 (social), and 0.11 to 0.70 (innovation) of the classifier's raw score.

IA.II.2 Model Selection

Each classifier is either one logistic regression with an L_2 penalty on the BGE-M3 embedding of the posting, or one such regression for each part of the work, whose probabilities p_k are combined into the probability that at least one part is present, $1 - \prod_k (1 - p_k)$. Five-fold

cross-validation on the training postings makes three choices: the structure, the strength of the penalty, through six values of the inverse penalty C from 0.03 to 10, and whether to weight the classes so as to balance positives and negatives. All postings of a firm fall into the same fold, so that every posting is scored by a model that has not seen its employer, and text that a firm repeats in all its postings cannot inflate the fit. The criterion is the area under the precision–recall curve of the postings left out, which rewards a classifier for placing the rare positives at the top of its ranking, whereas the area under the ROC curve can be high even when many negatives outrank them. The configuration with the highest value among the 24 is refitted on all the training postings.

Table IA.18 reports the selection. Separate regressions for the parts of the work rank postings better for environmental, social, and governance work, and a single regression does so, by 0.005, for innovation work. Weighting the classes never helps. The penalty matters most, and weak penalties do best. For environmental and social work, the selected value, $C = 10$, is the weakest in the grid, and the area still rises by 0.001 and 0.006 from $C = 3$, so that a weaker penalty might score marginally higher. The out-of-fold statistics exceed those on the simple random sample (Appendix Table B1) because the training samples over-represent postings likely to involve the work.

IA.II.3 How Much Training Data the Classifiers Need

Figure IA.1 refits each classifier on random subsets of its training postings and scores every fit on the simple random sample. The returns to more training postings are small. With a tenth of its training postings, between 1,539 and 3,199 postings of which 291 to 463 are positive, each classifier already ranks postings within the 95 percent interval of the full classifier's area under the precision–recall curve for environmental, social, and innovation work, and it does so with a quarter for governance work. From a tenth to all of the training postings, the area rises from 0.553 to 0.592 for environmental work, from 0.378 to 0.432 for social work, from 0.554 to 0.618 for governance work, and from 0.743 to 0.773 for innovation work, and the area under the ROC curve rises by at most 0.024. By this standard, the training samples are four to ten times larger than ranking postings by whether they involve the work at all requires.

The curves concern the overall score. Each regression that recognizes a part of the work learns only from the positives of that part, which number from 448, for information security and privacy, to 2,936, for clean technology (Table IA.17). A tenth of the training postings would leave the regression for information security and privacy with about 45 positives, and those for product and patient safety and for people and rights with fewer than 80. A learning curve also measures only the returns to more postings of the kinds already held. It cannot show what postings of a new kind would add, which is what the targeted samples supply.

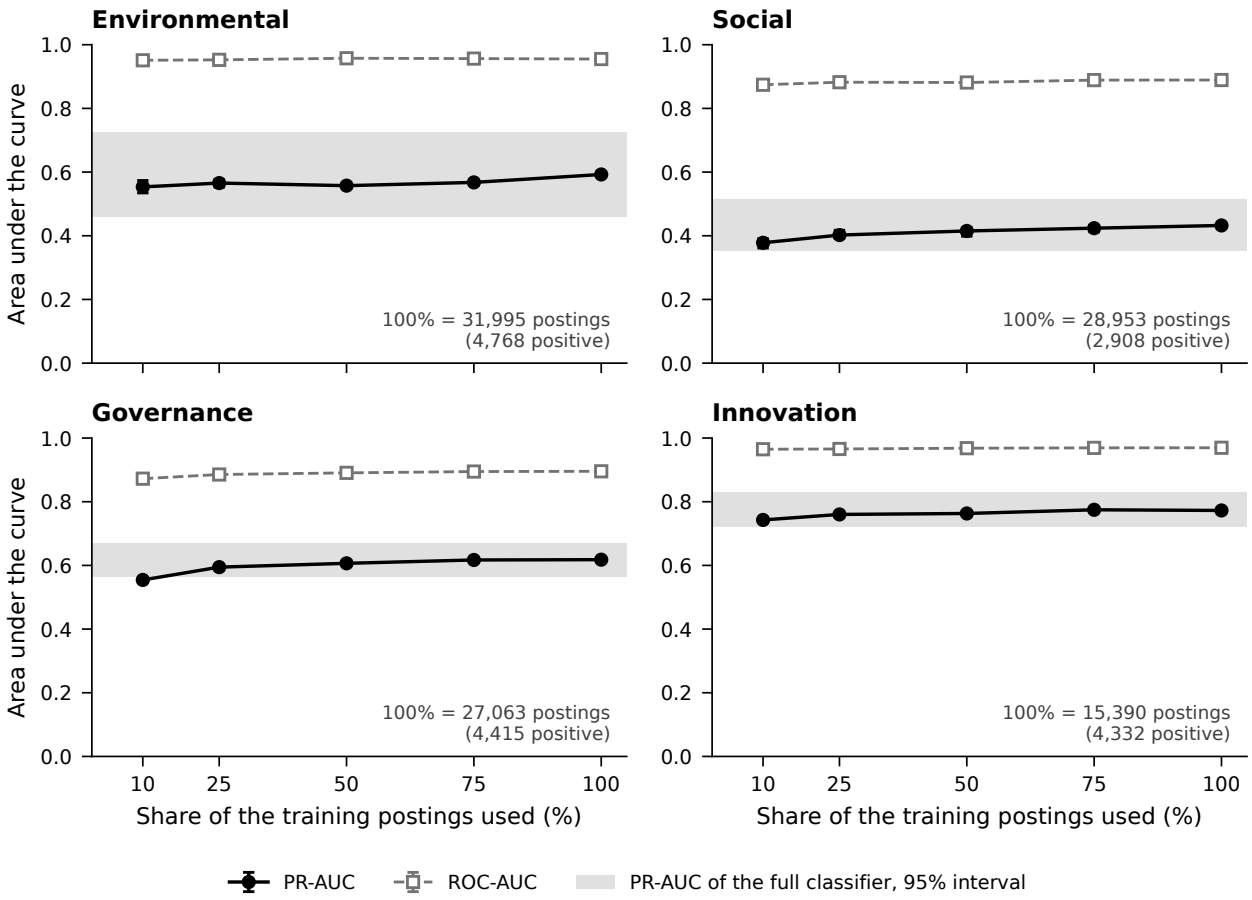


Figure IA.1. Learning Curves. Each classifier is refitted, with the structure, penalty, and class weight selected by cross-validation (Table IA.18), on random subsets of 10, 25, 50, and 75 percent of its training postings, three draws each, and on all of them, and each fit is scored on the simple random sample of 4,000 postings, which is not used for training, against the labels of the language model. PR-AUC is the area under the precision–recall curve, whose value for a classifier without information equals the base rate (1.6 percent of postings for environmental work, 4.5 for social, 10.5 for governance, and 8.8 for innovation work); ROC-AUC is the probability that a randomly chosen positive outranks a randomly chosen negative. Markers give the mean over the three draws and bars their range, with a single fit at 100 percent. The shaded band is the 95 percent interval of the PR-AUC of the full classifier, from a bootstrap of the random sample that resamples whole firms (500 draws). Both areas depend only on the ranking of postings, and are therefore the same for the raw score and the calibrated probability.

Table IA.18. Model Selection

	Environmental	Social	Governance	Innovation
A. Selected configuration				
Structure	Sub-function heads	Sub-function heads	Sub-function heads	One head
Logistic regressions (heads)	2	3	4	1
Inverse penalty C	10	10	3	3
Class weight	None	None	None	None
Training postings	31,995	28,953	27,063	15,390
of which positive	4,768	2,908	4,415	4,332
Firms	4,533	4,361	4,284	3,496
B. Out-of-fold performance of the selected configuration				
ROC-AUC	0.972	0.922	0.895	0.908
PR-AUC	0.875	0.656	0.701	0.800
PR-AUC, range across the five folds	0.853–0.887	0.634–0.672	0.662–0.742	0.769–0.828
Brier score	0.044	0.054	0.083	0.108
C. Best out-of-fold PR-AUC under each choice				
One head	0.864	0.626	0.694	**0.800**
Sub-function heads	**0.875**	**0.656**	**0.701**	0.795
No class weight	**0.875**	**0.656**	**0.701**	**0.800**
Balanced class weights	0.866	0.622	0.689	0.797
D. Out-of-fold PR-AUC by inverse penalty (selected structure and class weight)				
$C = 0.03$	0.786	0.496	0.628	0.736
$C = 0.1$	0.832	0.555	0.655	0.763
$C = 0.3$	0.858	0.605	0.683	0.783
$C = 1$	0.871	0.637	0.700	0.796
$C = 3$	0.874	0.649	**0.701**	**0.800**
$C = 10$	**0.875**	**0.656**	0.700	0.800

Notes. Five-fold cross-validation on the training postings of each kind of work, with folds grouped by firm, over two structures, six values of the inverse penalty C (0.03, 0.1, 0.3, 1, 3, and 10), and two class weights (none, or weights inversely proportional to the frequency of each class). The parts are clean technology and environmental management for environmental work; worker safety, product and patient safety, and people and rights for social work; quality assurance, risk and financial controls, information security and privacy, and compliance and integrity for governance work; and software and data, and research, products, and processes for innovation work. Panel B: out-of-fold statistics of the selected configuration; the Brier score is the mean squared difference between the out-of-fold score and the label. Panel C: the highest out-of-fold PR-AUC among the configurations that make each choice. Panel D: the out-of-fold PR-AUC along the penalty grid for the selected structure and class weight. Selected values are in bold.

IA.II.4 Calibration

Calibration makes the probabilities usable as frequencies, which the expected share of equation (3) requires. Figure IA.2 compares the calibrated probabilities of postings with the rates at which the language model labels them positive, on the half of the calibration sample on which the calibration was not fitted. Above probabilities of about one percent, the bins lie close to the 45-degree line for all four kinds of work. Below it, few sampled postings are positive, and the labelled rates are noisy. Figure IA.3 makes the same comparison for firm-year shares, which can fail to be calibrated even when single postings are, since the errors of postings by the same employer need not be independent. Figure IA.4 makes it by sector, where a classifier whose errors differed across sectors would bias comparisons between

industries.

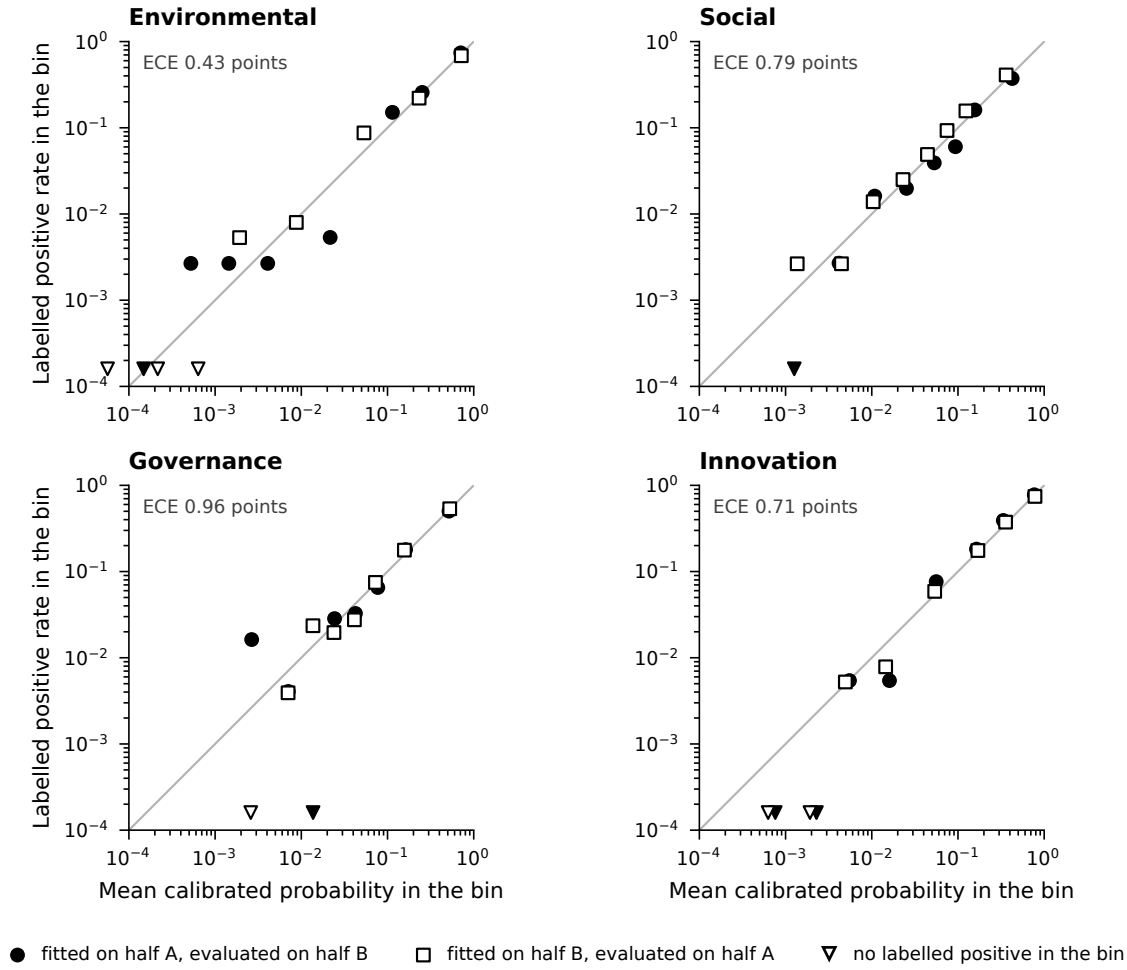


Figure IA.2. Calibration of Postings. The calibration sample, the simple random sample of 4,000 postings joined for environmental, social, and innovation work by 2,000 postings drawn at random within a band of the classifier's raw score, is split into two fixed halves, and the calibration of equation (1) is fitted on one half and applied to the other. The postings of the evaluation half are sorted by their calibrated probability into eight bins with equal numbers of sampled postings, and each marker plots the mean calibrated probability in a bin against the share of its postings that the language model labels positive, both weighted by the inverse of each posting's probability of selection. Circles are fitted on half A and evaluated on half B, squares the reverse, and a calibrated classifier lies on the 45-degree line. Both axes are logarithmic, and bins without a labelled positive are drawn as triangles at the foot of the panel. The expected calibration error (ECE) is the mean absolute difference, in percentage points, between the mean probability and the labelled rate across the eight bins, weighted by bin size and averaged over the two directions (Appendix Table B1, Panel A). Governance work has no score-band supplement.

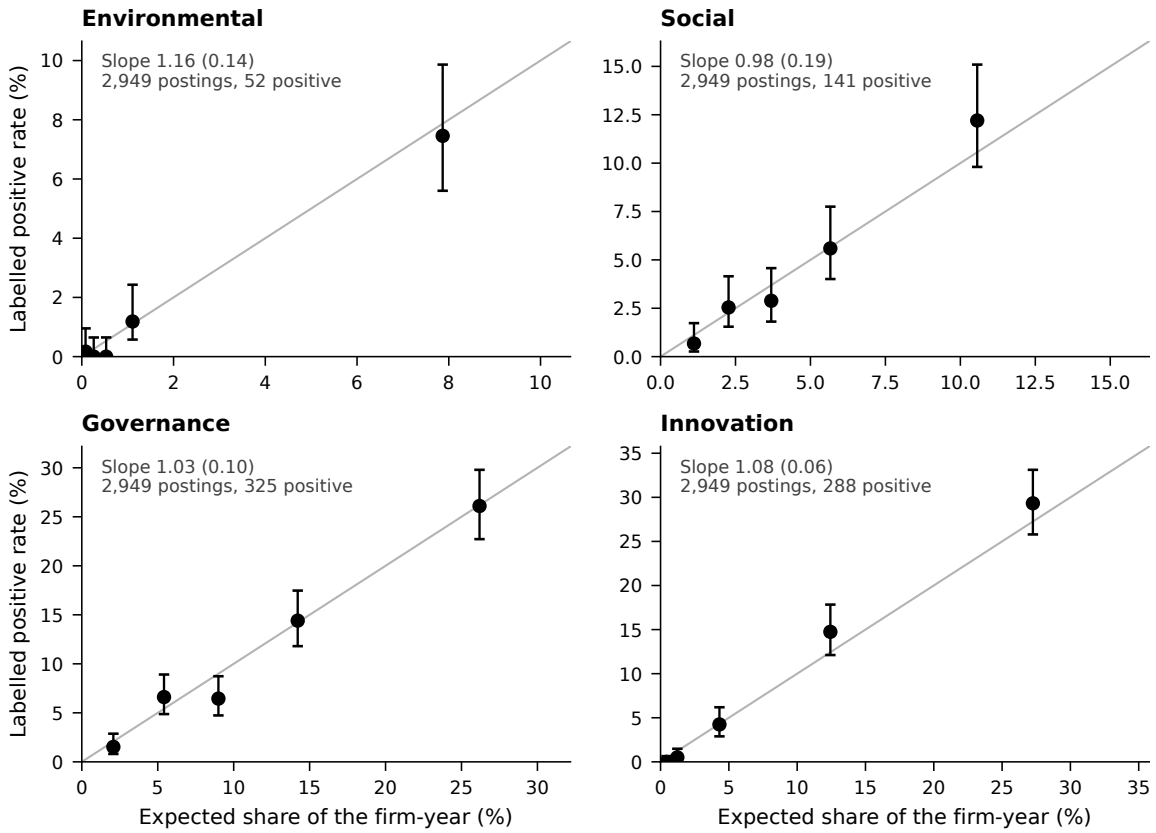


Figure IA.3. Calibration of Firm-Year Shares. The 2,949 postings of the simple random sample that satisfy the panel's rules and fall in firm-years of the estimation sample (1,060 firms and 2,026 firm-years) are sorted into five equal groups by the expected share of the firm-year that posted them, computed without the posting itself. Each marker plots the mean expected share of a group against the share of its postings that the language model labels positive, with a 95 percent Wilson confidence interval. If expected shares are calibrated across firm-years, a posting drawn at random from a firm-year involves the work with a probability equal to the firm-year's true share, and the markers lie on the 45-degree line. The slope is that of a regression of the labels on the leave-one-out expected share, with its standard error clustered by firm in parentheses (Appendix Table B1, Panel B).

IA.II.5 Ranking and the Threshold

The paper measures shares with probabilities rather than with flags. A threshold is nonetheless needed to compute statistics of agreement, such as Cohen's κ, and some measures built from postings count the postings that a classifier flags. Figure IA.5 shows how the classifiers rank the postings of the simple random sample. Figure IA.6 shows the agreement between classifier and language model as the threshold varies. The agreement changes little over a wide range of thresholds, so that the threshold that maximizes it, which the paper uses, is poorly determined. Table IA.19 compares the share of flagged postings with the expected share. The flagged share is zero in 3 to 47 percent of firm-years, and it rises by 1.21 to 1.52

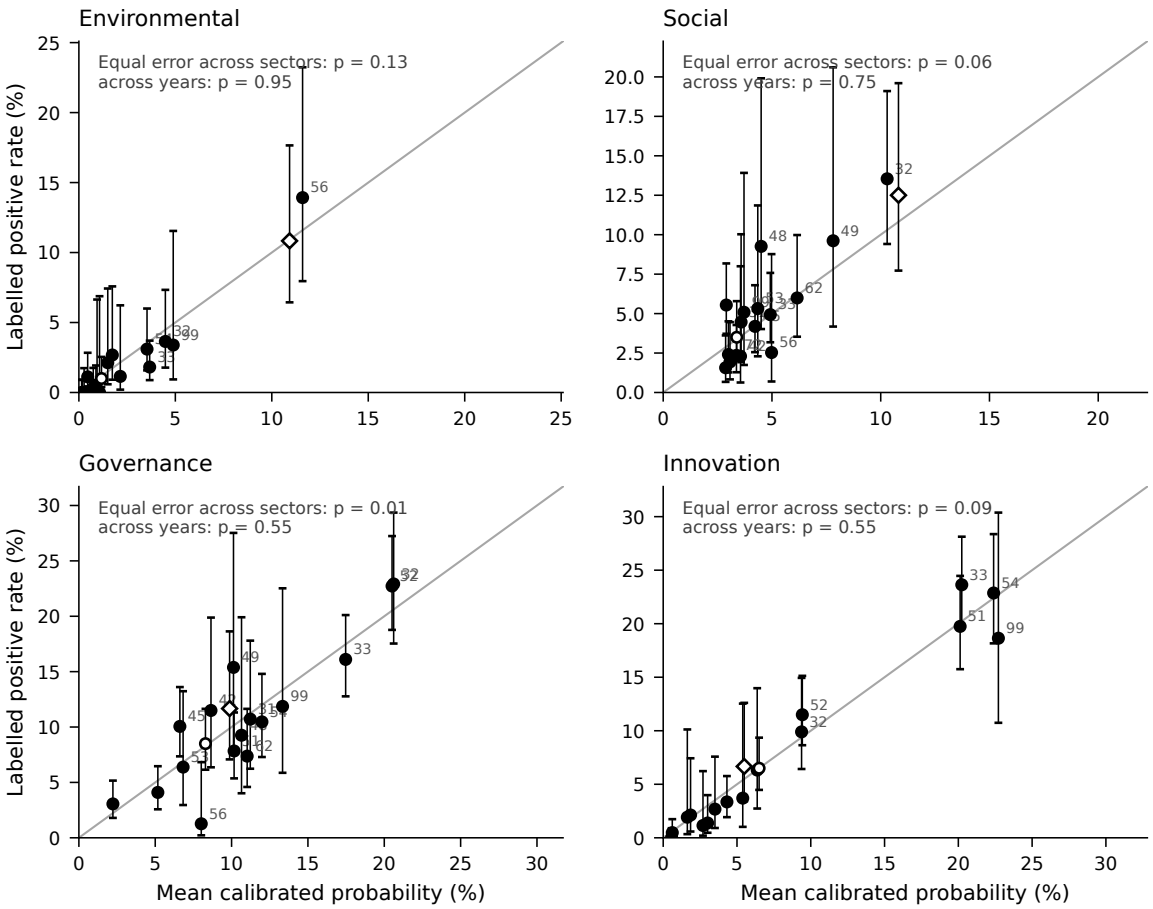


Figure IA.4. Calibration of Postings by Sector. The 4,000 postings of the simple random sample, grouped by the two-digit NAICS sector of the issuer to which the crosswalk assigns the posting's career site in its year. Each marker plots the mean calibrated probability of a sector's postings against the share that the language model labels positive, with a 95 percent Wilson confidence interval; sectors with fewer than 40 sampled postings are pooled (diamond), and postings of career sites without an issuer form their own group (open circle). Labels name the larger sectors. A calibrated classifier lies on the 45-degree line in every sector. The p-values test whether the mean difference between probability and label is equal across sectors and across years, by permuting the groups 2,000 times and comparing the between-group sum of squares with its permutation distribution.

for each unit of the expected share, so that a regression on flagged shares would rescale the coefficients that the expected share gives (Section 3.4).

IA.II.6 Robustness to the Training Data

Table IA.20 refits each classifier without its main draw, without each of its targeted samples in turn, and on five bootstrap draws of its training firms, and scores every posting of the estimation sample again. Leaving out the main draw, which removes most of the training postings, lowers the correlation of the refitted shares with those in use to between 0.88 and

Table IA.19. The Flagged Share and the Expected Share

	Environmental	Social	Governance	Innovation
A. The threshold and the error of a flag (random sample)				
Threshold τ on the raw score	0.368	0.245	0.236	0.375
bootstrap range (10th–90th percentile)	[0.36, 0.50]	[0.21, 0.31]	[0.18, 0.27]	[0.32, 0.39]
Sensitivity	0.63	0.39	0.58	0.74
Specificity	0.991	0.977	0.946	0.972
Attenuation factor (sensitivity + specificity − 1)	0.62	0.36	0.52	0.72
95% confidence interval	[0.48, 0.74]	[0.29, 0.44]	[0.47, 0.58]	[0.67, 0.76]
Flagged rate (%)	1.9	3.9	10.9	9.0
Labelled positive rate (%)	1.6	4.5	10.5	8.8
B. Firm-year shares (estimation sample, 35,772 firm-years)				
Mean expected share (%)	3.0	5.9	15.7	11.8
Mean flagged share (%)	2.9	6.6	17.7	12.7
Firm-years with an expected share of zero (%)	0	0	0	0
Firm-years with a flagged share of zero (%)	47	13	3	12
Correlation of flagged and expected shares	0.98	0.97	0.98	0.99
within firms	0.94	0.91	0.93	0.96
Slope of the flagged share on the expected share	1.32	1.52	1.43	1.21
within firms	1.39	1.59	1.49	1.25
Slope of the expected share on the flagged share	0.73	0.62	0.67	0.81
within firms	0.63	0.53	0.58	0.74

Notes. The flagged share of a firm-year is the share of its openings whose raw score is at least the threshold τ, and the expected share is the mean calibrated probability of its openings (equation (3)). Panel A: the threshold maximizes Cohen's κ with the language model on the calibration sample (Figure IA.6), and its bootstrap range spans the 10th to the 90th percentile of the threshold chosen in 200 bootstrap draws of that sample. Sensitivity is the share of labelled positives that the classifier flags, and specificity the share of labelled negatives that it does not flag, on the 4,000 postings of the simple random sample. If every posting were flagged with these probabilities whatever firm-year posted it, the expected flagged share of a firm-year with true share s would be $(1-\text{specificity})+(\text{sensitivity}+\text{specificity}-1)\,s$, and the attenuation factor, sensitivity plus specificity minus one, would scale the coefficients of a regression with the flagged share as the dependent variable. Its 95 percent confidence interval is from a bootstrap of the random sample that resamples whole firms (2,000 draws), with the threshold held fixed. The labelled rate is that of the random sample alone, without the score-band supplement of Appendix Table B1, Panel A. Panel B: the firm-years of the estimation sample; correlations and slopes within firms are computed after removing each firm's mean. Under the assumption of Panel A, and with calibrated expected shares, the slope of the flagged on the expected share would equal the attenuation factor. It exceeds one for every kind of work, so that the errors of the flags are not independent of the firm-year, and the threshold widens the differences between firm-years. When expected shares are calibrated given the scores of the postings, the slope of the expected on the flagged share is the factor by which a regression with the flagged share as an explanatory variable scales the coefficient on the true share, and the slope of the flagged on the expected share approximates the factor by which a regression with the flagged share as the dependent variable scales the coefficients obtained with the true share.

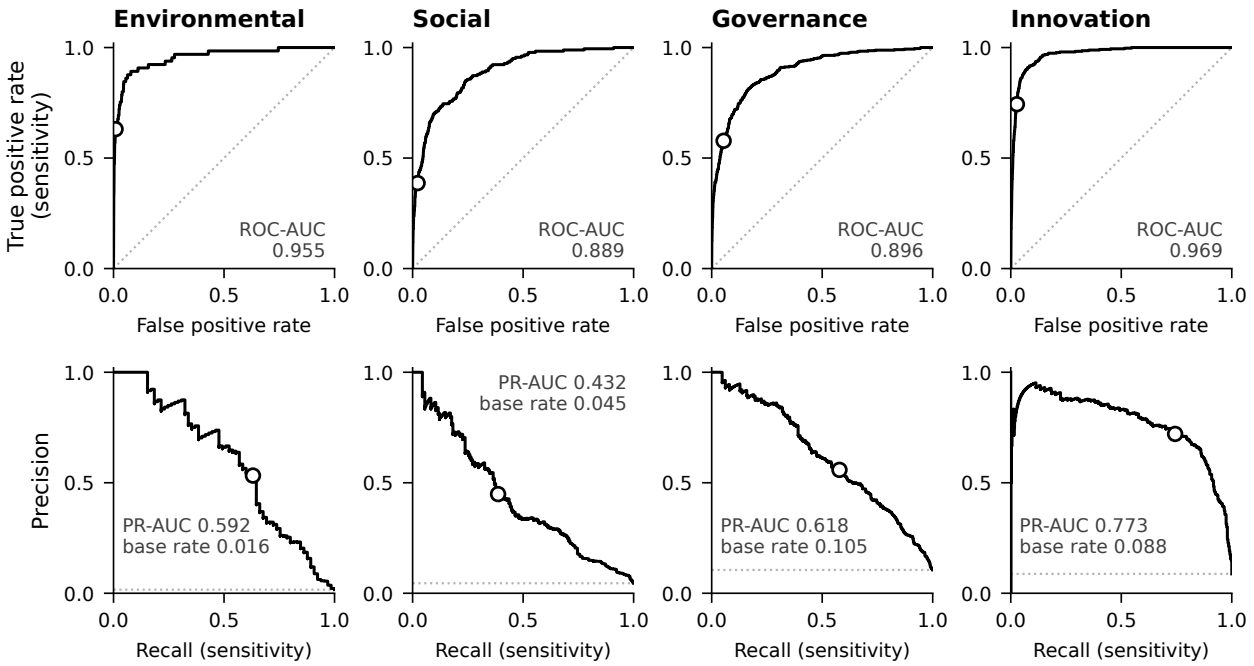


Figure IA.5. Ranking of Postings. The 4,000 postings of the simple random sample, which is not used for training, with the labels of the language model as reference; a posting is positive when the language model scores it 1 or 2. The top row shows receiver operating characteristic curves, the share of positives that the classifier flags (sensitivity) against the share of negatives that it flags (false positive rate) as the threshold on its score falls; the dotted diagonal is the curve of a classifier without information. The bottom row shows precision–recall curves, the share of flagged postings that are positive (precision) against the share of positives flagged (recall); the dotted line is the base rate, the precision of a classifier without information. Circles mark the threshold in use, which maximizes agreement with the language model (Figure IA.6). The areas under the curves are those of Appendix Table B1, Panel C, and are the same for the raw score and the calibrated probability, which ranks postings identically.

0.95. Leaving out a targeted sample keeps it at 0.95 or above, and so do the bootstrap draws. The shares thus change little unless most of the training postings are removed.

IA.II.7 The Terms and Occupations of the Comparison Measures

Appendix Table B2 compares the classifier with three simpler measures. The occupation average credits each posting with the mean calibrated probability of its six-digit occupation over the openings of consolidated firm-years, which is the most that occupation codes can tell. The learned dictionary holds, for each kind of work, the 100 words and two-word phrases most associated with positives in the classifier's training labels, ranked by a smoothed log-odds z-score among the terms that appear in at least 20 positive postings (Table IA.21). The ESG keyword dictionary was curated by hand for an earlier version of this project and plays no part in the measure, and for innovation work the pattern of research and development

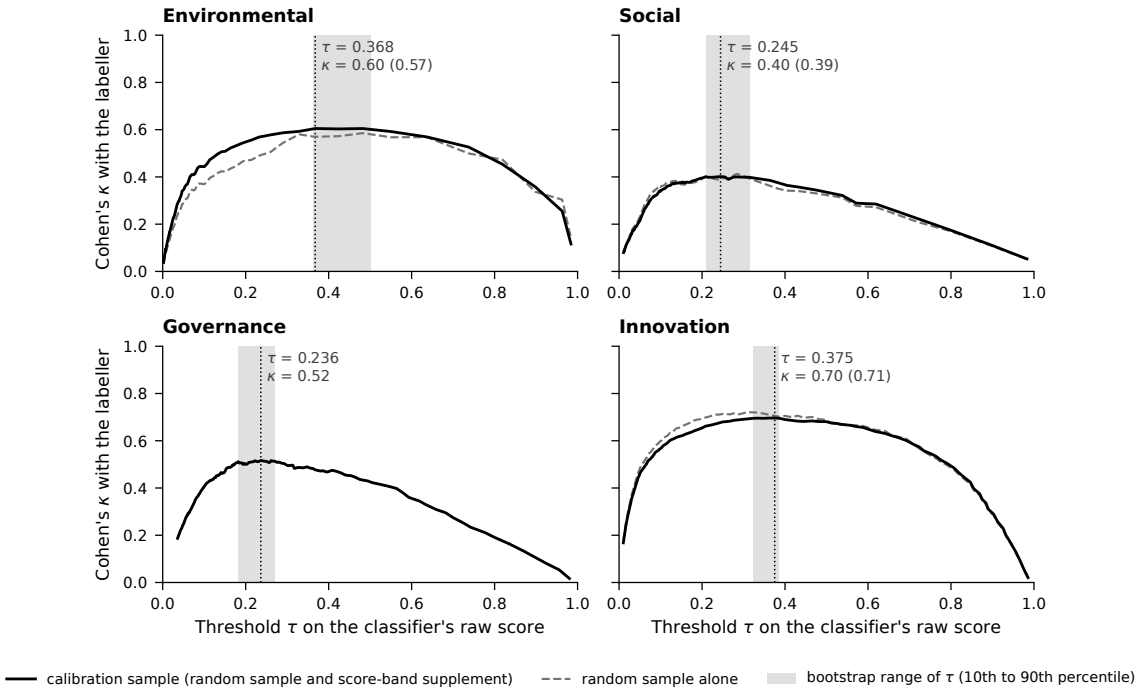


Figure IA.6. Agreement and the Threshold. Cohen's κ between the language model and the classifier, which flags a posting when its raw score is at least the threshold τ, for thresholds between the median and the 99.9th percentile of the raw score. With p_o the share of postings on which the two agree and p_e the agreement expected if each flagged postings at its own rate independently of the other, $\kappa = (p_o - p_e)/(1 - p_e)$. The solid line is computed on the calibration sample, the simple random sample of 4,000 postings joined for environmental, social, and innovation work by its score-band supplement of 2,000 postings, weighted by the inverse of each posting's probability of selection, and the threshold in use is the one that maximizes it (dotted line). The dashed line is computed on the random sample alone, which for governance work is the calibration sample, and the value in parentheses is κ at the threshold in use on the random sample (Appendix Table B1, Panel D). The shaded band spans the 10th to the 90th percentile of the threshold chosen in 200 bootstrap draws of the calibration sample, stratified by sample.

terms is the one that drew part of the innovation training sample (Table IA.22). The terms of both dictionaries match regardless of case and only as whole words or phrases, whereas the innovation pattern matches its strings anywhere in the text, also inside longer words. A posting's score is the number of distinct terms it contains.

The O*NET measures of Appendix Table C5 come from release 24.1 of O*NET, the last to include its green task statements. Green task intensity is the share of an occupation's task statements that O*NET classifies as green (Vona et al., 2018). Green occupations are those that O*NET classifies as new and emerging green occupations or as occupations with enhanced green skills; its third category, of occupations whose demand the green economy increases, leaves their tasks unchanged and is left out. O*NET classifies the occupations of its 2010 taxonomy, which I carry to the 2018 SOC codes of the postings through O*NET's crosswalk,

Table IA.20. Robustness to the Training Data

Refit	Training postings	Correlation with the shares in use: Levels	Ranks	Within firm	Mean share (in use = 1)
Environmental (in use: 31,995 training postings)					
Without the main draw	7,389	0.952	0.932	0.906	0.94
Without one targeted sample (2 refits)	26,081–30,520	0.949–0.963	0.960–0.971	0.907–0.943	1.00–1.02
Bootstrap of firms (5 draws)	30,401–34,865	0.974–0.988	0.974–0.987	0.945–0.975	0.94–1.01
Social (in use: 28,953 training postings)					
Without the main draw	4,362	0.886	0.883	0.856	1.04
Without one targeted sample (2 refits)	26,073–27,471	0.960–0.974	0.971	0.941–0.956	0.98–1.04
Bootstrap of firms (5 draws)	29,081–30,697	0.949–0.964	0.962–0.971	0.932–0.947	0.96–1.04
Governance (in use: 27,063 training postings)					
Without the main draw	2,469	0.880	0.872	0.844	0.85
Without the targeted sample	24,594	0.993	0.993	0.988	1.02
Bootstrap of firms (5 draws)	24,581–29,393	0.979–0.983	0.982–0.985	0.968–0.974	0.98–1.01
Innovation (in use: 15,390 training postings)					
Without the main draw	1,461	0.924	0.938	0.900	0.95
Without the targeted sample	13,929	0.998	0.998	0.996	1.01
Bootstrap of firms (5 draws)	15,003–16,981	0.991–0.993	0.992–0.993	0.982–0.985	0.98–1.00

Notes. Each row refits the classifier, with the structure and penalty selected by cross-validation and a calibration of its own, either without its main draw, without one of its targeted samples (Internet Appendix IA.II.1), or on a bootstrap draw of the training firms, and scores every posting of the estimation sample again. Where a row covers several refits, the range is shown. The correlations relate the refitted firm-year shares to those in use, in levels, in ranks, and within firms after removing firm means; the last column gives the mean refitted share relative to the mean share in use. The main draw is shared by the environmental, social, and governance classifiers.

averaging where several occupations of 2010 merge and over the detailed occupations of each six-digit code. Table IA.23 lists the 90 six-digit occupations with green tasks, 67 of which are green occupations, wholly or in part. The environmental occupations of Panel C of that table are the tenth of six-digit occupations, among those with at least 1,000 consolidated openings, with the highest mean environmental probability. Table IA.24 lists these 66 occupations, which hold 3.7 percent of the openings of the occupations with at least 1,000.

Table IA.21. The Dictionaries Learned from the Training Labels

Environmental. energy; environmental; engineering; sustainable; sustainability; renewable; power; battery; electrical; solar; renewable energy; electric; carbon; water; project; the energy; wind; engineer; construction; of energy; projects; waste; and energy; emissions; storage; in engineering; design; climate; energy storage; utility; engineering or; site; zero; renewables; generation; technical; hydrogen; future; plant; hybrid; mechanical; energy and; grid; electrification; industrial; and environmental; transition; electrical engineering; and sustainability; more sustainable; energy solutions; degree in; esg; development of; in electrical; gas; sustainability and; electricity; des; the environment; plants; climate change; ev; electric vehicle; energy efficiency; electric vehicles; nuclear; utilities; charging; fuel; the development; and construction; of environmental; sector; permitting; engineers; pv; development; green; infrastructure; mobility; vehicle; net zero; commissioning; sites; cell; chemical; studies; voltage; commercial; mechanical engineering; project development; energy industry; energy management; storage systems; turbine; safety; thermal; engineering and; der

Social. safety; safety and; compliance; investigations; and safety; compliance with; audits; employee relations; relations; of safety; regulatory; regulations; hr; osha; health safety; environmental; corrective; the safety; policies; programs; inspections; improvement; human resources; incident; ensure; plant; incidents; employees; hazard; training; initiatives; safety programs; employee; ehs; hse; implementation of; human; leadership; in safety; related; performance; procedures; safe; actions; hazards; conduct; accidents; implement; ensure compliance; to ensure; reports; and environmental; personnel; legislation; management; site; risk; the site; prevention; policies and; investigations and; safety management; labor; programs and; emergency response; occupational; reporting; pharmacovigilance; implementation; activities; manager; corrective action; emergency; practices; continuous improvement; review; coaching; hr business; assessments; accident; committee; effective; safety meetings; safety related; performance management; resources; monitor; to safety; regulations and; ensuring; regulatory requirements; environmental health; supervisory; standards; health; ensures; safety program; matters; operations; management system

Governance. compliance; regulatory; audit; audits; risk; controls; control; compliance with; review; risk management; assurance; processes; corrective; procedures; and compliance; internal; reporting; and regulatory; monitoring; documentation; regulations; regulatory requirements; ensure; ensure compliance; policies; process; quality assurance; compliance and; oversight; investigations; quality; improvement; to ensure; reports; reviews; data; analysis; activities; iso; risk and; issues; quality control; inspection; risks; review and; implementation; monitor; related; actions; manufacturing; assessments; the quality; degree; and risk; management; auditing; external; policies and; remediation; and quality; regulatory compliance; of quality; regulations and; implementation of; processes and; of risk; audits and; gmp; compliance to; corrective action; procedures and; quality management; information security; standards; guidance; corrective actions; requirements; governance; sops; fraud; analytical; accounting; findings; and procedures; legal; policy; controls and; documents; validation; and reporting; of regulatory; conduct; review of; operational; fda; compliant; regulatory and; auditors; support; written

Innovation. design; engineer; engineering; design and; software; python; technologies; programming; java; development; designs; engineers; science; testing; software development; computer science; developer; code; architecture; designing; experience with; test; sql; scientist; technical; languages; integration; javascript; software engineer; the design; agile; in computer; analysis; and design; development experience; engineering or; develop; validation; phd; machine learning; modeling; algorithms; git; techniques; cloud; applications; software engineering; aws; experiments; automation; models; frameworks; coding; distributed; systems; scalable; methods; to design; stack; components; data; technology; degree in; tools; object; tests; web; science or; scale; developers; design of; product development; machine; patterns; developing; and test; debugging; electrical engineering; next generation; scripting; hands on; cad; simulation; devops; of software; object oriented; linux; docker; development of; designs and; generation; development and; design development; testing and; experimental; solutions; platform; application; prototypes; in design

Notes. For each kind of work, the 100 words and two-word phrases with the highest smoothed log-odds z-score of appearing in a positive rather than a negative posting of the classifier's training sample, among the terms that appear in at least 20 positive postings, in decreasing order of that score. Texts are lowercased, and terms are sequences of letters and digits that begin with a letter.

Table IA.22. The ESG Keyword Dictionary and the Innovation Pattern

Environmental (118 terms). air pollution; air quality; biodiversity; carbon accounting; carbon credit; carbon footprint; carbon inventory; carbon neutral; carbon offset; carbon reduction; chemical management; circular economy; circularity; clean energy; climate action; climate adaptation; climate change; climate exposure; climate mitigation; climate program; climate resilience; climate risk; climate scenario; climate-related risk; co2; composting; conservation; corporate responsibility; corporate social responsibility; csr; decarbonisation; decarbonization; deforestation; ecological restoration; ecosystem; effluent; ehs; emission factor; emissions; emissions control; emissions reduction; energy audit; energy conservation; energy efficiency; energy management; energy optimization; energy performance; energy procurement; energy transition; environmental audit; environmental compliance; environmental contamination; environmental engineer; environmental health and safety; environmental impact assessment; environmental management; environmental monitoring; environmental permit; environmental social and governance; environmental sustainability; esg; esg reporting; esg strategy; ghg; ghg inventory; green energy; greenhouse gas; greenhouse gas emissions; habitat; hazardous waste; hse; iso 14001; land use; life cycle assessment; material efficiency; materials recovery; natural capital; net zero; net-zero; photovoltaic; physical risk; pollution prevention; power purchase agreement; recycling; reforestation; remediation; renewable energy; sbti; science-based target; scope 1; scope 2; scope 3; solar; spill response; sustainability; sustainable; sustainable development; sustainable packaging; sustainable procurement; tcfd; tnfd; transition risk; waste diversion; waste management; waste reduction; wastewater; water conservation; water efficiency; water management; water quality; water recycling; water reuse; water stewardship; water treatment; water withdrawal; wildlife; wind power; zero waste

Social (112 terms). ai governance; apprenticeship; belonging; child labor; child labour; climate action; climate change; collective bargaining; community development; community engagement; community investment; community relations; community transition; consumer protection; corporate responsibility; corporate social responsibility; csr; cybersecurity; data governance; data privacy; data protection; dei; diversity and inclusion; diversity equity and inclusion; diversity, equity, and inclusion; employee development; employee engagement; employee relations; employee resource group; employee well-being; employee wellbeing; environmental social and governance; esg; esg reporting; esg strategy; ethical sourcing; ethical supply chain; fair labor standards; fair trade; fair wages; food safety; forced labor; forced labour; free prior and informed consent; freedom of association; gdpr; gender equality; green jobs; grievance mechanism; health and safety; human rights; human rights due diligence; human rights impact; human trafficking; inclusive hiring; indigenous peoples; indigenous rights; industrial hygiene; information security; injury prevention; just transition; labor exploitation; labor relations; labor rights; labor standards; learning and development; living wage; modern slavery; near miss; occupational health; osha; patient safety; pay equity; personal protective equipment; privacy compliance; process safety; product quality assurance; product recall; product safety; product stewardship; reskilling; responsible ai; responsible business conduct; responsible marketing; responsible procurement; responsible sourcing; rights holders; safety compliance; safety management; safety training; salient human rights; social audit; social impact; social license; supplier audit; supplier code of conduct; supplier diversity; supply chain due diligence; sustainability; sustainable; sustainable development; trade union; underrepresented; upskilling; work-life balance; worker representation; workforce development; workforce diversity; workforce transition; working conditions; workplace safety; works council

Governance (80 terms). anti-bribery; anti-competitive; anti-corruption; anti-corruption compliance; anti-money laundering; antitrust; board committee; board diversity; board of directors; board oversight; bribery; business conduct; business ethics; chief sustainability officer; climate action; climate change; code of conduct; code of ethics; competition law; compliance program; compliance risk; conflict of interest; corporate governance; corporate responsibility; corporate social responsibility; country-by-country reporting; csr; double materiality; enterprise risk management; environmental social and governance; esg; esg disclosure; esg governance; esg integration; esg ratings; esg reporting; esg risk; esg strategy; ethics and compliance; export controls; facilitation payment; fair competition; fcpa; governance risk and compliance; government affairs; gri standards; integrity program; internal audit; internal controls; lobbying; lobbying disclosure; materiality assessment; political contributions; public policy engagement; responsible tax; risk and controls; sanctions compliance; sarbanes-oxley; sasb; stakeholder engagement; supplier risk management; supply chain governance; sustainability; sustainability committee; sustainability disclosure; sustainability governance; sustainability metrics; sustainability reporting; sustainability strategy; sustainable; sustainable development; tax governance; tax strategy; tax transparency; third party risk; third-party due diligence; trade compliance; transfer pricing; vendor risk management; whistleblower

Innovation (16 strings). research; prototyp; invent; novel; algorithm; patent; formulat; new product; new feature; develop* new; design* new; new method; new process; new model; next generation; proof of concept

Notes. Terms in alphabetical order. A term appears in several lists where it was assigned to several pillars, such as *sustainability*. Innovation: the strings of the pattern that drew 1,500 postings of the innovation training sample, which match anywhere in the lowercased text, also inside longer words, so that *invent* matches *inventory*; an asterisk stands for any letters, and *next-generation* and *proof-of-concept* also match.

Table IA.23. Occupations with Green Tasks in O*NET

SOC	Occupation	Green tasks	Green occupation
11-1011	Chief Executives	0.50	
11-1021	General and Operations Managers	0.12	yes
11-2011	Advertising and Promotions Managers	0.50	partly
11-2021	Marketing Managers	0.20	yes
11-3051	Industrial Production Managers	0.75	partly
11-3071	Transportation, Storage, and Distribution Managers	0.27	yes
11-9013	Farmers, Ranchers, and Other Agricultural Managers	0.05	partly
11-9021	Construction Managers	0.28	yes
11-9041	Architectural and Engineering Managers	0.60	yes
11-9121	Natural Sciences Managers	0.33	
11-9199	Managers, All Other	0.48	
13-1022	Wholesale and Retail Buyers, Except Farm Products	0.30	yes
13-1041	Compliance Officers	0.03	
13-1081	Logisticians	0.19	
13-1151	Training and Development Specialists	0.10	yes
13-1199	Business Operations Specialists, All Other	0.20	
13-2051	Financial and Investment Analysts	0.22	yes
13-2052	Personal Financial Advisors	0.14	yes
13-2054	Financial Risk Specialists	0.25	yes
13-2099	Financial Specialists, All Other	0.12	
15-1299	Computer Occupations, All Other	0.03	
17-1011	Architects, Except Landscape and Naval	0.29	yes
17-1012	Landscape Architects	0.26	yes
17-2011	Aerospace Engineers	0.50	yes
17-2031	Bioengineers and Biomedical Engineers	0.17	partly
17-2051	Civil Engineers	0.57	yes
17-2071	Electrical Engineers	0.18	yes
17-2072	Electronics Engineers, Except Computer	0.11	yes
17-2081	Environmental Engineers	1.00	yes
17-2112	Industrial Engineers	0.06	
17-2141	Mechanical Engineers	0.54	yes
17-2161	Nuclear Engineers	0.35	yes
17-2199	Engineers, All Other	0.49	
17-3023	Electrical and Electronic Engineering Technologists and Technicians	0.20	partly
17-3024	Electro-Mechanical and Mechatronics Technologists and Technicians	0.14	yes
17-3025	Environmental Engineering Technologists and Technicians	1.00	yes
17-3026	Industrial Engineering Technologists and Technicians	0.24	yes
17-3027	Mechanical Engineering Technologists and Technicians	0.17	partly
17-3029	Engineering Technologists and Technicians, Except Drafters, All Other	0.40	yes
19-1013	Soil and Plant Scientists	0.63	yes
19-1031	Conservation Scientists	0.33	yes
19-2021	Atmospheric and Space Scientists	0.44	yes
19-2041	Environmental Scientists and Specialists, Including Health	0.75	
19-2042	Geoscientists, Except Hydrologists and Geographers	0.48	yes
19-2099	Physical Scientists, All Other	0.08	
19-3011	Economists	0.50	
19-3051	Urban and Regional Planners	0.18	partly
19-3099	Social Scientists and Related Workers, All Other	0.14	
19-4012	Agricultural Technicians	0.22	yes
19-4042	Environmental Science and Protection Technicians, Including Health	1.00	yes
19-4043	Geological Technicians, Except Hydrologic Technicians	0.21	yes

Table IA.23, continued

SOC	Occupation	Green tasks	Green occupation
19-4051	Nuclear Technicians	0.17	yes
19-4099	Life, Physical, and Social Science Technicians, All Other	0.07	
19-5012	Occupational Health and Safety Technicians	0.35	yes
23-1022	Arbitrators, Mediators, and Conciliators	0.05	yes
27-3023	News Analysts, Reporters, and Journalists	0.02	partly
27-3031	Public Relations Specialists	0.22	yes
41-3031	Securities, Commodities, and Financial Services Sales Agents	0.27	partly
41-4011	Sales Representatives, Wholesale and Manufacturing, Technical and Scientific Products	0.57	yes
43-5011	Cargo and Freight Agents	0.10	
43-5071	Shipping, Receiving, and Inventory Clerks	0.09	yes
47-1011	First-Line Supervisors of Construction Trades and Extraction Workers	0.50	
47-2061	Construction Laborers	0.18	yes
47-2152	Plumbers, Pipefitters, and Steamfitters	0.64	yes
47-2181	Roofers	0.32	yes
47-2211	Sheet Metal Workers	0.25	yes
47-2231	Solar Photovoltaic Installers	1.00	yes
47-4011	Construction and Building Inspectors	0.63	yes
47-4041	Hazardous Materials Removal Workers	1.00	yes
47-4099	Construction and Related Workers, All Other	1.00	
47-5013	Service Unit Operators, Oil and Gas	0.05	yes
47-5041	Continuous Mining Machine Operators	0.13	yes
49-3023	Automotive Service Technicians and Mechanics	0.23	partly
49-3031	Bus and Truck Mechanics and Diesel Engine Specialists	0.16	yes
49-9021	Heating, Air Conditioning, and Refrigeration Mechanics and Installers	0.13	partly
49-9071	Maintenance and Repair Workers, General	0.13	yes
49-9081	Wind Turbine Service Technicians	1.00	yes
49-9099	Installation, Maintenance, and Repair Workers, All Other	0.50	
51-2011	Aircraft Structure, Surfaces, Rigging, and Systems Assemblers	0.13	yes
51-4041	Machinists	0.07	yes
51-8011	Nuclear Power Reactor Operators	0.30	yes
51-8013	Power Plant Operators	0.87	yes
51-8099	Plant and System Operators, All Other	1.00	
51-9012	Separating, Filtering, Clarifying, Precipitating, and Still Machine Setters, Operators, and Tenders	0.05	yes
51-9061	Inspectors, Testers, Sorters, Samplers, and Weighers	0.06	yes
53-1042	First-Line Supervisors of Helpers, Laborers, and Material Movers, Hand	0.50	
53-3032	Heavy and Tractor-Trailer Truck Drivers	0.10	yes
53-6051	Transportation Inspectors	0.14	
53-7062	Laborers and Freight, Stock, and Material Movers, Hand	0.50	
53-7081	Refuse and Recyclable Material Collectors	1.00	yes

Notes. Six-digit 2018 SOC codes with at least one green task statement in O*NET 24.1, carried from the O*NET-SOC 2010 taxonomy through O*NET's crosswalk. Green tasks: the share of the occupation's task statements that O*NET classifies as green, averaged over the detailed occupations of the code. Green occupation: the code's principal occupation is a green occupation with new or enhanced skills (yes), or merges occupations of the 2010 taxonomy of which some are (partly).

Table IA.24. The Environmental Occupations

SOC	Occupation	Prob. (%)	SOC	Occupation	Prob. (%)
49-9081	Wind Turbine Service Technicians	74.5	47-4041	Hazardous Materials Removal Workers	13.3
19-1023	Zoologists and Wildlife Biologists	74.1	49-9098	Helpers–Installation, Maintenance, and Repair Workers	12.7
47-2231	Solar Photovoltaic Installers	71.3	55-2013	First-Line Supervisors of All Other Tactical Operations Specialists	12.6
53-7081	Refuse and Recyclable Material Collectors	63.5	11-9041	Architectural and Engineering Managers	12.1
51-8031	Water and Wastewater Treatment Plant and System Operators	59.8	51-9011	Chemical Equipment Operators and Tenders	12.1
17-2081	Environmental Engineers	59.8	17-3022	Civil Engineering Technologists and Technicians	11.9
17-2161	Nuclear Engineers	54.7	49-2092	Electric Motor, Power Tool, and Related Repairers	11.8
19-2043	Hydrologists	50.5	51-8021	Stationary Engineers and Boiler Operators	11.8
19-2041	Environmental Scientists and Specialists, Including Health	45.4	51-8092	Gas Plant Operators	11.8
17-3025	Environmental Engineering Technologists and Technicians	35.1	19-1032	Foresters	11.6
51-8011	Nuclear Power Reactor Operators	35.1	51-8093	Petroleum Pump System Operators, Refinery Operators, and Gaugers	11.3
19-4042	Environmental Science and Protection Technicians, Including Health	33.8	19-2032	Materials Scientists	11.2
17-2199	Engineers, All Other	32.4	49-3092	Recreational Vehicle Service Technicians	11.0
19-4071	Forest and Conservation Technicians	28.8	49-2094	Electrical and Electronics Repairers, Commercial and Industrial Equipment	10.8
17-2071	Electrical Engineers	24.1	19-1031	Conservation Scientists	10.7
51-8013	Power Plant Operators	22.6	49-9043	Maintenance Workers, Machinery	10.4
51-8099	Plant and System Operators, All Other	22.3	51-8012	Power Distributors and Dispatchers	10.4
47-2073	Operating Engineers and Other Construction Equipment Operators	22.2	17-2171	Petroleum Engineers	10.2
19-4051	Nuclear Technicians	21.8	47-2071	Paving, Surfacing, and Tamping Equipment Operators	9.5
17-2111	Health and Safety Engineers, Except Mining Safety Engineers and Inspectors	21.6	17-2041	Chemical Engineers	9.5
19-3091	Anthropologists and Archeologists	20.2	11-9021	Construction Managers	9.4
17-2051	Civil Engineers	19.9	47-5099	Extraction Workers, All Other	9.2
19-5011	Occupational Health and Safety Specialists	19.6	17-1022	Surveyors	8.9
19-2042	Geoscientists, Except Hydrologists and Geographers	19.1	25-1032	Engineering Teachers, Postsecondary	8.8
47-2111	Electricians	18.5	43-5111	Weighers, Measurers, Checkers, and Samplers, Recordkeeping	8.8
49-2095	Electrical and Electronics Repairers, Powerhouse, Substation, and Relay	17.9	51-4035	Milling and Planing Machine Setters, Operators, and Tenders, Metal and Plastic	8.7
51-8091	Chemical Plant and System Operators	16.0	19-5012	Occupational Health and Safety Technicians	8.5
53-3032	Heavy and Tractor-Trailer Truck Drivers	15.7	37-2021	Pest Control Workers	8.4
11-1011	Chief Executives	15.3	47-4099	Construction and Related Workers, All Other	8.3
17-1012	Landscape Architects	14.4	37-3012	Pesticide Handlers, Sprayers, and Applicators, Vegetation	8.2
47-4011	Construction and Building Inspectors	14.2	19-3011	Economists	8.2
27-1012	Craft Artists	13.7	47-1011	First-Line Supervisors of Construction Trades and Extraction Workers	8.1
17-2141	Mechanical Engineers	13.4	17-2021	Agricultural Engineers	7.9

Notes. The tenth of six-digit occupations with the highest mean environmental probability over the consolidated openings of 2014 to 2023, among the 667 occupations with at least 1,000 such openings, in decreasing order of that probability. Prob.: the mean calibrated environmental probability of the occupation's openings. Titles are those of O*NET-SOC 2019.

IA.III The Data

This part gives the details of the data summarized in Section 2 of the paper. Table 1 of the paper traces the postings from LinkUp's records to the estimation sample. The subsections describe the postings and how they become openings, how career sites are linked to issuers, the estimation sample, how it compares with Compustat and with US labor statistics, the patents, the records of activity, the existing measures of climate exposure, and the market data.

IA.III.1 The LinkUp Data

LinkUp takes postings only from employers' own career websites and none from job boards, aggregators, or the listings that staffing agencies publish for other employers (LinkUp, 2020). It visits each career site at least every 48 hours and identifies each posting by its web address, so that a job advertised once is recorded once. It gives each career site an identifier of its own, so that a firm with several brands or subsidiaries can appear under several identifiers, which the crosswalk described in Section IA.III.3 brings back together. Because a posting is identified by its web address, a job whose address disappears and later reappears is restored rather than recorded anew, and each record carries the date on which LinkUp first saw the posting and the date on which it found the posting removed, together with the dates of its latest visit and of its latest recorded change (LinkUp, 2026). LinkUp aims to cover every public company in the United States that advertises positions on its own website, and it reports that about 15 percent of public companies do not (LinkUp, 2020). It also covers private employers and, increasingly over the decade, employers outside the United States.

For each posting, the files record its identifier and that of its career site, its title and the text of its description, its city, state, postal code, and country, the occupation code that LinkUp assigns from the O*NET-SOC taxonomy, the four dates, and its web address. The measure reads only the title and the description. The aggregation to firm-years uses the career site and the removal date, and the occupation code serves only to decompose the shares by occupation. Table IA.25 describes the length and language of the openings of the estimation sample year by year. The median opening runs to 489 words, and the median rises from 300 words in 2014 to 568 in 2022. Of the openings, 92.8 percent are written in English, 99.5 percent of those located in the United States and 80.2 percent of those located elsewhere, followed by French (2.0 percent), German (1.8), Chinese (1.1), and Spanish (0.9). The English share falls from 98.8 percent in 2014 to 89.0 percent in 2023, as the share of openings located outside the United States rises. LinkUp's software could not place 4.1 percent of the openings in any occupation and gives them a catch-all code, which happens

to 23 percent of the openings not written in English and to 2.5 percent of those written in English.

Postings nonetheless differ from hires in several respects. A removed posting was either filled or withdrawn, and the data cannot tell which, although removal is the observed event closest to a hire, and studies of vacancy duration measure it by the time for which an advertisement remains online (Bassier, Manning and Petrongolo, 2025). Collection from career sites removes the phantom vacancies that syndication creates, because a posting disappears when the employer takes it down, but it cannot rule out requisitions that an employer keeps open on purpose. LinkUp calls such postings evergreen and recommends treating any posting open for more than 180 days as one, and the restrictions of Section IA.III.2 exclude them. Postings also describe the work for which a firm recruits rather than the work that its current employees perform, so that the measure records the direction of a firm's hiring rather than the stock of its workforce. Finally, the collection itself can leave traces in the data. A career site that changes its format can drop out for a time, and a firm that changes its applicant-tracking system can produce a short burst of postings. Because the measure is a share of openings, such episodes affect it only insofar as the postings that are lost or repeated differ in their work from the firm's other postings.

Table IA.25. Length and Language of the Openings by Year

Removed	Openings	Length in words			Located in	Written in	No specific
in	(M)	median	P10	P90	the US (%)	English (%)	occupation (%)
2014	1.31	300	86	512	100.0	98.8	4.1
2015	2.73	351	106	604	91.9	97.9	2.1
2016	5.98	400	130	745	81.8	96.7	2.0
2017	7.70	407	138	750	73.5	95.7	2.8
2018	9.43	427	162	782	68.9	94.6	3.6
2019	10.85	453	175	815	64.1	91.5	4.4
2020	9.22	489	191	871	63.1	91.8	4.6
2021	16.01	547	225	976	62.3	91.9	4.2
2022	17.89	568	223	995	58.6	92.2	4.3
2023	10.65	561	204	989	57.0	89.0	5.6
All	91.8	489	173	896	65.6	92.8	4.1

Notes. Openings of the estimation sample, by year of removal; 2023 ends on 30 September. Length: words of the title and the description as cleaned, separated by white space; P10 and P90 are the 10th and 90th percentiles. Written in English: share of a random subsample of up to 3,000 openings per year (27,979 in all) whose first 2,000 characters the `langdetect` library identifies as English; the *All* row weights the years by their openings. No specific occupation: openings with LinkUp's catch-all O*NET-SOC code 99-9999.00 or without a code.

IA.III.2 From Postings to Openings

The corpus starts from LinkUp's files, which hold 257.9 million postings created between 2007 and 30 September 2023 on 61,519 career sites, 200.4 million of them with a description. I keep the postings with a description on the 16,482 career sites that carry, at some point, a primary ticker in LinkUp's company reference of 30 September 2023. I clean the title and the description of each posting, which the measure reads together, of markup, encoding errors, web and e-mail addresses, telephone numbers, and postal codes, and I keep the 126.3 million postings whose cleaned text is not empty. No posting is dropped for its length, language, country, or occupation code. LinkUp records a job advertised for several locations as one posting per location, and I count each location as one opening.

Identification by web address fails on career sites that change the address of a job at every visit. On these sites, the same job is recorded again at each crawl, and a single vacancy appears as a chain of postings, each open for a day or two. I treat a posting as a copy of an earlier job if it appears on the same career site within three days after the removal of an earlier posting with the same title and location, and if it is itself removed within three days. I merge each chain of copies into a single opening that runs from the creation of its first posting to the removal of its last. The rule merges 4.9 million copies, or 3.9 percent of the corpus. The copies are concentrated on a few career sites, and the rule leaves the counts of ordinary sites almost unchanged.[1]

I date each opening by the calendar year in which it is removed from the career site, which is the observed event closest to the filling or withdrawal of the position. Three restrictions follow from this rule and from LinkUp's own guidance. A posting must be removed by the end of the data. It must be open for at most 180 days, the threshold beyond which LinkUp treats a posting as an evergreen requisition rather than a vacancy. And it must be created in 2014 or later. No posting in the corpus is recorded as removed before 2014, so the 0.25 million postings created between 2007 and 2013 are only those still open in 2014, a selected set of long-running requisitions. Their median life is 618 days for those created in 2013 and more than 1,000 days for those created earlier, and 95 percent of the 2013 postings remain

[1]The 100 career sites with the most copies hold 74 percent of them. For example, Urban Outfitters' site added a session key to each address in 2017 and 2018, and 77 percent of its postings are copies. On the largest ordinary sites, such as those of Target, HCA, and Marriott, the rule merges at most 3.3 percent of the postings created in any year before 2023. Tegna's career site added a tracking parameter to each address from 2020 onward, and copies make up 81 percent of its postings. A further 264,179 postings that satisfy the rule could not be paired with a single earlier posting, because several appeared at once, and they are kept as openings. Five career sites were crawled for a time through job boards that list other employers' jobs, and I remove these 116,000 postings, 98 percent of which come from two boards, the my.jobs network on Mobile Mini's site in 2017 and 2018 and the VentureLoop board of venture-backed start-ups on Safeguard Scientifics' site in 2018.

open beyond 180 days.[2] These restrictions retain 110.6 million postings, or 87.6 percent of the corpus.

These choices matter little for the measure. Table IA.26 recomputes the four shares under five alternative rules, dating openings by the year of their creation, keeping postings of any duration, keeping only those open at most 90 days, adding the postings still open at the end of the data, and counting a job advertised for several locations once. Across firm-years, the shares under each rule correlate with the baseline shares at 0.99, and their changes from one year to the next at 0.83 to 0.99. The lowest correlations of changes arise when openings are dated by creation, which moves some of them to an earlier year.

Table IA.26. The Shares under Alternative Rules for Dating and Keeping Postings

	Firm-years	Correlation in levels				Correlation of yearly changes			
Rule	in the sample	E	S	G	I	E	S	G	I
Dated by creation	37,150	0.99	0.99	0.99	0.99	0.89	0.87	0.87	0.83
Any duration	36,364	1.00	1.00	1.00	1.00	0.98	0.97	0.97	0.95
At most 90 days	34,829	1.00	0.99	1.00	0.99	0.97	0.96	0.96	0.94
Open postings added	36,120	1.00	1.00	1.00	1.00	0.99	0.99	0.99	0.99
Locations collapsed	35,552	1.00	0.99	0.99	1.00	0.98	0.97	0.96	0.96

Notes. Each rule replaces one choice of the baseline, which dates openings by the year of their removal and keeps the postings removed by 30 September 2023 and open at most 180 days, counting each location of a job as one opening. Dated by creation: openings dated by the year of their creation, whether removed or not. Any duration: no limit on the days open. At most 90 days: a limit of 90 days instead of 180. Open postings added: the postings still open at the end of the data are kept and dated 2023. Locations collapsed: a job advertised for several locations on the same day counts as one opening. Firm-years in the sample: firm-years with at least 20 openings under the rule and Compustat fundamentals with positive total assets, 2014–2023. Correlations are computed over the firm-years that the rule's sample shares with the baseline sample, in levels and, for consecutive firm-years of the same firm, in changes from one year to the next. E, S, G, I: the environmental, social, governance, and innovation shares.

Table IA.27 applies these restrictions to each year's postings. The number of postings created in a year grows from 2.1 million in 2014 to 22.4 million in 2022, as LinkUp extends its coverage to more career sites and, increasingly, to jobs located outside the United States. A typical retained posting stays open for three to four weeks, and this barely changes over the decade, with a median of 22 to 29 days and a 90th percentile of 87 to 107 days in every year.

[2]Computed on the 245,967 postings created between 2007 and 2013, of which 182,775 were created in 2013. Among all postings created in 2014, before the 180-day restriction, the median life is 40 days and 16 percent remain open beyond 180 days.

Table IA.27. Postings by Year of Creation

Created in	Postings (M)	Career sites	Located in the US (%)	Copies (%)	Still open at the cut (%)	Open more than 180 days (%)	Removed in a later year (%)	Retained (M)
2013	0.18	2,344	99.7	0.0	0.0	95.1	100.0	0.00
2014	2.11	5,120	99.6	0.5	0.0	16.3	26.7	1.76
2015	4.42	5,767	90.0	1.0	0.0	14.1	29.5	3.76
2016	7.92	5,741	82.4	1.5	0.0	9.6	18.1	7.05
2017	11.62	7,309	77.0	6.7	0.0	7.7	17.1	10.00
2018	13.23	8,078	71.1	5.7	0.0	6.8	15.0	11.63
2019	14.86	8,610	66.9	2.8	0.4	7.1	15.0	13.36
2020	13.04	8,782	67.7	5.2	0.4	6.5	15.9	11.51
2021	22.27	8,874	63.3	3.6	0.4	6.1	16.0	20.08
2022	22.40	9,190	59.4	3.9	1.2	4.9	11.5	20.21
2023	14.20	9,728	59.5	3.9	17.2	0.6	0.0	11.23
All	126.2		68.0	4.0	2.3	6.5	14.2	110.6

Notes. All postings of the corpus, on the career sites that LinkUp links to a listed company's ticker, by calendar year of creation, each counted once. Copies: postings that record again a job already on the career site under a new web address, together with the postings that career sites drew from job boards (Section IA.III.2), in percent of the year's postings; the other shares are computed over the remaining postings. Located in the US: share located in the United States. Still open at the cut: share not yet removed on 30 September 2023, the end of the data. Open more than 180 days: share removed by then after more than 180 days. Removed in a later year: share removed in a calendar year after the year of creation. Retained: postings created in 2014 or later, removed by the cut, and open at most 180 days. The *All* row weights the shares by the postings that remain after copies.

IA.III.3 From Career Sites to Firms

LinkUp assigns a separate identifier to each career site and records, for each site, the listed company with which it is associated, through the history of the site's primary stock ticker and through its legal entity identifiers, Refinitiv PermIDs, names, and web domains. The ticker history dates the links, and the other records serve to identify the issuer. A site carries one symbol at a time, and I call the span over which it carries the same symbol a ticker period. The symbol changes when a company only renames its stock, but also when the company behind the site changes through an acquisition, a spin-off, or a listing, so that the ticker periods date these changes of ownership. The career site of MTS Systems, for example, carries the symbol of MTS Systems until April 2021, that of Amphenol, which acquired the company, until December 2021, and that of Illinois Tool Works, which bought its test business from Amphenol, from then on. Of the 16,482 career sites of the corpus, 16,345 have at least one ticker period that overlaps 2007–2023, 22,970 periods in all, and each of the other 137 sites receives a single period spanning its postings.

The crosswalk links each ticker period to at most one Compustat issuer. Four routes propose candidates. The first matches LinkUp's legal entity identifiers and PermIDs to Compustat through the ISINs and CUSIPs that Refinitiv and the Global Legal Entity Identifier Foundation associate with them, through S&P Capital IQ, and through the CRSP–Compustat Merged link. The second matches the period's ticker to the dated ticker histories of Capital IQ, Compustat, CRSP, Datastream, and Worldscope, the third to undated Compustat tickers and to the tickers that SEC EDGAR associates with each filer, and the fourth to legal names and registered web domains. A candidate must be an issuer that exists during the period and, under most rules, one that trades in the country of the period's stock exchange. Each candidate is then graded by evidence that does not depend on the route that proposed it, which is strong when the site and the issuer share a registrable web domain or have names that coincide or nearly coincide,[3] and it is accepted under one of a fixed set of rules, such as an exact match of identifiers, a dated ticker confirmed by names or web domains, or a ticker that identifies a single issuer in the country of listing (Table IA.28). Where the accepted issuer covers only part of a period, the rest of the period is resolved again without it.

An interval is a period during which a career site is linked to one issuer. Consecutive ticker periods linked to the same issuer are merged into one interval, gaps of up to 45 days

[3]Names are compared after they are lowercased and stripped of accents, punctuation, legal forms such as Inc., Corp., and Ltd., and words such as Holdings, Group, and The. Two names coincide when they are then identical. They nearly coincide when one contains the other, both being at least five characters long, or when their token-set similarity, a score from 0 to 100 that compares the sets of words in the two names regardless of their order, computed with the `rapidfuzz` library, is at least 90. The comparison takes the best pair among the site's names in LinkUp and the issuer's dated names in Compustat.

between them are closed, and each interval is clipped to the dates during which its issuer exists, so that the intervals of a site never overlap. After the hand rulings and the review described below, the crosswalk comprises 18,412 intervals that link 15,585 career sites to 10,514 issuers. Most sites have a single interval, and 2,448 have between two and six.

Of these intervals (Table IA.28), 3,312 rest on an exact match of identifiers and carry 58.4 percent of the postings assigned to issuers, 7,449 rest on a dated ticker confirmed by names or web domains (27.3 percent), 4,780 on a ticker that is unique within the firm's country (9.9 percent), and 2,340 on other automatic rules (2.8 percent), while 531 were set by hand (1.5 percent). The crosswalk labels a link as the issuer's own site when the evidence of identity is strong, which is the case for 10,499 intervals after the hand rulings, and as an affiliated site otherwise, for 7,451 intervals, a label that records the strength of the evidence rather than an ownership relation. The subsidiary lists that issuers file as Exhibit 21 and the parent relations recorded by the Global Legal Entity Identifier Foundation and Capital IQ do not create links, but they serve to check them. I ruled by hand on 3,763 of the issuer–site pairs that rest on a ticker without corroborating names or domains, and a final review read the links most exposed to error against corporate events and against each site's own postings.[4]

I assign each posting to the issuer whose interval contains its removal date. A site's openings thus follow its owner through splits, spin-offs, acquisitions, and sales, as Figure IA.7 shows for six career sites. I exclude postings when the issuer does not consolidate the employer, as for minority stakes, joint ventures, operators that lease their facilities from a real estate investment trust, and the portfolio companies of investment funds. Franchise systems call for a finer rule, because a franchisor's career site lists the positions of independent franchisees together with its own. I exclude any posting that states that its employer is an independent franchisee. On the sites of systems whose units are mostly franchised, I also exclude the positions that staff those units in every year, while the franchisor's corporate positions remain assigned to it.[5] These exclusions remove 5.9 million postings of 62 issuers, 5.6 million of which are franchisees' openings, and fifteen issuers account for 99 percent of them (Table IA.29).[6]

[4]The final review corrected the links of 74 career sites, which carry about 0.25 percent of all the postings that LinkUp records as removed over 2014–2023. The links of eighteen career sites outlived a divestiture, pointed to an issuer that never owned the employer, or fell on the wrong side of a spin-off or merger, such as the career site of TK Elevator after thyssenkrupp sold its elevator business in July 2020. Those of the other 56 pointed to issuers that hold the employer without consolidating it.

[5]Most franchisors print such statements only from some year onward, so that excluding the declared postings alone would make a franchisor's hiring change in the year its site begins to print them. The unit positions are those in which franchisees' declared postings fall, such as food preparation, restaurant management, and cashiering for restaurants. The rule covers the career sites of 18 restaurant, salon, fitness, and chiropractic systems, among them those of McDonald's, Yum! Brands, Wendy's, and Domino's.

[6]Domino's illustrates why the rule covers the unit positions as well as the declared postings. Its career site, which carried at most a few hundred openings a year until 2021, lists the store openings of its franchisees from 2022 onward, and the rule excludes 494,879 openings, against the 6,955 that remain assigned to Domino's, an

Table 1 of the paper traces the postings from LinkUp's records to the estimation sample. The corpus keeps the postings on the 16,482 career sites that LinkUp links, at some point, to the ticker of a listed company, nearly all of which the final crosswalk links. Of the 110.6 million postings that satisfy the dating restrictions, 100.4 million are assigned to a Compustat issuer, and 94.5 million remain after the exclusion of non-consolidated relations. The postings that are not assigned belong mostly to employers that were not part of a listed issuer at the time, on linked career sites outside any ownership interval, and about a fifth of them lie on the career sites that the final crosswalk does not link.

Table IA.28. The Crosswalk from Career Sites to Issuers

A. By acceptance rule

		Labelled as					Assigned
Rule	Intervals	own site	affiliated	manual	Career sites	Issuers	postings (%)
Exact match of identifiers	3,312	3,219	93	0	3,269	3,157	58.4
Dated ticker confirmed by names or web domains	7,449	7,022	427	0	7,320	6,635	27.3
Ticker unique within the firm's country	4,780	1	4,779	0	4,338	2,267	9.9
Other automatic rules	2,340	234	2,106	0	2,222	1,087	2.8
Set by hand	531	23	46	462	504	374	1.5
All	18,412	10,499	7,451	462	15,585	10,514	100.0

B. By label

	Intervals	Not consolidated	Career sites	Issuers	Assigned postings (%)
Own site	10,499	5	10,289	9,456	84.0
Affiliated site	7,451	114	6,479	3,217	15.7
Manual	462	2	445	329	0.3
All	18,412	121	15,585	10,514	100.0

Notes. Dated intervals linking a career site to a Compustat issuer. Panel A classifies each interval by the rule under which it was accepted. Exact match of identifiers: legal entity identifiers and PermIDs matched to Compustat through the ISINs and CUSIPs associated with them. Dated ticker confirmed by names or web domains: the site's dated ticker matched to an issuer whose name coincides or nearly coincides with the site's, or with which it shares a registrable web domain. Ticker unique within the firm's country: a dated ticker that identifies a single issuer in the country of listing. Other automatic rules: tickers of authoritative US listings (1,717 intervals), tickers found in a single family of sources (270), firm-level fallbacks (234), dominant tickers among ambiguous candidates (112), and tickers in two families with weak identity (7). Set by hand: 523 manual overrides and 8 intervals that the final review added or re-dated directly in the crosswalk. Labelled as: the label that the crosswalk records after the hand rulings, the issuer's own site when the evidence of identity is strong and an affiliated site otherwise, a label that records the strength of the evidence rather than an ownership relation; 61 of the 523 overrides were relabelled as own (21) or affiliated (40) sites once their evidence was ruled on, so that 462 intervals keep the label "manual". Panel B classifies the same intervals by label. A career site can have intervals under several rules or labels, for example before and after an acquisition, so that the counts of sites and issuers do not add up to the totals. Not consolidated: intervals whose issuer does not consolidate the employer, excluded from the sample. Assigned postings: the retained postings whose removal date falls in the interval, in percent of the 100.4 million so assigned (including those of non-consolidated intervals).

employer of some 11,000 people.

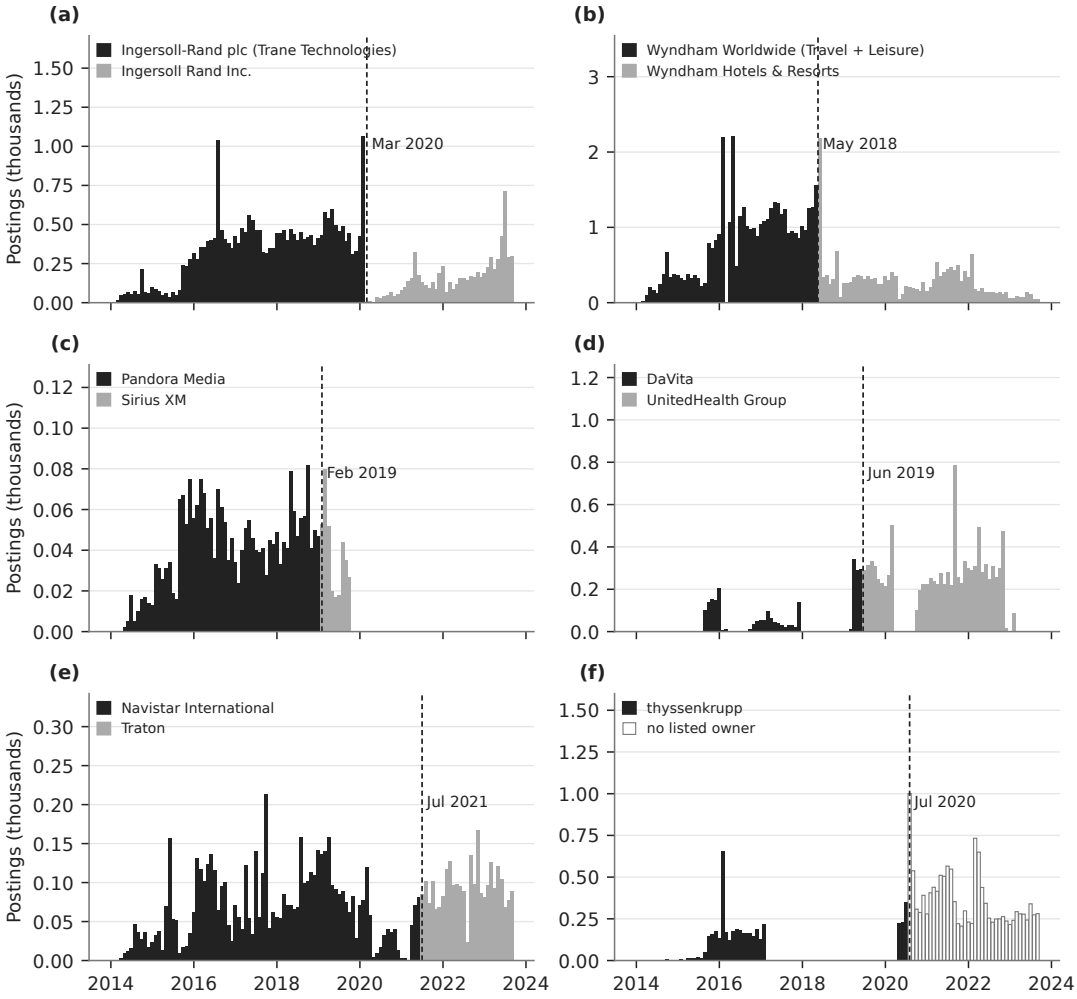


Figure IA.7. Dated Links around Corporate Events. Retained postings on six career sites by month of removal, shaded by the issuer to which the crosswalk assigns them; white bars are postings removed after the employer left listed ownership, which are not assigned. The dashed line marks the event: (a) the split of Ingersoll-Rand plc (now Trane Technologies) and Ingersoll Rand Inc. in March 2020; (b) the spin-off of Wyndham Hotels & Resorts from Wyndham Worldwide (now Travel + Leisure) in May 2018; (c) the acquisition of Pandora by Sirius XM in February 2019; (d) the sale of DaVita's medical group (HealthCare Partners) to UnitedHealth Group in June 2019; (e) the acquisition of Navistar by Traton in July 2021; (f) the sale of thyssenkrupp's elevator business to private equity owners in July 2020. The issuer of a month is that whose interval contains the middle of the month; each posting is assigned by its removal date.

Table IA.29. Openings Excluded Because the Issuer Does Not Consolidate the Employer

Issuer	Excluded openings belong to	Excluded openings	Kept openings	Excluded share (%)
Yum! Brands	Franchisees	3,981,214	37,915	99.1
McDonald's	Franchisees	578,294	24,732	95.9
Domino's Pizza	Franchisees	494,879	6,955	98.6
Regis	Franchisees	180,091	40,112	81.8
Ventas	Operators of leased facilities	151,112	594	99.6
Jack in the Box	Franchisees	148,358	24,411	85.9
Papa John's	Franchisees	51,874	1,862	96.5
Wendy's	Franchisees	51,029	1,028	98.0
Planet Fitness	Franchisees	45,808	1,059	97.7
Welltower	Operators of leased facilities	44,409	1,735	96.2
Safeguard Scientifics	Minority holdings	39,313	0	100.0
Denny's	Franchisees	35,755	544	98.5
MTY Food Group	Franchisees	26,302	498	98.1
Marriott International	Franchisees	13,372	913,257	1.4
Sabra Health Care REIT	Operators of leased facilities	11,265	0	100.0
Other issuers (47)		57,166	1,648,639	3.4
All (62 issuers)		5,910,241	2,703,341	
of which franchisees' openings		5,619,885		

Notes. Retained openings, 2014–2023, assigned by the crosswalk to an issuer that does not consolidate the employer, by issuer. Franchisees: openings on a franchisor's career site that state that the employer is an independent franchisee, or that fall in the unit positions of a mostly franchised system (Section IA.III.3). Operators of leased facilities: health-care operators whose facilities a real estate investment trust owns. Minority holdings: companies in which the issuer holds a stake. Kept openings: the issuer's openings that remain in the sample. The fifteen issuers shown account for 99 percent of excluded openings.

IA.III.4 The Estimation Sample

The estimation sample retains the firm-years with at least 20 openings and with Compustat fundamentals.[7] It comprises 35,772 firm-years of 7,128 firms over 2014–2023, with 91.8 million openings. The knowledge and organization capital of Peters and Taylor (2017), which the validation of the innovation share uses beside R&D, comes from the data that WRDS distributes and is matched through the Compustat identifier. Table IA.30 describes it year by year, complementing Table 2, and Table IA.31 by the country of the issuer's headquarters, which is the United States for 62 percent of firm-years and 72 percent of openings. Of the openings themselves, 65.6 percent are located in the United States, from nearly all in 2014 to 57.0 percent in 2023. The largest other locations are the United Kingdom (4.5 percent), India (3.4), and Canada (3.2), and 8.8 percent record no country (Figure IA.8). Figure IA.9 shows that the threshold of 20 openings removes 13.8 percent of issuer-years but only 0.05 percent of openings, and Figure IA.10 that a firm's openings scale with its employment, with an elasticity of 0.67 across firms and of 0.77 from one year to the next within firms, as they should if they record the firm's own hiring. Table IA.32 lists the twenty issuers with the most openings, which together hold 18.3 percent of them.

Most firms are observed without interruption (Table IA.33). Of the 7,128 firms, 5,259 appear in several years with no year missing, 1,136 in a single year, and 733 with at least one year missing between their first and last. Of the 1,228 missing years, 724 record no opening at all and 496 record fewer than 20, while only 8 lack Compustat fundamentals. Comparisons within firms are therefore little affected, since 97.3 percent of the firm-years after a firm's first year follow a year in the sample. A firm's first year, however, records fewer openings than its later ones, as it would if LinkUp began to collect the firm's career sites during that year (Figure IA.11). Net of firm and year effects, openings per employee in the first year are 0.31 log points below their level three or more years later. The shortfall reaches 0.86 log points among large firms, whose median of 500 or more openings in their other years rules out the threshold of 20 openings as its cause, and 0.16 log points remain in their second year. The last year shows no shortfall of comparable size. Because the measure is a share, a partial year raises its sampling error without necessarily changing its level, and the environmental, social, and governance shares in a firm's first year differ from their later levels by at most 0.15 percentage points, none of them significantly. The innovation share, by contrast, is 1.0 percentage point higher in a firm's first year and 0.6 points higher in its second, about 8 and 5 percent of its mean.

[7] If openings were independent draws, the sampling error of an environmental share at its median of 0.8 percent would exceed the share itself with 20 openings. The 2,296 firm-years with 10 to 19 openings, which a lower threshold would add, hold 0.04 percent of all openings.

Table IA.30. The Estimation Sample by Year

				Openings per firm-year			Openings per	Career sites
Year	Firm-years	Firms entering	Openings (M)	P10	median	P90	100 employees	per firm-year
2014	1,632	1,632	1.3	29	104	1,142	1.9	1.36
2015	2,300	813	2.7	35	175	1,774	3.8	1.37
2016	2,561	471	6.0	48	350	4,109	7.5	1.40
2017	3,105	676	7.7	41	341	5,024	8.8	1.47
2018	3,831	905	9.4	44	336	4,933	11.5	1.43
2019	3,963	428	10.8	44	378	5,419	12.2	1.48
2020	4,052	481	9.2	39	302	4,037	10.4	1.43
2021	4,620	748	16.0	47	451	6,493	15.7	1.38
2022	4,858	540	17.9	49	482	7,268	19.0	1.37
2023	4,850	434	10.6	39	329	4,689	11.1	1.32
2014–2023	35,772	7,128	91.8	41	328	4,859	10.5	1.40

Notes. Firm-years with at least 20 openings and Compustat fundamentals with positive total assets, including issuers in Compustat Global. Firms entering: issuers whose first firm-year in the sample falls in the year. Openings per 100 employees: openings in the firm-year divided by Compustat employees, median over firm-years with positive employment. Career sites per firm-year: sites with at least one assigned opening in the year, mean. Openings are dated by the year of removal, and the year 2023 ends on 30 September. Table 2 reports the split between Compustat North America and Global and the coverage of Compustat North America.

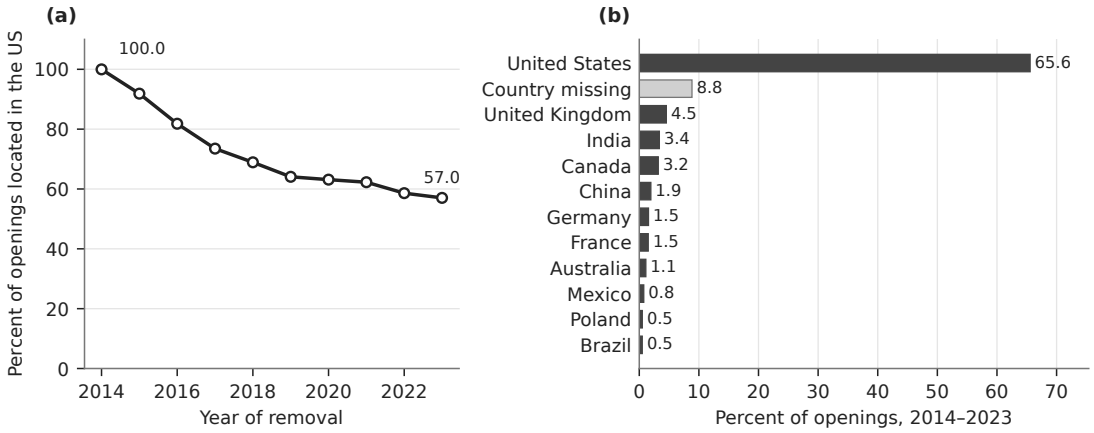


Figure IA.8. Geography of the Openings. (a) Share of the openings of the estimation sample located in the United States, by year of removal. (b) The twelve largest countries over 2014–2023, in percent of the openings; "country missing": no country recorded.

Table IA.31. The Estimation Sample by Country of Headquarters

Country of headquarters	Firm-years	%	Firms	Openings (M)	% of openings
United States	22,181	62.0	3,992	66.14	72.1
Canada	1,893	5.3	399	2.59	2.8
United Kingdom	1,833	5.1	403	5.66	6.2
Japan	1,207	3.4	225	1.22	1.3
France	1,068	3.0	206	3.16	3.4
Australia	1,055	2.9	238	0.92	1.0
Germany	730	2.0	160	2.49	2.7
China	515	1.4	216	0.95	1.0
Switzerland	498	1.4	88	1.68	1.8
Netherlands	397	1.1	78	0.92	1.0
India	359	1.0	96	0.59	0.6
Sweden	321	0.9	74	0.62	0.7
Ireland	316	0.9	50	1.85	2.0
New Zealand	232	0.6	54	0.08	0.1
Singapore	223	0.6	59	0.15	0.2
Israel	207	0.6	48	0.11	0.1
Belgium	205	0.6	46	0.12	0.1
Hong Kong	183	0.5	54	0.14	0.2
Brazil	182	0.5	58	0.22	0.2
Denmark	174	0.5	37	0.28	0.3
South Korea	169	0.5	41	0.14	0.1
Finland	150	0.4	36	0.22	0.2
Italy	148	0.4	39	0.10	0.1
Taiwan	138	0.4	43	0.04	0.0
Bermuda	135	0.4	23	0.19	0.2
Norway	131	0.4	37	0.06	0.1
Spain	127	0.4	34	0.28	0.3
Mexico	124	0.3	24	0.15	0.2
South Africa	119	0.3	33	0.11	0.1
Luxembourg	118	0.3	24	0.14	0.1
Other (42 countries)	634	1.8	213	0.44	0.5
All	35,772	100.0	7,128	91.77	100.0

Notes. Country of the issuer's headquarters in Compustat (`loc`). Countries with fewer than 100 firm-years are combined. Firms: all issuers of the estimation sample, whether in Compustat North America or Global; 3,987 of the 3,992 issuers headquartered in the United States are in Compustat North America. Openings are counted wherever the job is located.

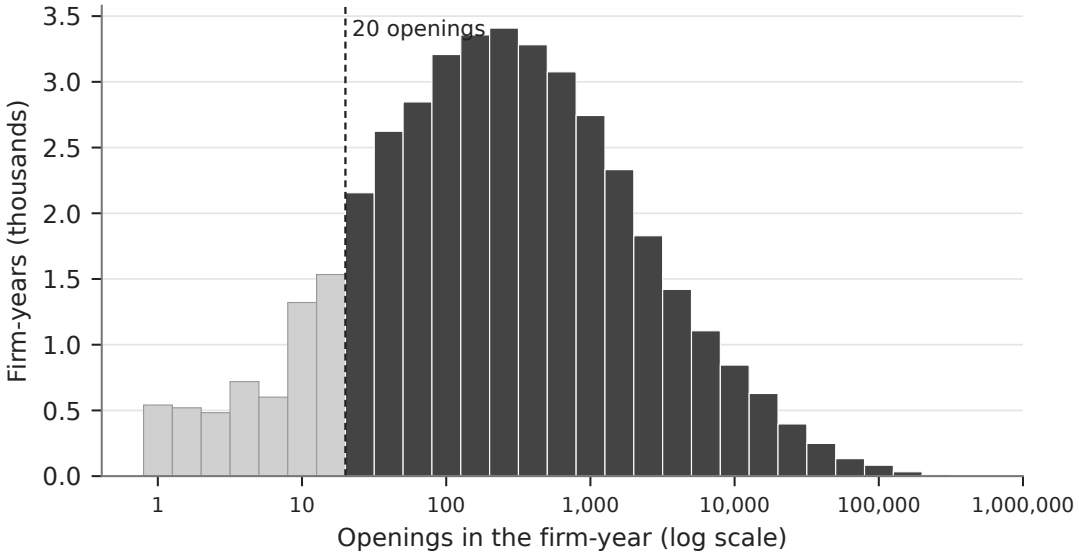


Figure IA.9. Openings per Firm-Year. Openings of the issuer-years with Compustat fundamentals in 2014–2023, on a logarithmic scale; the light bars fall below the threshold of 20 openings and leave the sample. The number of years in which each firm appears is in Table IA.33.

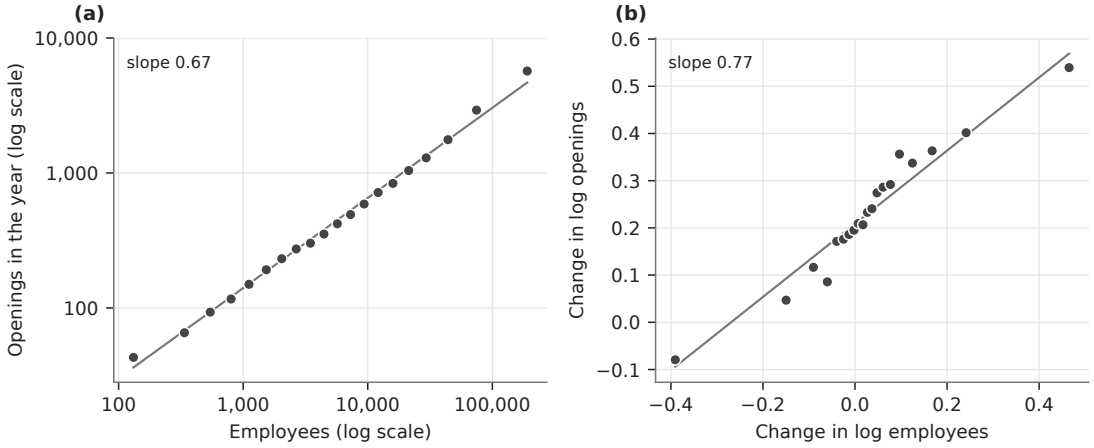


Figure IA.10. Openings and Employment. (a) Openings in the firm-year against the issuer's employees in Compustat, both in logarithms and net of year effects, averaged within 20 bins of equal size; the line is the fitted elasticity of 0.67 (standard error 0.01, clustered by firm). (b) The change in log openings from one year to the next against the change in log employees, for consecutive firm-years of the same issuer, net of year effects and winsorized at the 1st and 99th percentiles; slope 0.77 (0.03). Estimation sample, firm-years with positive employment.

Table IA.32. Issuers with the Most Openings

	Issuer	Sector	Openings (M)	Firm-years	Openings per 100 employees	Career sites
1	Lowe's	Retail	1.71	10	50.8	3
2	Amazon	Retail	1.24	10	13.9	22
3	Walgreens Boots Alliance	Retail	1.23	10	35.7	6
4	Walmart	Retail	1.09	10	3.7	18
5	Accenture	Professional services	0.92	10	13.4	24
6	Marriott International	Accommodation and food	0.91	10	61.2	1
7	Macy's	Retail	0.87	10	72.3	3
8	Target	Retail	0.83	10	20.2	3
9	FedEx	Transportation	0.83	10	14.5	8
10	HCA Healthcare	Health care	0.82	10	30.3	19
11	Wells Fargo	Finance	0.77	8	29.5	3
12	Compass Group	Accommodation and food	0.74	10	14.0	13
13	CVS Health	Health care	0.73	10	28.1	9
14	Kroger	Retail	0.72	10	17.3	7
15	Best Buy	Retail	0.66	10	57.5	8
16	Burlington Stores	Retail	0.57	10	107.9	2
17	JPMorgan Chase	Finance	0.56	10	23.4	6
18	Healthcare Services Group	Health care	0.56	9	68.9	1
19	Adecco	Administrative services	0.55	10	81.8	10
20	IBM	Information	0.47	9	13.4	25
	Share of all openings in the sample		18.3%			

Notes. The twenty issuers with the most openings in the estimation sample over 2014–2023. Openings per 100 employees: openings in the firm-year divided by Compustat employees, median over the issuer's firm-years. Career sites: sites with at least one opening assigned to the issuer over the period.

Table IA.33. Continuity of the Firms in the Estimation Sample

	Number	Percent
A. Firms		
All firms	7,128	100.0
In the sample for a single year	1,136	15.9
For several years, without a missing year	5,259	73.8
For several years, with at least one missing year	733	10.3
In the sample in all ten years	1,012	14.2
First year in the sample after 2014	5,496	77.1
Last year in the sample before 2023	2,278	32.0
B. Firm-years		
All firm-years	35,772	100.0
With the previous year in the sample	27,883	77.9
C. Years missing between a firm's first and last year		
All missing years	1,228	100.0
Fewer than 20 openings	496	40.4
No Compustat fundamentals	8	0.7
No opening recorded	724	59.0

Notes. Estimation sample: 35,772 firm-years of 7,128 firms, 2014–2023. A missing year is a year between a firm's first and last year in the sample in which the firm is not in the sample, because fewer than 20 consolidated openings were assigned to it, because it had no Compustat fundamentals with positive total assets, or because no opening was assigned to it at all.

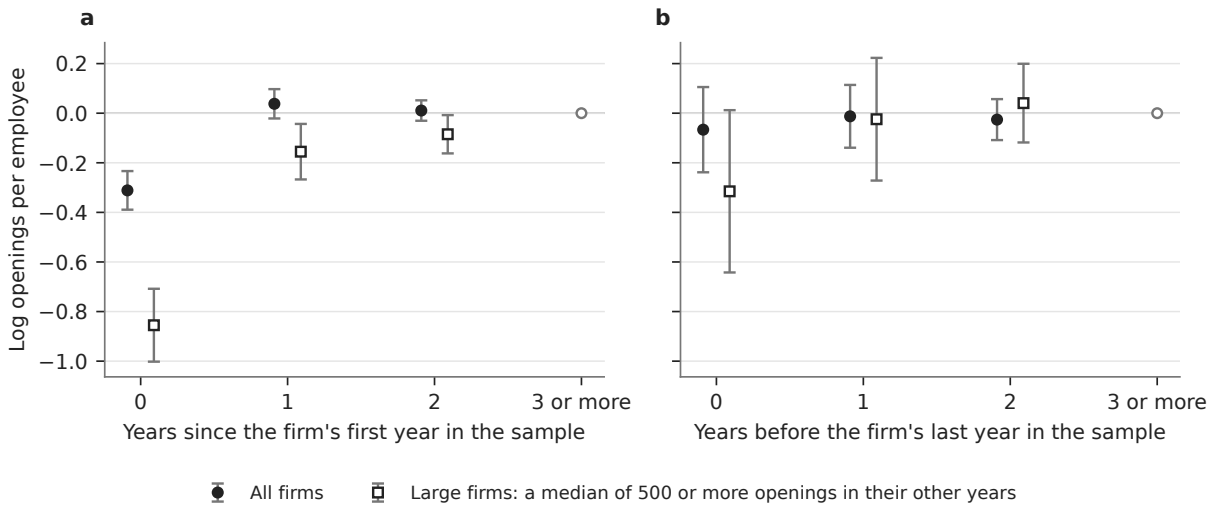


Figure IA.11. Openings around a Firm's First and Last Year in the Sample. Coefficients of a regression of log openings per employee in the firm-year on indicators for the years since the firm's first year in the sample (a) or before its last year (b), with firm and year effects; the reference is three or more years (open circle), and the bars are 95 percent confidence intervals with standard errors clustered by firm. (a) Firms whose first year in the sample is 2015 or later (4,195 firms); (b) firms whose last year is 2022 or earlier (1,471 firms). Squares: large firms, with a median of at least 500 openings in their other years in the sample (1,278 and 292 firms). Firm-years with positive employment.

IA.III.5 How the Sample Compares with Compustat and US Labor Statistics

Against the full universe of Compustat North America, 82,595 firm-years with positive assets over 2014–2023, the estimation sample includes a third of firm-years but 80 percent of employment and 88 percent of market value, because coverage rises steeply with size. It includes 79 percent of the firm-years of firms with 10,000 employees or more and 63 percent of those with 1,000 to 9,999, but 2.5 percent of those with fewer than 100 (Figure IA.12). Table IA.34 compares the firm-years that the sample includes with those it leaves out, which are smaller, younger, and less profitable, and Table IA.35 reports coverage by sector. Coverage is highest in retail and lowest in mining, oil, and gas, where seven in ten firm-years belong to companies incorporated in Canada, most of them small, so that the sample includes 9 percent of the sector's firm-years but 61 percent of its employment.

The openings of the estimation sample can also be compared with US labor-market statistics. Within the United States, they are distributed across states almost exactly as nonfarm employment, with a correlation of 0.985 between the shares (Figure IA.13). Across occupations, the correlation with employment is 0.68 (Figure IA.14). The openings over-represent computer occupations, at 3.3 times their share of employment, sales (2.2), engineering (2.0), and management occupations (1.6), and they under-represent education (0.05), construction (0.11), personal care (0.19), and production occupations (0.46), for the reasons discussed in Section 2.1 of the paper. Over time, the open US postings of the 1,488 sample firms that post in the United States in every year from 2015 move with the job openings of the Job Openings and Labor Turnover Survey, with correlations of 0.87 in levels and 0.85 in twelve-month changes (Figure IA.15). The correlations are the same for all sample firms, and 0.84 and 0.85 for the 1,142 of the constant firms headquartered in the United States.

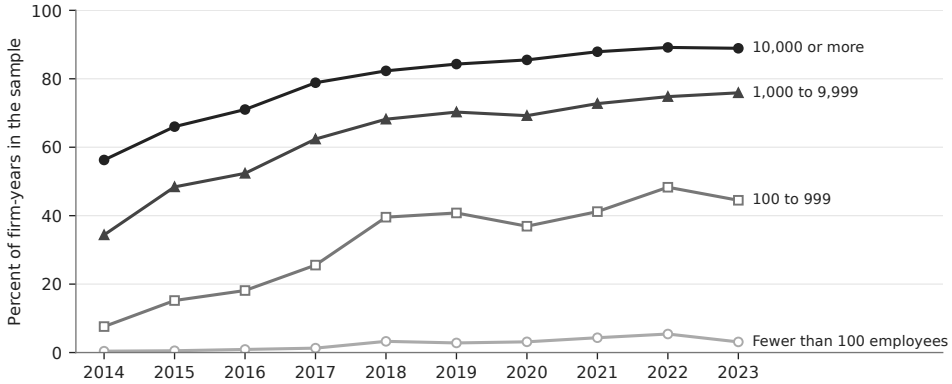


Figure IA.12. Coverage of Compustat North America by Firm Size. Share of the firm-years of Compustat North America with positive total assets that the estimation sample includes, by fiscal year and by the number of employees reported in Compustat. The 14,495 firm-years without a reported number of employees, 407 of them in the sample, are left out.

Table IA.34. Firm-Years of Compustat North America In and Out of the Estimation Sample

	In the estimation sample (N = 27,237)			Not in the sample (N = 55,358)			Difference
	Mean	SD	Median	Mean	SD	Median	in means
Total assets ($ billion)	35.71	188.99	3.09	4.99	53.76	0.11	30.72***
Sales ($ billion)	9.08	27.84	1.57	1.10	8.75	0.03	7.97***
Employees (thousands)	22.1	70.0	4.1	3.5	23.8	0.1	18.5***
Market value ($ billion)	16.24	64.46	2.76	1.30	8.07	0.09	14.94***
Age (years in Compustat)	24.7	18.5	21.0	13.0	13.4	9.0	11.7***
Sales growth	0.162	0.729	0.066	0.310	1.363	0.058	-0.148***
Tobin's q	2.12	2.10	1.46	9.47	36.86	1.38	-7.35***
Leverage	0.296	0.253	0.263	0.530	1.635	0.146	-0.235***
Return on assets	0.062	0.232	0.091	-0.705	2.812	0.006	0.768***
R&D / assets	0.049	0.115	0.000	0.110	0.323	0.000	-0.061***
SG&A / assets	0.214	0.292	0.125	0.750	2.391	0.107	-0.536***
Capex / assets	0.037	0.049	0.022	0.051	0.109	0.011	-0.014***

Notes. Firm-years of Compustat North America in 2014–2023 with positive total assets, by whether the estimation sample includes them. Dollar amounts in US dollars, converted from Canadian dollars at the exchange rate of the fiscal year-end month. Age: years since the firm's first fiscal year in Compustat. Sales growth, Tobin's q, leverage, return on assets, and R&D, SG&A, and capital expenditure over lagged assets are winsorized at the 1st and 99th percentiles within year; missing R&D and SG&A are set to zero. Difference in means: coefficient on an indicator for the sample in a regression without controls, with standard errors clustered by firm. *, **, and *** denote significance at the 10, 5, and 1 percent levels.

Table IA.35. The Estimation Sample by Industry

		Estimation sample			Compustat North America			
		Firm-	Share (%) of		Firm-	Share in the sample (%)		
NAICS	Sector	years	firm-years	openings	years	firm-years	employees	mkt. value
11	Agriculture	66	0.2	0.0	216	15.3	84.1	87.5
21	Mining, oil, and gas	1,268	3.5	0.6	10,918	9.2	61.2	70.7
22	Utilities	1,014	2.8	1.0	2,726	29.5	62.0	82.7
23	Construction	652	1.8	0.9	986	37.5	75.6	76.7
31-33	Manufacturing	13,700	38.3	22.0	27,958	35.7	83.2	92.3
42	Wholesale trade	1,100	3.1	2.1	1,779	43.7	81.5	80.2
44-45	Retail trade	2,184	6.1	22.0	2,452	59.1	88.1	92.7
48-49	Transportation	1,026	2.9	2.5	2,234	31.0	61.7	75.7
51	Information	4,616	12.9	9.4	8,269	43.6	70.6	86.2
52	Finance and insurance	4,395	12.3	10.5	12,771	34.3	79.1	84.1
53	Real estate	1,098	3.1	2.3	4,096	25.5	75.2	71.4
54	Professional services	1,805	5.0	7.3	2,321	43.3	94.1	95.7
56	Administrative services	751	2.1	3.4	1,266	41.5	69.2	91.6
61	Education	177	0.5	0.1	513	27.5	46.2	41.7
62	Health care	639	1.8	6.6	1,026	48.8	94.4	94.4
71	Arts and recreation	319	0.9	0.5	545	38.2	79.5	80.4
72	Accommodation and food	715	2.0	6.8	1,111	48.2	85.6	90.4
81	Other	247	0.7	1.8	1,408	12.6	76.5	94.6
	All	35,772	100.0	100.0	82,595	33.0	80.3	87.7

Notes. Firm-years of the estimation sample, including issuers in Compustat Global, by the NAICS sector of the issuer, with their shares of all firm-years and of all openings in the sample. The last four columns give the firm-years of Compustat North America (2014–2023, positive total assets) in each sector and the shares of their number, employees, and market value that the estimation sample includes, pooled over the years. Sectors 31–33, 44–45, and 48–49 are combined, and "Other" combines NAICS 55, 81, 92, and 99 with issuers without a NAICS code. Sector: historical NAICS code in Compustat, else the current one.

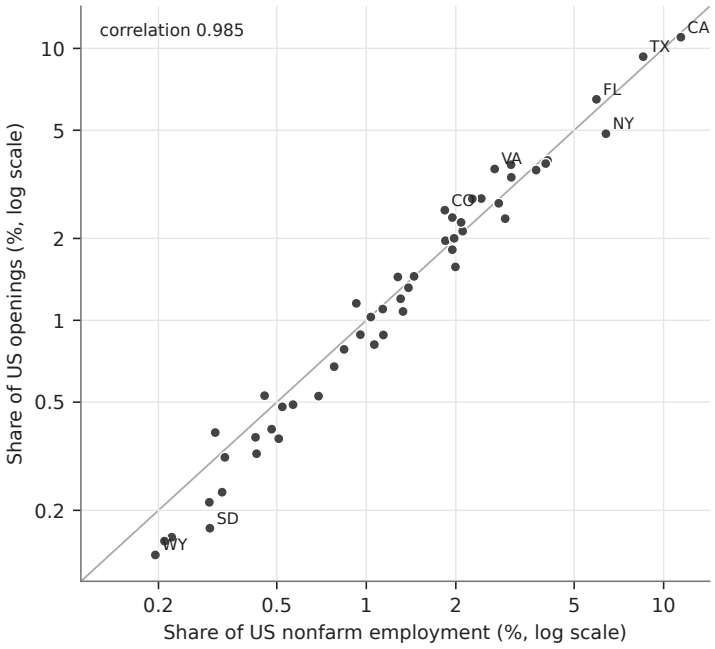


Figure IA.13. US Openings and Employment by State. Each state's share of the US openings of the estimation sample over 2014–2023, against its share of total nonfarm employment, averaged over the same years, in the Current Employment Statistics of the Bureau of Labor Statistics (monthly state series obtained from FRED), on logarithmic scales; the line marks equal shares. The 50 states and the District of Columbia; openings without a state or in the territories are left out.

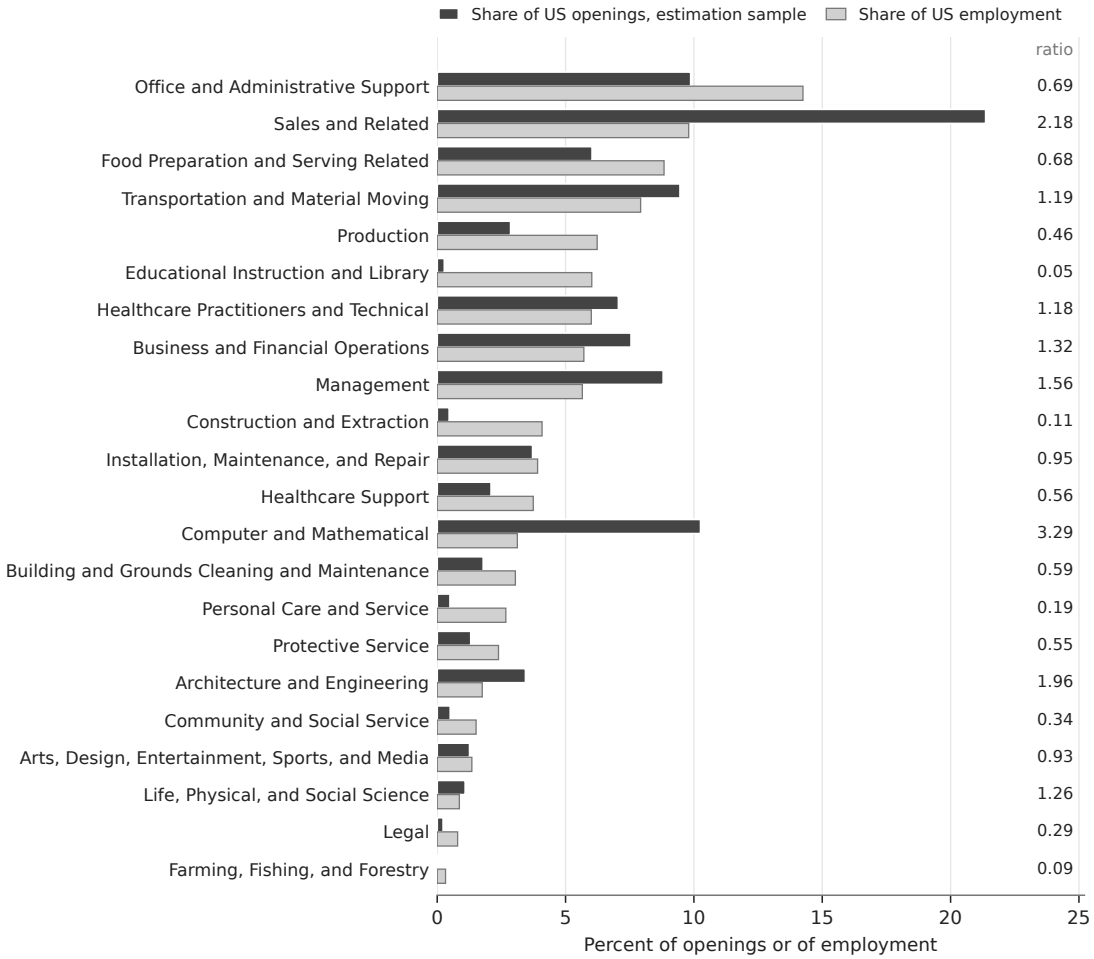


Figure IA.14. Occupations of the Openings and of US Employment. For each SOC major group, the share of the US openings of the estimation sample in 2014–2023 (59.2 million with a specific occupation code) and the share of US employment in the national May estimates of the Occupational Employment and Wage Statistics of the Bureau of Labor Statistics (mean of 2014–2023), sorted by employment share. At the right, the ratio of the two shares. The correlation of the shares is 0.68, and the dissimilarity index is 0.28.

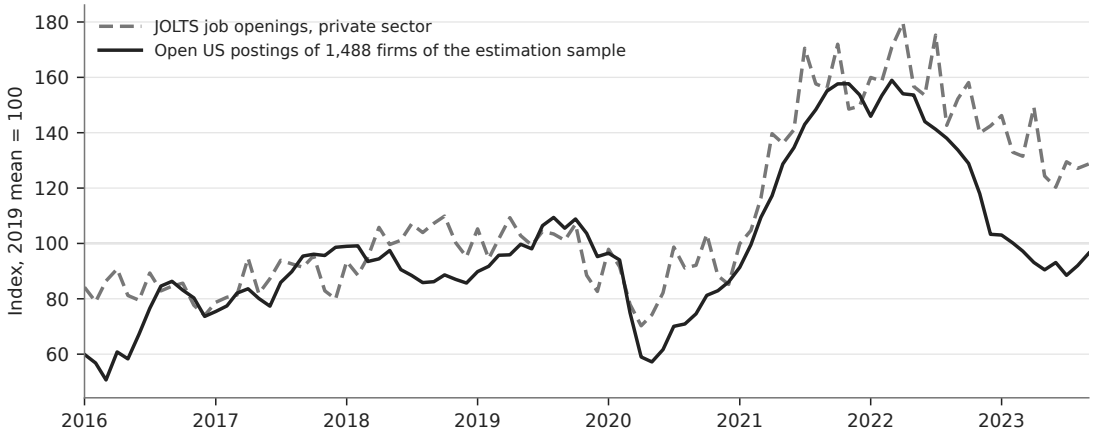


Figure IA.15. Open Postings and Job Openings in the United States. Open US postings of the 1,488 firms of the estimation sample with at least one US posting created in every year from 2015 to 2023, at the end of each month, against job openings in the private sector reported by the Job Openings and Labor Turnover Survey of the Bureau of Labor Statistics (series JTU1000JOL, not seasonally adjusted, obtained from FRED), each indexed to its 2019 mean, January 2016 to September 2023. A posting counts as open from its creation until its removal or its 180th day, whichever comes first, and belongs to a sample firm if its consolidating issuer is in the estimation sample in the year of its removal, or in 2023 if it is still open at the end of the data. The two series correlate at 0.87 in levels and at 0.85 in twelve-month log changes.

IA.III.6 Patents

Patent data come from Google Patents, which compiles the publications of patent offices worldwide. I use the applications published by the US Patent and Trademark Office, together with their assignees and technology classes. A single office counts each invention once, rather than once for every office in which it is filed, and the three existing patent–firm links through which I attribute applications to issuers, those of Ma (2026), Dyèvre and Seager (2023), and WRDS, pooled, are keyed to granted US patents. For issuers listed outside North America, the counts therefore cover their US filings only. A link counts only if the application's filing date falls within an interval in which the crosswalk from career sites to issuers (Section IA.III.3) links the issuer to a career site, and within the period in which LinkUp observes that site, so that patents and postings are attributed to issuers through the same dated links. Each link is also checked against the assignee name printed on the patent. Applications not yet granted, and grants that the links miss, are matched to the dated names of issuers and their subsidiaries (the name panel of Section IA.III.7), and granted patents that the links place on an issuer in years in which its career site did not yet post are recovered when they print a Compustat or CRSP name of the issuer, or a name already accepted for it. Of the 2.7 million issuer–application pairs filed between 2007 and September 2023, which involve 4,437 issuers, 58 percent come from the links, 23 percent from names, and 19 percent from recovered grants. I count applications by year of filing in four groups: all applications; applications in the Y02 class of the Cooperative Patent Classification; applications in the classes of the technologies that the IRA subsidizes (Section 2.4); and applications in the environment-related technologies of the OECD's ENV-TECH classification (Haščič and Migotto, 2015; Favot et al., 2023). ENV-TECH covers most of Y02 and adds the technologies of pollution abatement, waste management, water, and environmental monitoring, which are the technologies of environmental management work. I code firm-years without an attributed application as zero. At least one application is filed in 42 percent of firm-years, a Y02 application in 18 percent, and an IRA-class application in 10 percent. The economic value of granted patents is the stock market reaction to their grant (Kogan et al., 2017), from the extended data of Kogan et al. (2017), which cover firms with a CRSP security; I use it for applications filed through 2020.

IA.III.7 Records of Environmental, Workplace, and Legal Activity

Names of establishments. Records of establishments carry only the names of their operators. For these records, I use a panel of dated names that links 860,016 name variants to 9,785 issuers. It is built from the historical names of issuers in Compustat and CRSP, the

names of their career sites in LinkUp, and the subsidiaries that issuers list in their Exhibit 21 filings, each dated by the filings in which it appears. Names that point to more than one issuer are dropped, generic names are not used, and hand rulings and a list of rejected names override the automatic matches.

Environmental records. Four sources record a firm's environmental activity. Environmental incidents come from RepRisk, which screens news and other public sources for reports of firms' involvement in environmental controversies and rates the severity of each. I match RepRisk's entities to issuers through their ISIN or CUSIP, requiring their names to agree, and attribute incidents at subsidiaries through their career sites, to the issuer that the crosswalk from career sites to issuers (Section IA.III.3) links to the site on the date of the incident, counting each story once. An incident is severe if RepRisk rates its severity at two or more on its three-point scale. The Toxics Release Inventory records the pounds of listed toxic chemicals that industrial facilities release each year. I match facilities through the parent company that each facility reports in each year in the TRI facility history, by its standardized and raw names and then by the name stripped of legal suffixes, by the current name, and by the name of a foreign parent, against the dated name panel described above. The match links 91,734 of 377,510 facility-years, 24 percent of them, accounting for 51 percent of the pounds released, to 1,318 issuers. Formal enforcement cases at the matched facilities come from the agency's Enforcement and Compliance History Online, linked through the program identifiers of its Facility Registry Service. Finally, environmental suits are the federal cases against SEC registrants that Audit Analytics classifies as environmental, linked to issuers through the defendant's SEC identifier alone. In each source, I code covered firm-years without an event as zero.

Tax credits in 10-K filings. Two records concern the activity that the clean-energy tax credits pay for, and neither comes from hiring; Section 7.2 uses them to ask whether the firms whose hiring was oriented toward clean work later undertook that activity. The full-text search of SEC EDGAR indexes the text of filings since 2001, including their exhibits. I query it for 10-K filings with exact phrases, one calendar year at a time, so that no query reaches the search's limit of 10,000 results, and I link filings to firms through the Central Index Key in Compustat. For filings from January 2023 to October 2026, the phrases are the section numbers of nine credits that the Inflation Reduction Act created or extended, such as "45X", and the name "advanced manufacturing production credit". Table IA.36 lists the credits and the number of sample firms that mention each. For filings from January 2019 to July 2022, which measure earlier activity, the phrases are "production tax credit", "investment

tax credit", their plurals, and "45Q". Of the 2,466 US firms in the sample, 217 (8.8 percent) mention one of them before August 2022.

Section numbers can appear in other contexts, so I read the list of firms that mention the manufacturing credit. Of these 26 firms, 17 make solar, wind, battery, fuel-cell, or electric-vehicle products or their inputs, among them First Solar, Enphase Energy, Array Technologies, TPI Composites, EnerSys, and Rivian, and four produce metals and engineered materials (Alcoa, ATI, Materion, and Kennametal). The other five are three coal producers, a bank, and a maker of medical devices. Of the 70 firms that mention any of the nine credits, 45 are manufacturers, 8 are utilities, and 2 are in finance and insurance.

The law also lets firms sell their credits to other taxpayers (section 6418), and four further phrases capture such transfers. They reach a different set of firms. Of the 80 firms that mention a transfer but none of the nine credits, 24 are in finance and insurance, among them Citigroup, Prudential Financial, and Visa, and 19 are utilities. Because buyers of credits do not carry out the activity that the credits pay for, column (1) of Table 15 counts the nine credits only. With the transfer phrases included, the mean rises to 6.3 percent and the coefficient on the environmental share is 6.48 ($t = 5.6$). A mention is not a receipt: a filing may discuss eligibility, expectations, or credits that the firm bought.

Clean facilities. The Clean Investment Monitor, release 2026_Q2 of August 2026, lists 10,721 clean manufacturing, energy, and industrial facilities in the United States, with the company, the announcement date, the estimated capital expenditure in 2024 dollars, and the current status of each. Of these facilities, 711 are manufacturing plants. I match each facility's company to the dated names of issuers and of the subsidiaries listed in their Exhibit 21 filings that were valid in 2018–2023, after removing the plant-operator codes of the Energy Information Administration, as I match patent applications not yet granted (Section IA.III.6). A name must point to a single issuer, or be that issuer's own name. Names shorter than four characters, generic names, and legal forms alone are dropped, and one match that the audit found wrong is removed.

This name match assigns 1,254 facilities to issuers. Its largest misses were wholly owned subsidiaries of sample firms whose names the name lists lack, such as Florida Power & Light (NextEra Energy), AES Clean Energy (AES), and Southern Power (Southern Company), or whose key is a common word, such as Ford. I therefore read the 400 unmatched companies with the largest announced capital expenditure and match by hand 26 company names to the sample firm that wholly owns them or that they are. Joint ventures stay unmatched because they have no single owner, as do companies whose owner changed during the outcome window, such as Consolidated Edison Development, sold to RWE in 2023. With these matches, 1,709

facilities (15.9 percent) are assigned to 207 issuers, and they account for 23.5 percent of announced capital expenditure. The largest unmatched owners are private developers (such as Invenergy and Intersect Power), foreign firms outside the sample (such as Toyota and RWE), and joint ventures (such as BlueOvalSK, owned by Ford and SK On). Before the hand matches, the coefficients of Table 15, Panel A, were 2.24 ($t = 2.5$) for any facility, 3.16 ($t = 3.5$) for manufacturing facilities, and 0.209 ($t = 2.4$) for announced capital expenditure.

From August 2022 to June 2026, 79 US sample firms announce 536 facilities, 54 of them manufacturing plants.

Table IA.36. The Credits Searched in 10-K Filings

	Firms mentioning	Percent of firms	Mean clean share (SD units)
Advanced manufacturing production credit (45X)	26	1.05	3.08
Clean electricity production credit (45Y)	6	0.24	5.05
Clean electricity investment credit (48E)	14	0.57	5.11
Clean hydrogen production credit (45V)	8	0.32	3.83
Carbon oxide sequestration credit (45Q)	13	0.53	1.38
Clean fuel production credit (45Z)	18	0.73	2.01
Qualifying advanced energy project credit (48C)	11	0.45	4.73
Qualified commercial clean vehicle credit (45W)	6	0.24	5.82
Zero-emission nuclear power production credit (45U)	3	0.12	4.10
Transfer of credits (section 6418)	97	3.93	0.95
Any of the nine credits (Table 15, column 1)	70	2.84	2.41
Any credit or transfer	150	6.08	1.36

Notes. US-located sample firms (2,466) and their 10-K filings, including exhibits, filed from January 2023 to October 2026, from the full-text search of SEC EDGAR. A firm mentions a credit if any of its filings contains the section number in quotation marks, or, for section 45X, also the phrase "advanced manufacturing production credit". Transfer of credits: "transferable tax credits", "tax credit transfer", "sale of tax credits", or "transfer of tax credits". Mean clean share: the mean of the standardized clean-technology share of 2019–2021 among the firms that mention the credit.

Workplace and legal records. The Injury Tracking Application of the Occupational Safety and Health Administration records the injuries, illnesses, and hours worked that establishments report each year from 2016. I link establishments first through the employer identification number that they report from 2019, which Compustat records for each issuer, then through their names, and finally through hand rulings, releasing identification numbers that several issuers share and the establishments of franchisees, and I drop 6,414 records with implausible hours worked. The link covers 511,953 of 2.73 million establishment-years, 19 percent of them, accounting for 24 percent of hours worked. Inspections and violations come from the enforcement data of the Department of Labor, matched through establishment names

and the addresses of the establishments linked above, which covers 4 percent of inspections. Securities, class-action, derivative, and fraud suits come from Audit Analytics, linked through the defendant's SEC identifier and assigned to the fiscal year that contains the filing date. The yearly number of securities class actions in these data is between 0.96 and 1.20 times that of the Stanford Securities Class Action Clearinghouse over 2014–2023, with a correlation of 0.988.

IA.III.8 Existing Measures of Climate Exposure

Greenhouse gas emissions come from Trucost, matched through the Compustat identifier that S&P Global records for every company and assigned to the Compustat fiscal year that ends within 14 days of Trucost's. Trucost estimates Scope 1 emissions where firms do not report them, reported values are available mostly from 2020 onward, and its coverage of the estimation sample rises in 2016, from 59 to 91 percent of firm-years. LSEG's scores are matched through ISIN or CUSIP and, for subsidiaries, through career sites, and assigned to the Compustat fiscal year; the climate-change exposure of Sautner et al. (2023) is matched through the Compustat identifier that their data carry, or else through ISIN, by calendar year.

The clean revenue share uses the business segments of Compustat, or the operating segments where a firm reports no business segments, as first reported. I classify a segment as clean technology from its name and industry code (Section 2.6) and divide the revenue of clean segments by the revenue of all segments of the firm. Where a firm has no segment data, the share is set to zero with an indicator.

IA.III.9 Market Data, Forecasts, Options, and Ownership

Daily returns come from CRSP, and abnormal returns are measured against the five factors of Fama and French (2015) and momentum (Carhart, 1997), from the factor files that WRDS distributes. Returns after December 2024, which the windows after the 2024 election and the placebo days of that year require, come from the CRSP daily file in its new format, whose returns are identical to those of the earlier file for 98 percent of the firm-days the two share and correlate with them at 1.000. Price, shares outstanding, and exchange and share codes at the end of each month come from the CRSP monthly file, and opening prices from the daily file split the return of a session into its overnight and intraday parts. I link each firm to its CRSP security through the link valid on the day of each event. The controls of equation (9) are measured before the event. Market equity is that at the end of the month before the event. Book-to-market divides the most recent book equity reported at least four months

before the event by that market equity. Momentum is the cumulative return over months -12 to -2. Market beta and idiosyncratic volatility, the standard deviation of the residuals, come from the factor regression used to compute abnormal returns (Appendix D). The hiring shares are those of the calendar year before the event, and so are the logarithms of the firm's openings, employees, and assets, which control for size and for the coverage of the hiring data. The exposures to taxes and trade, which control for the repricing of firms through these exposures that the same news may cause, follow Wagner, Zeckhauser and Ziegler (2018) and Ramelli et al. (2021). They are the cash effective tax rate, which is income taxes paid over pretax income, the foreign share of pretax income, and the foreign share of sales from the geographic segments.

Analysts' forecasts come from the monthly consensus of IBES, linked to CRSP through the link that WRDS maintains for it, retaining only its matches of the highest quality. I use the mean forecasts of earnings per share for the next three fiscal years and of long-term growth, and the number of analysts who revise their forecasts up or down. For each day, I compare the last consensus computed before the event with the first one computed after it, for the same fiscal period, and I scale revisions of earnings forecasts by the share price, adjusted for later stock splits because IBES restates past forecasts to the current number of shares. Option prices come from the standardized volatility surfaces of OptionMetrics, linked to CRSP in the same way, at maturities of 30 and 91 days, which report the premium and strike of out-of-the-money options at each delta, and from the closing prices of the underlying stocks, from which I compute option-implied expected returns (Martin and Wagner, 2019).

Ownership comes from two files that WRDS distributes, the holdings of mutual funds in the CRSP Mutual Fund database, with each fund's name as in force at each report, and the holdings that institutions report on Form 13F, from the Thomson Reuters file, matched to CRSP through historical CUSIPs. Product markets come from the text-based network industry classification of Hoberg and Phillips (2016), which is keyed to Compustat issuers. The sensitivity of each firm to interest rates and to the price of oil is measured against the daily 10-year Treasury yield of the Federal Reserve and the daily price of Brent crude oil, both from FRED.

IA.IV The Accounting of Labor Costs

Firms develop capabilities largely through their own employees, and the resources that a firm commits to a capability appear in its financial statements, but not as such, because accounting classifies costs by their nature and function rather than by the activity that they serve. Table IA.37 shows what this implies for environmental work, whose labor cost lands in different lines of the accounts depending on what the work produces. The wages of engineers who build a solar plant for the firm's own use are capitalized in property, plant, and equipment. Those of technicians who produce battery cells enter inventory and then the cost of goods sold. Those of researchers who develop electrolyzers are expensed as R&D, those of the staff who prepare emissions reports enter SG&A, and the costs of remediating past contamination are accrued as a liability. Each line therefore mixes environmental work with every other activity that it finances. Even where wages are capitalized, they count as investment only as part of the cost of the asset that they help produce, so that an engineer who builds a solar plant and one who builds a warehouse are recorded identically. International standards differ only at the margin. Under IFRS, development costs may be capitalized only under strict conditions, and internally generated brands and customer lists are not recognized as assets (IAS 38).

Only a few broad activities, such as research and development and advertising, have expense lines of their own, and even these record how much a firm spends on an activity rather than which capabilities it develops. They are also often missing, and what they contain varies across firms.[8]

[8]Among nonfinancial firms in Compustat North America over fiscal years 2014–2023, R&D expenditure is not reported in 43 percent of firm-years and SG&A expenses in 15 percent, or 52 and 18 percent when financial firms are included, and the same expenditure may be classified as cost of goods sold by one firm and as SG&A by another (Lev, 2019).

Table IA.37. Where the Labor Cost of Environmental Work Is Recorded

Work	Example from the postings	Line of the accounts	Standard
A. Clean technology			
Building clean-energy assets for the firm's own use	A project engineer who builds a utility's solar plant	Property, plant, and equipment (capitalized)	ASC 360
Producing clean products	A technician on a battery-cell production line	Inventory, then cost of goods sold	ASC 330
Developing clean technology	An engineer who develops hydrogen electrolyzers	R&D expense	ASC 730-10-25-1
B. Environmental management			
Permits and monitoring at a plant	A plant's environmental, health, and safety specialist	Production overhead in inventory at a manufacturing plant; otherwise operating expense	ASC 330
Reporting and permitting at headquarters	A sustainability reporting analyst	SG&A	Regulation S-X, Rule 5-03
Emissions accounting software built in-house	A developer of an internal greenhouse-gas inventory tool	Capitalized internal-use software (development costs)	ASC 350-40
Remediating past contamination	A remediation project manager	Accrued environmental liability	ASC 410-30

Notes. The lines in which US GAAP records the wages of each kind of work, under the standards in the last column. The examples describe typical positions, not specific postings. In Compustat, SG&A includes R&D. No line records environmental work as such, and each line also records the same kind of cost for every other activity of the firm.

IA.V The Policy-Day Tests: Data and Documentation

This part documents the data behind the tests of Sections 6 to 8 that the main text and Appendix E do not describe: the days of policy news, the sample of firm-events, the days of large market moves, the estimates behind Figure 2 for ratings and emissions, the firms behind the dollar amounts, the clean business segments, the ownership of firms by funds that invest by environmental, social, and governance criteria, the option-implied expected returns, and the measures of tax capacity.

IA.V.1 The Policy Days

The application relates returns to the orientation of hiring on days on which news about US climate policy reached the market. The days follow earlier studies of that policy, which identified them as the main moments at which its prospects changed (Kruse, Mohnen and Sato, 2024; Ramelli et al., 2021; Bauer, Offner and Rudebusch, 2026; Martins et al., 2025), and each is signed by the content of its news, positive when the news raised the expected support for clean activity and negative when it lowered it. Table IA.38 lists the days, and the paragraphs below describe what each meant for the policy, how much of its news the market could have anticipated, and what other news it carried.

The Paris Agreement, 14 December 2015. On Saturday, 12 December 2015, the parties to the United Nations Framework Convention on Climate Change adopted the Paris Agreement, under which countries committed to limit global warming through national contributions. The agreement raised the expected support for clean activity over the following decades, and its first trading day was Monday, 14 December (Kruse, Mohnen and Sato, 2024).

The 2016 election, 9 November 2016. Donald Trump, who had promised during the campaign to withdraw the United States from the Paris Agreement, was elected president on 8 November 2016, and the outcome became known overnight. The election was a large surprise, since betting markets and forecasting models gave him 17 and 28 percent probabilities of winning on election day (Ramelli et al., 2021). It also carried news about taxes and trade (Wagner, Zeckhauser and Ziegler, 2018), for which the paper controls on the election days.

The withdrawal from the Paris Agreement, 1 June 2017. The president announced the withdrawal of the United States from the Paris Agreement on 1 June 2017. The announcement lowered the expected support for clean activity, but the withdrawal had been promised during the campaign (Kruse, Mohnen and Sato, 2024), so that much of its news was anticipated.

The 2020 election, 9 November 2020. The major networks called the presidential election for Joe Biden on Saturday, 7 November 2020, and the first trading day after the call was Monday, 9 November. The outcome reversed the direction of 2016 (Ramelli et al., 2021), but it was widely expected, with probabilities of 78 and 89 percent for Biden on election day. On the same Monday, Pfizer and BioNTech announced the first results of their COVID-19 vaccine trial, which moved returns across the whole cross-section, so that the paper reports this day but does not interpret its coefficient.

Senator Manchin's withdrawal, 14 July 2022. On the evening of 14 July 2022, after the close of trading, press reports stated that Senator Manchin, the pivotal vote in the Senate, would not support new climate spending, which brought the probability of the legislation close to zero (Bauer, Offner and Rudebusch, 2026). The paper reports every result with and without this day.

The agreement on the IRA, 27 July 2022. On 27 July 2022, at 5:03 pm, after the close of trading, Senator Manchin and the Democratic leadership announced their agreement on the Inflation Reduction Act, which raised the probability of its passage close to one (Bauer, Offner and Rudebusch, 2026). The act, signed in August 2022 (U.S. Congress, 2022), pays firms for clean activity mostly through uncapped tax credits per unit of output or investment, which account for $271 billion of the $392 billion that its climate provisions were expected to cost through 2031 (Bistline, Mehrotra and Wolfram, 2023, pp. 78–86). Because it concerned a single policy and came as a surprise, it is the cleanest of the days. The window includes the session of 27 July, on which the Federal Reserve raised interest rates, and the first full session after the announcement, 28 July.

The 2024 election, 6 November 2024. Donald Trump was re-elected on 5 November 2024, and the outcome became known overnight (Martins et al., 2025). The election lowered the expected support for clean activity, and in July 2025, after the last of the days studied, a budget reconciliation law curtailed several of the credits of the IRA (U.S. Congress, 2025). The hiring shares used for this day are those of 2023, the last year of the hiring data.

Two further days. Two other days carried news about the same policy. The Georgia Senate runoffs of 5 January 2021 gave Democrats control of the Senate, which later made the IRA possible; the first session after the result, 6 January 2021, is also the day of the attack on the Capitol, and the outcome carried news about fiscal policy broadly. On 19 December 2021, Senator Manchin said that he could not vote for the Build Back Better Act and its

climate credits, which ended their prospects at the time; the first session after his statement, 20 December 2021, fell in a week of news about the Omicron variant. The paper reports both days as further evidence (Section 6.6).

The regulation days. Two days concern the regulation of emissions and climate disclosure rather than the payments for clean activity, and the paper uses them to compare clean technology with environmental management. On 30 June 2022, the Supreme Court limited the authority of the Environmental Protection Agency to regulate the emissions of power plants (Supreme Court of the United States, 2022), which lowered the expected demand for environmental management work. On 6 March 2024, the Securities and Exchange Commission adopted its rules on climate-related disclosure (U.S. Securities and Exchange Commission, 2024), which raised it.

Table IA.38. The Days of Policy News

Day 0	News	Direction	When the news reached the market	Source
A. Climate-policy days (Sections 5 to 7)				
14 Dec 2015	The parties to the UN Framework Convention on Climate Change adopt the Paris Agreement	+	Saturday, 12 December	Kruse, Mohnen and Sato (2024)
9 Nov 2016	Donald Trump is elected president	−	Overnight after the vote of 8 November	Ramelli et al. (2021)
1 Jun 2017	The president announces the withdrawal from the Paris Agreement	−	Day 0	Kruse, Mohnen and Sato (2024)
9 Nov 2020	Joe Biden is elected president	+	Networks call the race on Saturday, 7 November	Ramelli et al. (2021)
14 Jul 2022†	Senator Manchin will not support new climate spending	−	Press reports at 9:29 pm, after the close	Bauer, Offner and Rudebusch (2026)
27 Jul 2022	Senator Manchin and the Democratic leadership announce their agreement on the IRA	+	Announcement at 5:03 pm, after the close	Bauer, Offner and Rudebusch (2026)
6 Nov 2024	Donald Trump is re-elected president	−	Overnight after the vote of 5 November	Martins et al. (2025)
B. Regulation days (Section 6)				
30 Jun 2022	The Supreme Court limits the EPA's authority to regulate power-plant emissions (*West Virginia v. EPA*)	−	Day 0	Supreme Court of the United States (2022)
6 Mar 2024	The SEC adopts rules on climate-related disclosure	+	Day 0	U.S. Securities and Exchange Commission (2024)

Notes. Day 0 is the day on which the news broke, or the next trading day if it broke on a weekend; returns are measured over trading days −1 to +1. Direction: the sign d_e of equation (9), + for news that raised the expected support for clean activity. On the regulation days, the direction refers to the expected demand for environmental management work. †Reported with and without, as in Table 11.

IA.V.2 The Event Sample

The firm-events of each policy day are the firms in Compustat North America with a CRSP security, linked through the CRSP–Compustat Merged link valid on the day of the event, and with an environmental share in the calendar year before the event. Table IA.39 reports their number on each day, the number that enter the regressions of Section 6, which require every control, and the numbers with the data used in Section 7. Abnormal returns use the daily CRSP stock file through 31 December 2024. The windows that extend into 2025, those after the 2024 election and those of the candidate days of the placebo tests, use the CRSP daily file in its new format (`dsf_v2`) from 1 January 2025. On the three months in which both files exist, October to December 2024, 98.1 percent of the 193,451 common firm-days have identical returns, the largest difference is 0.15 percentage points, and the two series correlate at 1.000. Price and market equity for the screens of Appendix Table E6 come from the CRSP monthly file at the end of the month before each event, and the NYSE breakpoint is the 20th percentile of the market equity of ordinary common shares listed on the NYSE in that month, between $597 million (2020) and $936 million (2024).

Table IA.39. Firm-Events by Policy Day

	Firm-events stored (1)	In the regression (2)	With option data (3)	With ESG-fund holdings (4)
Paris Agreement, 14 Dec 2015	1,336	1,271	1,285	1,206
Trump elected, 9 Nov 2016	1,742	1,647	1,683	1,425
Paris withdrawal, 1 Jun 2017	1,897	1,774	1,814	1,671
Biden elected, 9 Nov 2020	2,568	2,409	2,331	2,230
Manchin withdraws, 14 Jul 2022	2,904	2,721	2,722	2,473
IRA agreement, 27 Jul 2022	2,908	2,723	2,725	2,473
Trump re-elected, 6 Nov 2024	2,810	2,616	2,650	2,445

Notes. Column 1: firms with a CRSP security, an environmental share, and an abnormal return. Column 2: with every control of equation (9). Column 3: with 30-day option-implied expected returns on at least three of the four trading days before and after the event. Column 4: held by at least one ESG fund at its last report before the day.

IA.V.3 The Days of Large Market Moves

Appendix Table E8 compares the climate-policy days with days of large market moves. In each calendar year from 2015 to 2024, I select the two largest rises and the two largest falls of the daily market excess return, taken in order of the size of the move, excluding days within seven calendar days of any of the policy days or of a day already selected, so that no two

windows overlap. Table IA.40 lists the 40 days. Returns on these days are computed exactly as on the policy days and signed by the direction of the market.

Table IA.40. The Days of Large Market Moves

Day	Direction	Market (%)	Day	Direction	Market (%)
24 Aug 2015	fall	-3.91	16 Mar 2020	fall	-12.01
01 Sep 2015	fall	-2.91	24 Mar 2020	rise	+9.34
05 Oct 2015	rise	+1.93	06 Apr 2020	rise	+7.10
04 Dec 2015	rise	+1.87	11 Jun 2020	fall	-5.91
13 Jan 2016	fall	-2.66	27 Jan 2021	fall	-2.53
29 Jan 2016	rise	+2.57	25 Feb 2021	fall	-2.72
01 Mar 2016	rise	+2.34	07 Dec 2021	rise	+2.22
24 Jun 2016	fall	-3.69	21 Dec 2021	rise	+2.06
01 Mar 2017	rise	+1.47	13 Jun 2022	fall	-4.12
24 Apr 2017	rise	+1.18	13 Sep 2022	fall	-4.29
17 May 2017	fall	-1.97	04 Oct 2022	rise	+3.29
17 Aug 2017	fall	-1.60	10 Nov 2022	rise	+5.68
05 Feb 2018	fall	-4.03	06 Jan 2023	rise	+2.21
26 Mar 2018	rise	+2.67	21 Feb 2023	fall	-2.16
04 Dec 2018	fall	-3.46	25 Apr 2023	fall	-1.76
26 Dec 2018	rise	+5.07	14 Nov 2023	rise	+2.18
04 Jan 2019	rise	+3.55	22 Feb 2024	rise	+2.02
04 Jun 2019	rise	+2.32	05 Aug 2024	fall	-3.02
05 Aug 2019	fall	-3.07	15 Aug 2024	rise	+1.77
14 Aug 2019	fall	-2.95	18 Dec 2024	fall	-3.17

Notes. Market: the Fama–French daily market excess return on the day, in percent.

IA.V.4 Ratings and Carbon Intensity Alone, Day by Day

Table IA.41 reports the coefficients plotted in Figure 2. Columns (1) and (2) give the LSEG environmental score and carbon intensity, each alone. Carbon intensity has the sign that a carbon price would imply on four days: negative on the Paris Agreement and the 2020 election, positive on the 2024 election and on 14 July 2022. Columns (3) to (6) enter the two hiring shares, the score, and carbon intensity together, on the firms that have both a score and emissions data. The clean-technology share keeps its coefficient on the agreement on the IRA (0.87) and on the 2024 election (-1.43). On the 2016 election, these firms are larger than average, and the coefficient is -0.64 (Appendix Table E20). The score keeps a negative coefficient on five days beside the hiring shares.

Table IA.41. Each Day: Environmental Ratings and Carbon Intensity

	Alone		Together, on the firms with both				
	LSEG score	Carbon intensity	Clean technology	Environ-mental management	LSEG score	Carbon intensity	Firm-events
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Paris Agreement, 14 Dec 2015	−0.268*	−0.075	−0.006	−0.006	−0.313**	0.048	660
	(0.160)	(0.226)	(0.377)	(0.138)	(0.158)	(0.233)	
Trump elected, 9 Nov 2016	−0.626***	−1.128***	−0.637**	0.465**	−0.469*	−0.976***	835
	(0.231)	(0.304)	(0.251)	(0.229)	(0.250)	(0.332)	
Paris withdrawal, 1 Jun 2017	−0.002	−0.328***	−0.076	−0.067	0.019	−0.332***	1,487
	(0.087)	(0.120)	(0.102)	(0.073)	(0.090)	(0.122)	
Biden elected, 9 Nov 2020	0.395*	−1.817***	0.188	−0.104	0.236	−1.925***	2,105
	(0.216)	(0.341)	(0.200)	(0.195)	(0.221)	(0.368)	
Manchin withdraws†, 14 Jul 2022	−0.207*	0.037	−0.401***	−0.053	−0.133	0.052	2,300
	(0.115)	(0.138)	(0.117)	(0.134)	(0.117)	(0.148)	
IRA agreement, 27 Jul 2022	−0.141	0.209	0.873***	0.208*	−0.256*	0.306*	2,300
	(0.143)	(0.180)	(0.126)	(0.111)	(0.142)	(0.177)	
Trump re-elected, 6 Nov 2024	−0.402**	0.476*	−1.426***	−0.164	−0.180	0.605**	1,977
	(0.204)	(0.283)	(0.241)	(0.154)	(0.215)	(0.284)	

Notes. One regression per day and column, as in Appendix Table E1. Columns (1) and (2): the LSEG environmental score or carbon intensity, the logarithm of Trucost's Scope 1 and 2 emissions over market equity (Ramelli et al., 2021), each standardized within event and entered alone with the event controls and NAICS-3 effects, on the firms for which it exists (713 to 2,397 firm-events for the score and 737 to 2,490 for carbon intensity); these are the estimates of Figure 2. Columns (3) to (6): the clean-technology and environmental management shares, the score, and carbon intensity together, on the firms with both measures (column 7). †Not one of the six main days. Standard errors clustered by firm in parentheses. *, **, and *** denote significance at the 10, 5, and 1 percent levels.

IA.V.5 The Concentration of the Repricing in Dollars

Table IA.42 shows how the dollar amounts of Table 12 are distributed across firms. On every day, the tenth of firms furthest above their industry's mean accounts for 78 to 89 percent of the amount. On the 2016 election, the amount is more dispersed than on the IRA agreement and the 2024 election, since the five largest firms of the top decile account for $2.0 billion of the $19.7 billion.

IA.V.6 Clean Business Segments

Table IA.43 describes the clean-technology segments used in Section 6.6. The most frequent names are those of electric vehicles, renewable energy, solar power, ethanol, wind power, and energy storage. With 14 July 2022 added to the six days, the results of Appendix Table E13 are similar: the coefficient of the environmental share is 0.59 in the segment sample and 0.44 beside the clean revenue share, and 0.35 among firms that report no clean segment.

Table IA.42. The Repricing in Dollars: Concentration

	Firms above industry mean ($bn) (1)	Top decile ($bn) (2)	Top decile, share of (1) (%) (3)	Five largest firms of the top decile ($bn) (4)
Paris 2015	5.6	4.5	81	1.1
Trump 2016	−19.7	−16.2	82	−2.0
Paris exit 2017	−1.4	−1.2	81	−0.2
Biden 2020	−3.4	−2.9	87	−0.9
Manchin 2022†	−16.2	−14.3	89	−4.7
IRA 2022	56.0	49.7	89	16.4
Trump 2024	−101.7	−79.5	78	−38.3

Notes. The amounts of Table 12, in billions of US dollars, relative to industry peers. They are the repricing that the regression attributes to the share, with all else held fixed, and not the observed returns of any firm. Column (1): the sum over the firms whose environmental share exceeds their NAICS-3 industry's mean. Column (2): the sum over the tenth of firms furthest above their industry's mean, and column (3) its share of column (1). Column (4): the sum over the five firms of the top decile with the largest market value at the end of the month before the event. †Not one of the six main days.

Table IA.43. Clean-Technology Business Segments

A. Coverage	
Segment firm-years	80,794
with a clean-technology segment	1,770 (2.2%)
Firm-events of the climate-policy days with segment data	15,719 of 16,165
with a clean-technology segment	338

B. Most frequent names of clean-technology segments

Segment name	Firms
Electric Vehicles	17
Renewables	15
Renewable Energy	8
Solar	7
Solar Energy	7
Ethanol	6
Wind	6
Energy Storage	5
Renewable Power	4
Ethanol Production	4
Energy	4
Solar Energy Systems	4

Notes. Compustat business segments, or operating segments where a firm reports no business segments, as first reported. A segment is clean technology if its name refers to renewable power, solar, wind, batteries or energy storage, electric vehicles or charging, hydrogen or fuel cells, biofuels, renewable natural gas, carbon capture, or clean or low-carbon energy, or if its industry code is that of renewable power generation, storage batteries, or ethanol (Section 2.6); generic names such as "Energy" enter through the segment's industry code. Panel B: the twelve names held by the most firms.

IA.V.7 Ownership by ESG Funds and by Institutions

I identify funds that invest by environmental, social, and governance criteria from their names in the CRSP Mutual Fund database. A fund is classified as such when its name contains one of a list of terms fixed before the estimation, namely ESG, sustainable, responsible, SRI, impact, climate, clean, green, environmental, low carbon, fossil free, renewable, solar, wind, or alternative energy, matched as whole words where a shorter word could otherwise match. A name counts only while it is in force, from the date on which the fund adopted it to the date on which it changed, so that a fund renamed in the wave of relabeling after 2019 is not counted before its new name. The list identifies 714 portfolios over the period, among them index funds and exchange-traded funds such as iShares ESG Advanced MSCI EAFE and Xtrackers S&P 500 ESG, active funds such as those of Calvert, Pax, and Domini, and thematic funds such as First Trust NASDAQ Clean Edge and PowerShares Global Wind Energy. For each day, I take each portfolio's last holdings report in the 183 days before the event and divide the shares that these portfolios hold by the firm's shares outstanding at the end of the month before. Table IA.44 shows how much these funds grew. They held on average 0.04 percent of a firm's shares before 2020 and about 0.2 percent from 2020 onward. Institutional ownership comes from the holdings that institutions report on Form 13F, in the Thomson Reuters (now LSEG) file distributed by WRDS. For each stock and quarter, I take each manager's largest reported position, which removes the duplicates of late amendments, sum these positions over managers, and divide the sum by the shares outstanding in CRSP at the end of the quarter. Securities are matched to CRSP through their historical CUSIP, and the few values above 1.5, which arise from overlapping filings, are set to 1.5. Each firm-event takes the last quarter-end before the event.

Table IA.44. ESG-Fund Ownership by Policy Day

	ESG	Ownership (% of shares)			Firms with
	portfolios	Mean	Median	90th pct.	none (%)
	(1)	(2)	(3)	(4)	(5)
Paris Agreement, 14 Dec 2015	58	0.04	0.01	0.07	8
Trump elected, 9 Nov 2016	74	0.04	0.01	0.07	17
Paris withdrawal, 1 Jun 2017	90	0.04	0.01	0.09	10
Biden elected, 9 Nov 2020	229	0.15	0.05	0.28	12
Manchin withdraws, 14 Jul 2022	366	0.23	0.10	0.43	13
IRA agreement, 27 Jul 2022	366	0.23	0.10	0.43	13
Trump re-elected, 6 Nov 2024	349	0.21	0.11	0.42	11

Notes. Column 1: ESG portfolios whose name was ESG at their last report in the 183 days before the day. Columns 2–5: firm-events of the regressions; ownership in percent of shares outstanding.

IA.V.8 Option-Implied Expected Returns

Martin and Wagner (2019) show that the expected excess return of a stock can be written in terms of risk-neutral variances that option prices reveal. In their formula, the expected excess return of stock i over a horizon T equals the risk-free rate times $\mathrm{SVIX}_t^2 + \frac{1}{2}(\mathrm{SVIX}_{i,t}^2 - \overline{\mathrm{SVIX}}_t^2)$, where SVIX_t^2 is the risk-neutral variance of the market and $\overline{\mathrm{SVIX}}_t^2$ the value-weighted mean of the stocks' risk-neutral variances. On a given day these market terms are the same for every firm, so that across firms the expected excess return differs by one half of $\mathrm{SVIX}_{i,t}^2$, and the industry-by-event effects of the regressions absorb the rest. I compute $\mathrm{SVIX}_{i,t}^2 = 2\left[\int_0^F P_{i,t}(K)\,dK + \int_F^\infty C_{i,t}(K)\,dK\right] / \left(TR_{f,t}S_{i,t}^2\right)$, annualized, from the standardized volatility surfaces of OptionMetrics at maturities of 30 and 91 days, using the out-of-the-money puts with deltas from -0.50 to -0.10 and calls with deltas from 0.10 to 0.50, with the premium and strike that the surface reports for each delta, integrated over strikes by the trapezoidal rule. $S_{i,t}$ is the closing price, and $R_{f,t}$ the gross risk-free return over the horizon.

Because the surface stops at deltas of 0.10, the integral omits the deep tails, which lowers the level of the index but should not affect its changes around a given day. To check the integration, I apply it to the options on the S&P 500 with the weights of the VIX, $1/K^2$, on the 110 trading days of the windows around the six days. The index rebuilt in this way correlates with the CBOE VIX at 0.995, at 89 percent of its level, between 82 and 95 percent (Figure IA.16). Across the firm-events of the six days, one half of the 30-day $\mathrm{SVIX}_{i,t}^2$ averages 21.8 percentage points a year, with a median of 11.6, and 15.7 and 8.7 at 91 days. The outcome of Appendix Table E28 is its change from the four trading days before each event to the four trading days after it, for the firm-events with at least three days on each side (12,488 of the 13,261 on the six days).

IA.V.9 Tax Capacity

The measures of Appendix Table E27 use the Compustat annual file for the fiscal year before each event. A firm has a tax loss carryforward when Compustat reports a positive value of `tlcf`, with missing values treated as none, and a loss when pretax income (`pi`) is negative. Low tax capacity combines the two. Carryforwards are common, since they include small and old losses, so that about two thirds of the firm-events have one, whereas 16 to 31 percent have a loss in the year before, depending on the day.

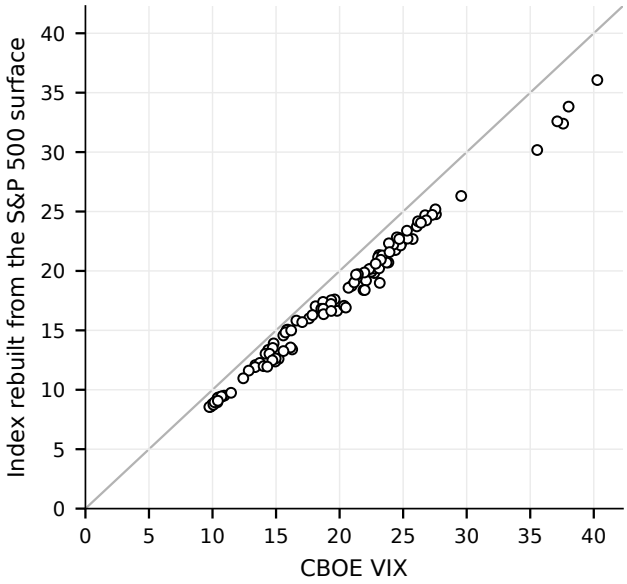


Figure IA.16. The Integration of the Option Surfaces against the VIX. Each point is a trading day in the windows around the six days: the CBOE VIX (FRED, VIXCLS) and the index computed from the 30-day OptionMetrics surface of the S&P 500 with the integration of this section and the weights $1/K^2$; the line marks equality.

IA.VI Risk and Pricing Details

This part of the Internet Appendix reports an interpretive exercise that the paper does not use as evidence, the complementarity of environmental capability with tangible capacity.

IA.VI.1 Complementarity with Tangible Capacity

The conclusion offers an interpretation in which environmental capability complements tangible capacity in production, while its value is most exposed to the policy where it is a large part of the firm, and Appendix B.3 reports the association of clean-technology hiring with capital expenditures. These tests ask whether the capability complements tangible capital, as organizational investment complements computers (Brynjolfsson, Hitt and Yang, 2002), and I report them here in full. Table IA.45 interacts the two parts of the share with tangible intensity. The association of the clean-technology share with capital expenditures (Appendix Table C7) does not vary with tangible intensity, in the same year or the next, and neither does its association with Y02 applications. In value, the interactions point the other way. Tobin's q does not rise with the clean-technology share at higher tangible intensity, and within firms the interaction is negative (-0.05, $t = -2.0$), as is the interaction on the policy days in Appendix Table E32.

Table IA.46 asks whether past hiring builds into technology. Among firms with above-median tangible intensity before the sample period, the clean-technology shares of the three preceding years predict Y02 applications, with a summed coefficient of 0.34 ($t = 3.1$) across firms and 0.26 ($t = 2.4$) within firms. Among the other firms, the summed coefficients are -0.29 and -0.08, although the share of the current year has positive coefficients within firms in both groups. The difference does not reflect persistence: the clean-technology share is equally persistent in both groups (Panel B). The lagged innovation shares, by contrast, have positive coefficients in both groups. The samples of the two parts are small, from 1,025 to 3,348 firm-years, and the environmental management share shows no consistent pattern. I therefore treat these results as consistent with complementarity rather than as a test of it.

Table IA.45. Environmental Hiring, Tangible Capacity, Value, and Investment

	Tobin's q		Y02 applications		Capital expenditures	
	Across (1)	Within (2)	Across (3)	Within (4)	Across (5)	Within (6)
A. Tangible intensity in the year						
Clean technology	−0.002	0.025	0.125***	0.076**	0.270***	−0.058
	(0.037)	(0.045)	(0.039)	(0.035)	(0.056)	(0.093)
Environmental management	−0.067**	0.036	−0.000	0.190**	0.019	−0.099
	(0.030)	(0.027)	(0.160)	(0.075)	(0.049)	(0.066)
Tangible intensity	−0.145***	−0.329***	0.260**	−0.070	2.786***	1.719***
	(0.038)	(0.053)	(0.122)	(0.171)	(0.101)	(0.181)
Clean technology × tangible	−0.011	−0.046*	−0.081	−0.021	−0.007	0.020
	(0.025)	(0.024)	(0.056)	(0.025)	(0.061)	(0.049)
Environmental management × tangible	0.027**	−0.022	−0.028	−0.103	−0.072**	0.034
	(0.012)	(0.015)	(0.074)	(0.073)	(0.036)	(0.049)
Firm-years	23,732	23,169	20,010	6,842	25,902	25,248
B. Tangible intensity before the sample period (2011–2013)						
Clean technology	−0.008	−0.065	0.115	0.100**	0.213***	−0.109
	(0.041)	(0.060)	(0.076)	(0.045)	(0.071)	(0.098)
Environmental management	−0.082***	0.060*	−0.005	0.368***	−0.016	−0.138**
	(0.029)	(0.031)	(0.185)	(0.110)	(0.042)	(0.064)
Tangible intensity	−0.079	−0.953*	0.169	−1.777	2.161***	6.270***
	(0.049)	(0.511)	(0.160)	(1.358)	(0.124)	(1.935)
Clean technology × tangible	0.006	0.052	−0.101	0.008	−0.070	0.027
	(0.021)	(0.033)	(0.101)	(0.049)	(0.088)	(0.059)
Environmental management × tangible	0.040***	−0.044**	0.030	−0.240**	0.004	0.070
	(0.014)	(0.018)	(0.083)	(0.106)	(0.038)	(0.053)
Firm-years	19,972	19,670	14,485	5,411	22,005	21,650

Notes. Firm-years 2014–2023. Tobin's q (market sample), Y02 applications in $t+1$ to $t+2$ (Poisson pseudo-maximum likelihood), and capital expenditures over lagged assets in year t (percentage points, North America) on the clean-technology and environmental management shares of year t, tangible intensity, and their interactions, each share and tangible intensity standardized within year. Tangible intensity is property, plant, and equipment over assets in the year (Panel A) or over fiscal 2011–2013 (Panel B). Controls: the social, governance, and innovation shares and the logarithms of assets, employees, and openings. Across: NAICS-3-by-year effects; within: firm effects added. Standard errors clustered by firm in parentheses. *, **, and *** denote significance at the 10, 5, and 1 percent levels.

Table IA.46. Does Past Hiring Build into Technology?

	Above-median tangible intensity				Below-median tangible intensity			
	Across		Within		Across		Within	
	Year t (1)	Lags (2)	Year t (3)	Lags (4)	Year t (5)	Lags (6)	Year t (7)	Lags (8)
A. Patent applications in $t+1$ to $t+2$ on the share in t and the sum of its three lags								
Clean technology, Y02	-0.318^{*}	0.341^{***}	0.102^{**}	0.258^{**}	0.649^{***}	-0.291	0.264^{***}	-0.085
	(0.176)	(0.109)	(0.043)	(0.109)	(0.190)	(0.245)	(0.082)	(0.104)
Firm-years	3,110		1,443		3,348		1,025	
Environmental management, Y02	0.472^{***}	-0.405	0.302^{**}	-0.066	-1.144^{*}	0.426	0.383	-0.616
	(0.155)	(0.263)	(0.150)	(0.168)	(0.678)	(0.361)	(0.363)	(0.443)
Firm-years	3,110		1,443		3,348		1,025	
Innovation, all applications	0.027	0.298^{**}	-0.023	0.124	0.450^{***}	0.111	0.084^{**}	0.145^{**}
	(0.121)	(0.122)	(0.039)	(0.105)	(0.146)	(0.120)	(0.037)	(0.068)
Firm-years	3,777		2,497		3,637		2,175	
B. Persistence: the share in $t+1$ on the share in t								
	Share in t		× above median					
Clean technology	0.914^{***}		-0.009		Firm-years: 18,302			
	(0.046)		(0.047)					
Environmental management	0.906^{***}		-0.031		Firm-years: 18,302			
	(0.037)		(0.039)					
Innovation	0.880^{***}		-0.006		Firm-years: 18,302			
	(0.008)		(0.010)					

Notes. Firm-years 2017–2023, split by whether the firm's tangible intensity over fiscal 2011–2013 is above the median of the year. Panel A: patent applications in $t+1$ to $t+2$ (Y02 for the two parts of environmental work, all classes for the innovation share), by Poisson pseudo-maximum likelihood, on the share of year t and its values in $t-1$, $t-2$, and $t-3$; "Lags" is the sum of the three lagged coefficients. Controls: the other shares and the logarithms of assets, employees, and openings. Across: NAICS-3-by-year effects; within: firm effects added. Panel B: the share in $t+1$ on the share in t, its interaction with above-median tangible intensity, and the indicator, with NAICS-3-by-year effects, over the firm-years with both years. Standard errors clustered by firm in parentheses. *, **, and *** denote significance at the 10, 5, and 1 percent levels.

References

Bassier, Ihsaan, Alan Manning, and Barbara Petrongolo. 2025. "Vacancy Duration and Wages." *Review of Economics and Statistics*. Forthcoming.

Bauer, Michael D., Eric A. Offner, and Glenn D. Rudebusch. 2026. "The Effect of US Climate Policy on Financial Markets: An Event Study of the Inflation Reduction Act." In *Econometrics of Climate, Energy, and Green Transition. Advances in Econometrics*, 57–90. Emerald Publishing Limited.

Bistline, John E. T., Neil R. Mehrotra, and Catherine Wolfram. 2023. "Economic Implications of the Climate Provisions of the Inflation Reduction Act." *Brookings Papers on Economic Activity*, 2023(1): 77–182.

Brynjolfsson, Erik, Lorin M. Hitt, and Shinkyu Yang. 2002. "Intangible Assets: Computers and Organizational Capital." *Brookings Papers on Economic Activity*, 2002(1): 137–198.

Carhart, Mark M. 1997. "On Persistence in Mutual Fund Performance." *Journal of Finance*, 52(1): 57–82.

Dyèvre, Arnaud, and Oliver Seager. 2023. "70 Years of Patents Matched to Compustat Firms: Methodology and Insights About Firm Heterogeneity." Working paper. Data: https://github.com/arnauddyevre/compustat-patents.

Fama, Eugene F., and Kenneth R. French. 2015. "A Five-Factor Asset Pricing Model." *Journal of Financial Economics*, 116(1): 1–22.

Favot, Marinella, Leyla Vesnic, Riccardo Priore, Andrea Bincoletto, and Fabio Morea. 2023. "Green Patents and Green Codes: How Different Methodologies Lead to Different Results." *Resources, Conservation and Recycling Advances*, 18: 200132.

Haščič, Ivan, and Mauro Migotto. 2015. "Measuring Environmental Innovation Using Patent Data." OECD Publishing OECD Environment Working Paper 89, Paris.

Hoberg, Gerard, and Gordon M. Phillips. 2016. "Text-Based Network Industries and Endogenous Product Differentiation." *Journal of Political Economy*, 124(5): 1423–1465.

Kogan, Leonid, Dimitris Papanikolaou, Amit Seru, and Noah Stoffman. 2017. "Technological Innovation, Resource Allocation, and Growth." *Quarterly Journal of Economics*, 132(2): 665–712.

Kruse, Tobias, Myra Mohnen, and Misato Sato. 2024. "Do Financial Markets Respond to Green Opportunities?" *Journal of the Association of Environmental and Resource Economists*, 11(3): 549–576.

Lev, Baruch. 2019. "Ending the Accounting-for-Intangibles Status Quo." *European Accounting Review*, 28(4): 713–736.

LinkUp. 2020. "LinkUp Raw: Overview and User Guide." LinkUp, Data documentation, December 2020.

LinkUp. 2026. "Data Dictionary: RAW Data Package Feeds." *LinkUp Data Support knowledge base, article 56*, Accessed 13 October 2026.

Ma, Song. 2026. "PatentsView to Compustat/CRSP Firm-Name Match (patent-firm-link-share)." *GitHub repository, https://github.com/songma/patent-firm-link-share*, Patent-level links to Compustat and CRSP accompanying Ma, "Technological Obsolescence," *Review of Financial Studies* (2025); grants through 30 December 2025.

Martin, Ian W. R., and Christian Wagner. 2019. "What Is the Expected Return on a Stock?" *Journal of Finance*, 74(4): 1887–1929.

Martins, António Miguel, Bruno Albuquerque, Luís Sardinha, and Nuno Moutinho. 2025. "2024 U.S. Presidential Elections: An Event Study for U.S. And Non-U.S. Fossil Fuel and Renewable Listed Firms." *International Review of Financial Analysis*, 105: 104430.

Peters, Ryan H., and Lucian A. Taylor. 2017. "Intangible Capital and the Investment-q Relation." *Journal of Financial Economics*, 123(2): 251–272.

Ramelli, Stefano, Alexander F. Wagner, Richard J. Zeckhauser, and Alexandre Ziegler. 2021. "Investor Rewards to Climate Responsibility: Stock-Price Responses to the Opposite Shocks of the 2016 and 2020 U.S. Elections." *Review of Corporate Finance Studies*, 10(4): 748–787.

Sautner, Zacharias, Laurence van Lent, Grigory Vilkov, and Ruishen Zhang. 2023. "Firm-Level Climate Change Exposure." *Journal of Finance*, 78(3): 1449–1498.

Supreme Court of the United States. 2022. "West Virginia v. Environmental Protection Agency." *597 U.S. 697, No. 20-1530, decided 30 June 2022.*

Törnberg, Petter. 2024. "Best Practices for Text Annotation with Large Language Models." *Sociologica*, 18(2): 67–85. arXiv:2402.05129.

U.S. Congress. 2022. "Inflation Reduction Act of 2022." *Public Law 117-169, 136 Stat. 1818, 16 August 2022*.

U.S. Congress. 2025. "An Act to Provide for Reconciliation Pursuant to Title II of H. Con. Res. 14." *Public Law 119-21, 139 Stat. 72, 4 July 2025*, commonly known as the One Big Beautiful Bill Act.

U.S. Securities and Exchange Commission. 2024. "The Enhancement and Standardization of Climate-Related Disclosures for Investors." *Final rules, Release Nos. 33-11275 and 34-99678, File No. S7-10-22, 6 March 2024*.

Vona, Francesco, Giovanni Marin, Davide Consoli, and David Popp. 2018. "Environmental Regulation and Green Skills: An Empirical Exploration." *Journal of the Association of Environmental and Resource Economists*, 5(4): 713–753.

Wagner, Alexander F., Richard J. Zeckhauser, and Alexandre Ziegler. 2018. "Company Stock Price Reactions to the 2016 Election Shock: Trump, Taxes, and Trade." *Journal of Financial Economics*, 130(2): 428–451.